Two-year Programs and College Financial Aid*

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Abstract

We evaluate recent policy proposals to make two-year programs tuition-free in a life cycle model featuring two-year (AA) and four-year (BA) programs. We document program differences in accessibility and quality and incorporate them into the model: the AA program is easier to get into and cheaper, but has a lower consumption value, graduation likelihood, and labor market returns. We find that making the AA program tuition-free increases postsecondary attainment cost-effectively, but reduces productivity and output. Furthermore, general equilibrium effects and transition dynamics erase any welfare gains. A spending-equivalent BA subsidy increases productivity and output, and leads to substantial welfare gains.

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1 Introduction

In the United States, two-year programs award Associate of Arts (AA) degrees and often serve a "transfer function" to four-year programs where graduates are predominantly awarded bachelor's (BA) degrees (Kane and Rouse, 1999; Lovenheim and Smith, 2023).[1] Several recent federal policy proposals have pushed to make two-year programs tuition-free at the national level, although these proposals have not yet been passed into law. Examples include the *America's College Promise Act of 2015* (U.S. Congress, 2015) and the *Free Community College* program in the Department of Education's 2024 fiscal year budget proposal (U.S. Department of Education, 2023). These proposed policies aim to increase the share of the population with a postsecondary education (PSE) degree in a cost-effective way. However, when subsidizing tuition for two-year programs as opposed to four-year programs, there is a tradeoff between subsidizing accessibility and subsidizing quality: compared to four-year programs, two-year programs are more accessible in that they are easier to get into and are cheaper, but are also of lower quality in terms of graduation likelihood, labor market returns, and the consumption value of on-campus experiences.

We contribute to the policy discussion in two ways. First, we document tradeoffs across the two programs and patterns in paths to highest postsecondary enrollment. Second, we build a quantitative life cycle model of postsecondary education that features both a two-year (AA) and four-year (BA) program and incorporates the tradeoffs we document empirically. We use our model to analyze a counterfactual experiment in which the AA program is made tuition-free, juxtaposed with a second experiment in which a spending-equivalent subsidy is given to BA program enrollees.

Our main finding is that making AA programs tuition-free is a cost-effective way to increase the share of graduates with a postsecondary education degree. However, the AA subsidy lowers average labor productivity and output. Furthermore, once we take general equilibrium effects and transition dynamics into account, the AA subsidy leads to minimal welfare gains or even losses. In contrast, the spending-equivalent BA subsidy leads to an increase in labor productivity and output and substantial welfare gains.

In the first part of our paper, we present empirical evidence of the tradeoffs between AA and BA programs, as well as between AA-to-BA and straight-to-BA paths to BA enrollment. Our findings rely mostly on the 1997 National Longitudinal Survey of Youth (NLSY97) and the High School Longitudinal Study of 2009 (HSLS:09), two nationally-representative panel datasets collected in the United States. The tradeoffs we document are divided into those related to accessibility and quality, described above. These tradeoffs are conceptualized and examined in similar forms in other studies using different datasets and methodologies—see for example Kane and Rouse (1995), Reynolds (2012), and Jenkins and Fink (2016). Our results articulate these tradeoffs and document supporting evidence in a way that is useful for our quantitative

[1]We use AA interchangeably with two-year and BA with four-year. AA refers to two-year degrees generally, which are usually either associate of arts or associate of sciences. BA refers to four-year bachelor's degrees generally, which are usually either bachelor's of arts or bachelor's of science. Besides AA degrees, two-year institutions may also offer non-degree credentials such as short certificates or long certificates or diplomas, which we abstract from. Short certificates last less than one year, are often pursued mid-career, and are associated with small earnings gains (Lovenheim and Smith, 2023). Long certificates or diplomas, which can last between one and two years, are much less frequently awarded by two-year institutions compared to AA degrees (Jenkins, Fink, and Velasco, 2025).

model's design and parameterization.

Next, we analyze the relationship between individual attributes and realized paths to highest postsecondary enrollment using the NLSY97. Our approach builds on work using the same dataset by Belley and Lochner (2007) to motivate our analysis of AA and BA programs in a model framework with borrowing constraints. We offer new findings on how demographic attributes predict postsecondary enrollment paths that include AA enrollment. In particular, we show that among high school graduates and postsecondary enrollees, those with low skill, low parental income, and low parental education are more likely to enroll in AA programs. However, among high school graduates who either enroll in an AA or do not enroll in postsecondary education, those with low skill, low parental income, and low parental education are less likely to enroll in AA programs. Additionally, among BA enrollees, the same attributes predict a higher likelihood of taking the AA-to-BA path. In supplementary exercises, we offer supporting evidence for these NLSY97 results using the HSLS:09.

Motivated by our empirical evidence, in the second part of our paper we build a life cycle model with both an AA and a BA program. The model features borrowing constraints, as well as key sources of postsecondary education financing such as Pell grants and other public and private grants, subsidized and unsubsidized student loans, parental transfers, and earnings from employment. In comparison to the BA program, the AA program is more accessible in terms of admissions, program duration (two instead of four years), annual tuition costs, and required effort. However, the AA program is also of lower quality in that it features a lower consumption value, lower continuation likelihood, and leads to lower returns on the labor market for graduates.[2] We also allow for transfers from the AA to the BA program. Consistent with our empirical findings, those who transfer experience a scarring effect in terms of BA graduation likelihood, labor market returns, and consumption value compared to direct enrollees in the BA program.[3]

We validate the model along three dimensions important to our main experiment. First, the calibrated model accounts for observed correlations of youth skill and parental income and education with AA, BA, and AA-to-BA enrollment from our NLSY97 regressions, discussed above. Second, the model generates estimates for average tuition net of grants in both AA and BA programs that are in line with the data. Third, the model generates an AA enrollment rate response to an AA subsidy that is consistent with the data from a quasi-natural experiment, reported in Denning (2017).

We analyze the aggregate and welfare implications of making the AA program tuition-free, and compare the results with a spending-equivalent subsidy given to those enrolled in the BA program. Specifically, we choose the BA subsidy so that the change in government spending as a share of output is equal to the change generated by making the AA program tuition-free. The resulting BA tuition subsidy is significantly smaller

[2] The admissions policy is modeled exogenously. The AA program follows an open admissions policy, consistent with the admissions policies for community colleges. In contrast, the BA program follows a selective admissions policy making it harder for those with lower skill to be admitted into the BA program. The selective admissions captures unmet admissions requirements, as well as capacity constraints and other frictions. For recent quantitative work on endogenous admissions, see Chade, Lewis, and Smith (2014), Fu (2014), Marto and Wittman (2024), and Hendricks, Koreshkova, and Leukhina (2024, 2025); **?**.

[3] This scarring effect captures both the lower quality of instruction in AA programs and the potentially limited ability for AA enrollees to transfer to a high quality BA program.

per recipient than the AA tuition subsidy.[4] Before using the model to analyze the two experiments, we show that our model's prediction for the rise in public education grant spending, which results from making the AA program tuition-free, aligns quantitatively with the annual amount of $9 billion requested for the *Free Community College* proposal in the FY 2024 Congressional Justification by the U.S. Department of Education (2023).[5]

In line with a goal of existing policy proposals, we find that making AA programs tuition-free is a cost-effective way to increase the share of the population with a postsecondary degree. The AA subsidy expansion leads to a large rise in AA enrollment driven by a "democratization effect" in which non-enrollees transition to AA enrollment, rather than a "diversion effect" in which those who would have gone straight to BA enrollment now enroll in the AA program. As a result, the share of the population with a postsecondary degree increases by 2.16 percentage points. In contrast, the BA subsidy expansion is not as effective in increasing the postsecondary enrollment rate or the share of the population with a postsecondary degree (the latter increases by only 0.83 percentage points).

While the AA subsidy generates a significant increase in the share of postsecondary graduates, it also leads to a fall in labor productivity and, therefore, output. The reason is the diversion effect in which those who would have enrolled in a BA now enroll in an AA, which is associated with a lower graduation likelihood and lower efficiency premium. Through numerical exercises, we show that labor productivity decreases with the diversion effect but increases without it. Therefore, in regard to productivity, the diversion effect dominates the democratization effect. We find this result despite the fact that in regard to enrollment, the democratization effect dominates the diversion effect. While the net impact on postsecondary enrollment is a popular outcome measure in reduced-form work examining the democratization and diversion effects, our quantitative analysis highlights how the same policy that raises enrollment can nevertheless reduce labor productivity. In contrast to the AA subsidy, the BA subsidy increases labor productivity and output. This is because the BA subsidy increases BA attainment among those who previously enrolled in an AA or did not enroll in postsecondary education.

Both subsidy expansions lead to welfare gains that amount to more than a one-time transfer worth 1 percent of initial steady state GDP in a partial equilibrium in which the prices, the tax rate, and transfers are held fixed at their initial steady state values. However, the welfare gains observed after the AA subsidy expansion in partial equilibrium are eliminated once we take general-equilibrium effects and transition dynamics into account. This is true regardless of whether the government increases the tax rate to balance the budget in every period or borrows in the early periods of the transition and shifts the tax burden to future generations. Specifically, the partial-equilibrium welfare gains are eliminated due to the fall in labor productivity following the AA subsidy expansion and the impact of the transition on consumers who are alive in the periods of the transition. By contrast, even after general-equilibrium effects and transition dynamics are taken into

[4]The smaller BA tuition subsidy results in the same increase in total government spending to output because more students enroll in a BA (extensive margin) and the cost per-enrollee in a BA program is higher as it lasts for longer and features a higher annual continuation likelihood (intensive margin).

[5]We do not allow the price of tuition to respond to changes in enrollment. This is not a concern for either policy: the BA subsidy leads to a small increase in BA enrollment; the AA subsidy leads to a large increase in AA enrollment, but most two-year programs in the United States are public and the government has some control over prices.

account, the BA subsidy expansion leads to gains amounting to a one-time transfer worth more than one percent of initial steady-state GDP.

Our structural analysis includes distinct AA and BA programs, which allows us to evaluate AA-specific policies. This distinguishes us relative to most of the macroeconomic literature on postsecondary education financial aid, which focuses only on BA programs. Examples include Caucutt and Kumar (2003), Ionescu (2009), Chatterjee and Ionescu (2012), Krueger and Ludwig (2016), Ionescu and Simpson (2016), Abbott, Gallipoli, Meghir, and Violante (2019), Matsuda (2020, 2022), Colas, Findeisen, and Sachs (2021), Luo and Mongey (2024), Krueger, Ludwig, and Popova (2025), Kim and Kim (2025), Wright (2025), Moschini, Raveendranathan, and Xu (2026), Vardishvili (2026), and Moschini and Raveendranathan (2026).

Complementing our analysis, a few recent structural studies in the macroeconomic literature on postsecondary education have incorporated representations of AA programs: Trachter (2015), Hendricks, Koreshkova, and Leukhina (2024, 2025, 2026), Park (2025), and Krueger, Ludwig, and Popova (2026). Our model shares fundamental attributes of the AA program with prior studies, in that the AA program is cheaper but leads to lower labor market returns. Our model additionally incorporates all of the following elements: the option to transfer from the AA to the BA program, borrowing constraints, dropout risk, and general-equilibrium effects. Our framework is suited to our distinct research focus on evaluating policy proposals to make AA programs tuition-free.

Specifically, Trachter (2015) models AA programs as a stepping stone to a BA in a framework without borrowing constraints; the model is then applied to the U.S. economy in the 1970s where this assumption is a reasonable one. In Trachter (2015)'s model, high school graduates face uncertainty about their ability to accumulate human capital. This makes the cheaper AA program attractive because enrollees can drop out or transfer to a BA program once more information is known. Trachter (2015)'s main experiment is to quantify the contribution of each of these options to the total value of the AA program.

In a framework with endogenous college admissions and capacity constraints, Hendricks, Koreshkova, and Leukhina (2024, 2025, 2026) approximate the AA program as a low-quality college. In their models, the AA increases human capital of an enrollee who will drop out after two years without a degree but does not give enrollees the option to transfer to a BA. As an illustrative example of their policy exercises, Hendricks, Koreshkova, and Leukhina (2026) examine the effects of large-scale capacity expansions at high-quality universities. Similarly, Krueger, Ludwig, and Popova (2026) build a life cycle model of the college market with heterogeneous college quality and focus on demographic changes.

Park (2025) builds a model framework that includes an AA and BA program, as well as heterogeneous parental altruism; the last feature generates high variation in inter vivos transfers, controlling for parental wealth. The model framework is intentionally stylized, in that it abstracts from dropout risk, the AA-to-BA transfer decision, and general-equilibrium effects. This leads to a highly tractable framework, which Park (2025) then uses to study the design of optimal need-based aid.

This paper proceeds as follows. In Section 2 we present our main empirical findings from the NLSY97 and the HSLS:09. Section 3 presents our model framework, and Section 4 the model parameterization to the U.S. economy. Section 5 analyzes the enrollment policy functions and validates the model. Section 6 reports the results of our main experiment and Section 7 concludes.

2 Data

In this section we offer empirical evidence of tradeoffs faced by young adults choosing between AA and BA programs as well as AA-to-BA and straight-to-BA paths to BA enrollment. We then document the relationship between these enrollment outcomes and demographic attributes.

Our two main sources of data are the 1997 National Longitudinal Survey of Youth (Bureau of Labor Statistics, U.S. Department of Labor, 2019) and the public-use High School Longitudinal Survey of 2009 (National Center for Education Statistics, U.S. Department of Education, 2020), supplemented by information on institutional structure from the literature and tabulations from the U.S. Department of Education's National Center of Education Statistics (NCES). The NLSY97 and the HSLS:09 are both panel surveys collected in the United States.

The NLSY97 follows a nationally representative sample of young adults from 1997 until 2019. The HSLS:09 follows a nationally representative sample of 9th graders from 2009 to 2016 (about three years after high school graduation for most sample members), with some information on postsecondary enrollees also collected for the 2017 academic year. The NLSY97 has a longer panel dimension, making it useful for measuring postsecondary enrollment, degree completion, and earnings later in life. On the other hand, the HSLS:09 features information on postsecondary application and acceptance outcomes, as well as financial aid receipt and pecuniary costs of college collected directly from postsecondary institutions, for a more recent cohort of young adults than the NLSY97.

Estimation samples Our main NLSY97 findings are computed using two cleaned samples: the postsecondary outcomes estimation sample and the earnings profile estimation sample. We use the former to examine the relationship between education outcomes and demographic attributes, and the latter to examine the relationship between education attainment and earnings. Following Abbott, Gallipoli, Meghir, and Violante (2019), our NLSY97 analyses do not use survey weights.

The postsecondary outcomes estimation sample is at the individual level. It requires that the respondent's education record indicate they earned a high school degree aged 17-20, and that they remain in the sample until they turn 30. We require that we observe the Armed Services Vocational Aptitude Battery (ASVAB) score, our preferred measure of skill, as well as parental education and parental income in 1997. We also require that the variable reporting enrollment and degree attainment status from high school and two- or four-year college degree programs in each wave not imply contradictory information. These requirements leave us with 2,257 individual observations in the postsecondary outcomes estimation sample.

The earnings profile estimation sample is at the individual-year level, where the year refers to the year of the survey round's interview. This sample requires that the respondent be aged 25-39, have completed a high school degree when they were 17-20 years old, and have valid values for skill. In each wave, we also require that the respondent not be enrolled and have valid observations of marital status, number of children, and the real wage at the respondent's main job (within bounds drawn from Guvenen (2009)). These requirements leave us with 4,130 individuals (25,680 individual-year observations) in the earnings profile estimation sample.

Our main HSLS:09 findings are computed using a cleaned sample in which we require that students earn a high school diploma in 2013. We also require that the respondent live with at least one parent at some point during the first two waves of data collection, and that we observe honors-weighted high school GPA, race, gender, parental education, and parental income bin. High school GPA is our measure of skill in the HSLS:09 because there is no standardized test like the ASVAB of the NLSY97. In addition, we require that we have sufficient information to construct the respondent's highest postsecondary education enrollment outcome and that this information be consistent. This leaves us with a sample of 7,214 individuals, subsets of which additionally satisfy specific table requirements (e.g., being enrolled in postsecondary education). Our HSLS:09 analyses use survey weights.[6]

Postsecondary education outcomes Using our NLSY97 postsecondary outcomes estimation sample, we construct a set of indicator variables that record the path each respondent takes to their highest postsecondary enrollment after high school. Specifically, we flag whether the respondent enrolled in an AA or BA program after high school completion, measured at any survey round collected up to and including when the respondent is 24 years old. Because we are interested in paths to the highest postsecondary enrollment by age 24, we separate AA enrollees into those who continue to a BA after AA enrollment and those who do not. This allows us to identify BA enrollees with a history of prior enrollment in an AA program ("AA to BA") and those who enroll directly in a BA ("straight to BA"). If the respondent enrolls first in a BA program after high school and, later, an AA program, then they are counted as a straight-to-BA enrollee and not counted as an AA enrollee. Analogous to how we define enrollment types, we define categories of degree attainment by flagging the highest degree received by age 30 (AA or BA) and the enrollment path taken to that degree. If no postsecondary degree is completed, the respondent has a high school degree.[7]

In the NLSY97 earnings profile estimation sample, in each survey round a respondent is assigned to one of five mutually exclusive postsecondary education attainment categories: AA degree, two or more years of BA enrollment with no degree, BA degree via the AA-to-BA path, BA degree via the straight-to-BA enrollment path, or high school degree (which includes AA dropouts and those with one year of BA enrollment, as well as those with no postsecondary enrollment history).

In the HSLS:09, postsecondary enrollment flags are constructed in an analogous fashion to the NLSY97

[6]Further details of how we identify our main estimation samples for the NLSY97 and HSLS:09 are available in Appendix A.1.

[7]To be conceptually consistent with tracking the path-to-highest enrollment, we count BA-to-AA enrollees as straight-to-BA enrollees and not AA enrollees. Among the group of BA-to-AA enrollees, earning a BA degree is more common than earning an AA, so our assignment choice also maximizes consistency between enrollment and degree attainment outcomes. Our lens on the data implies that, among straight-to-BA enrollees, some may go on to earn an AA degree but not a BA degree. This would result in observations where the highest degree earned is an AA but there is no record of AA enrollment because it did not occur on the path to the highest enrollment outcome. We therefore also set the indicator for AA degree attainment to zero in the BA-to-AA enrollment path case, to keep both enrollment and attainment classifications consistent with the path-to-highest-degree model lens (this affects less than 30 observations in the postsecondary outcomes estimation sample).

postsecondary outcomes estimation sample, although the enrollment horizon is limited to three years after high school completion instead of by age 24. We are not able to construct analogous degree attainment flags in the HSLS:09 due to its short panel dimension.

In Section 2.1 we document the main tradeoffs between AA and BA programs. In Section 2.2, we compare tradeoffs of taking the AA-to-BA or straight-to-BA path to BA enrollment. Section 2.3 documents postsecondary enrollment patterns of young adults who take the tradeoffs documented in Sections 2.1 and 2.2 as given.

2.1 Tradeoffs of AA versus BA programs

We organize the differences between AA and BA programs into differences in accessibility and quality. Regarding accessibility, besides being shorter in duration (usually two versus four years) AA programs are more accessible than BA programs in that they are easier to get into, they are cheaper per year, and they require lower effort. Regarding quality, AA programs are lower in quality than BA programs in that AA programs are associated with a lower completion likelihood than BA programs and AA degrees offer lower labor market returns than BA degrees.[8] AA programs also offer a lower consumption value for enrollees, where consumption value is measured using consumption amenity spending by AA versus BA institutions as in Jacob, McCall, and Stange (2018).

Accessibility Both four-year BA programs and two-year AA programs require aspiring enrollees to submit an application and be accepted in order to enroll. However, 92 percent of two-year AA programs have no admission criteria for applicants and follow an "open enrollment" policy, while only 25 percent of four-year BA programs have no admission criteria (Lovenheim and Smith, 2023).[9]

The lower pecuniary cost of AA programs versus BA programs holds true both before and after taking grant aid into account. Restricting to survey respondents who enroll in the academic year immediately following high school completion (that is, in the fall of the 2013-2014 academic year), Table 1 reports the average sticker price in annual tuition and fees, as well as net tuition and fees after total grant aid is taken into account, among those enrolled in a two-year AA program and a four-year BA program. On an annual basis, the average AA enrollee pays much less than the average BA enrollee, and grant aid does not close this gap.

Next, we provide evidence consistent with lower or more flexible time requirements of AA programs: AA enrollees work more than BA enrollees. This is true even conditioning on parental income, which could affect the incentive to work while enrolled. Table 2 shows labor supply statistics for HSLS:09 sample members who enroll in an AA or BA program immediately after high school, broken down by parental

[8] Using tabulations from the NCES, Table A14 of Appendix A.3.2 shows that instructional spending is twice as high for four-year programs compared to two-year programs. This offers suggestive evidence that differences in wage premiums and completion rates may arise due to differences in teaching inputs.

[9] Consistent with these differences, in Table A10 of Appendix A.2.2 we show that, comparing across skill, those with lower skill are more likely to apply to an AA program and those with higher skill are more likely to apply to a BA program. Furthermore, among applicants with low- and medium-high school GPAs, acceptance rates to postsecondary programs for the first academic year after high school completion are higher at AA programs than BA programs. The difference is especially stark in the lowest GPA tercile.

Table 1: Tuition and fees by postsecondary program type

Program type	Total	Net of grants
AA	3,942	869
BA	17,943	8,993
Observations	3,494	

Notes: The table reports tuition and fees, both total and net of grants, faced by AA enrollees and BA enrollees in the 2013-2014 academic year, as reported by the institution they attended. Weights: PETS-SR longitudinal weights. Source: HSLS:09.

income. Within each parental income tercile, AA enrollees work more hours on average. This is a result of both an extensive margin effect (AA enrollees are more likely to be working) and an intensive margin effect (AA enrollees work more hours conditional on working).

Table 2: Labor supply during first year of postsecondary enrollment

	AA enrollees			BA enrollees		
Parental income	**Hours**	**Share working**	**Hours | hrs>0**	**Hours**	**Share working**	**Hours | hrs>0**
1	16	0.593	27	11	0.526	21
2	18	0.687	27	10	0.517	20
3	18	0.712	26	7	0.377	19
Total	17	0.654	27	9	0.446	20
Observations	1,202			3,779		

Notes: Within each parental income tercile, the table reports average hours worked per week, labor force participation, and average hours worked per week conditional on working, for first-year enrollees in BA or AA programs in the fall of 2013. Weights: Second follow-up student longitudinal weights. Source: HSLS:09.

Quality We document lower AA completion likelihoods and labor market returns using the NLSY97. For each skill tercile and overall, Table 3 reports PSE completion rates for those who enroll in an AA or BA program by age 24. Specifically, the first three columns report mutually exclusive outcomes for AA enrollees: the share who earn an AA by age 30 but do not continue to a BA, the share who transfer to a BA program without earning an AA degree, and the share who both earn an AA and transfer to a BA. The fourth column reports the share of AA enrollees who either earn a degree or transfer to a BA, a broader (and, in our view, context-appropriate) definition of AA program "completion." In contrast to the first four columns, the fifth column focuses on BA enrollees by age 24, and reports the share who go on to earn a BA degree by age 30 regardless of whether or not they took the AA-to-BA path to BA enrollment.

Comparing across the first four columns reveals that the broader definition of completion for the AA program ("Either" column, sum of "AA degree only", "AA to BA only", and "Both") implies a sizably larger rate of completion than considering AA degree attainment as the only measure of completion (sum of "AA degree only" and "Both"). Comparing the fourth and fifth columns shows that, even controlling for skill and allowing for transferring to a BA to count as a successful completion, AA enrollees are less likely than BA enrollees to complete their program.

In Table 4, for each skill tercile we report median wages and sample counts by highest degree attained using

Table 3: Postsecondary completion rates among enrollees

	AA enrollees				BA enrollees
Skill	**AA degree only**	**AA to BA only**	**Both**	**Either**	**BA degree**
1	0.203	0.093	0.106	0.403	0.585
2	0.192	0.179	0.127	0.498	0.689
3	0.223	0.277	0.187	0.687	0.830
Total	0.204	0.173	0.135	0.512	0.754
Observations	631	631	631	631	1,307

Notes: The table reports postsecondary completion rates among AA and BA enrollees within each skill tercile and overall. Source: NLSY97.

the earnings profile estimation sample; wages are converted to 2013 dollars using the Consumer Price Index (CPI) from the Bureau of Labor Statistics, U.S. Department of Labor (2024). The ratio of median wages for those with an AA or BA to those with only a high school degree yields a skill-specific measure of the wage premium associated with reaching each postsecondary degree attainment; these ratios are reported in the last two columns of the table. Within each skill tercile, the wage premium is lower for an AA than a BA.[10]

Table 4: Wage premiums of highest degree attainment by skill tercile

	High school		AA		BA		Wage premium	
Skill	**Wage**	**Obs**	**Wage**	**Obs**	**Wage**	**Obs**	**AA**	**BA**
1	13.269	6,020	15.490	719	18.114	1,102	1.167	1.365
2	15.414	3,662	18.252	672	21.172	3,036	1.184	1.374
3	15.535	1,440	18.698	675	23.885	6,183	1.204	1.537

Notes: The table reports the median wage within each skill tercile by education attainment status for those not currently enrolled in postsecondary education and aged 25-39; the last two columns are the ratio of AA to HS and BA to HS median wages, that is the AA and BA wage premiums respectively. Observation counts are at the individual-year level. Source: NLSY97.

To provide indirect evidence of differences in consumption value across two- and four-year programs, we follow Jacob, McCall, and Stange (2018) and define spending on consumption amenities as equal to expenditures on student services and auxiliary enterprises. These categories include spending on items that make the university experience more enjoyable or improve quality of life for enrollees, such as student activities, student organizations, cultural events, residence halls, food services, intercollegiate athletics, and college unions. We compute spending on consumption amenities using nationally-representative NCES tabulations, converted to 2013 dollars using the CPI. Details of our estimation, with further explanation, are reported in Table A13 of Appendix A.3.2. Between 2013 and 2015, the average two-year program spent \$2,120 on consumption amenities per full-time-equivalent student, while the average four-year program

[10]The findings of Table 3 indicate that a large portion of postsecondary enrollees drop out without a degree. In Tables 4 and 5, wage premiums are computed for degree completers relative to those with only a high school degree and at most one year of BA enrollment or AA enrollment without a degree. This means that those who have two or more years of BA enrollment, but do not obtain a postsecondary degree, are not in either the numerator or denominator of the wage premium ratio. However, although such a premium is not reported in Tables 4 and 5, our structural model specification will allow for a return to BA enrollment of two years or more without degree completion, as well as a return to AA and BA degrees. The extent to which each education path affects labor efficiency units is disciplined empirically via an earnings profile estimation explained in Section 4.1.

spent \$6,471—roughly three times higher. We interpret this as evidence that, on average, the consumption value offered by a four-year BA program is higher than that offered by a two-year AA program.

2.2 Tradeoffs of AA-to-BA versus straight-to-BA

What are the tradeoffs of taking the AA-to-BA route versus going straight to a BA? To fix ideas, in this section we articulate and examine several possible dimensions in which differences in outcomes might arise across enrollees who take either the AA-to-BA or straight-to-BA route. These differences are the total pecuniary cost of pursuing a BA degree, the likelihood of completing a BA degree conditional on BA enrollment, the labor market returns to the BA degree, and the consumption value of BA enrollment. In regard to the last point, previous studies have shown that social integration of transfer students is a point of concern for the students themselves and for college officials (Ellis, 2012; Inside Higher Ed, 2020). In what follows, we offer new evidence that speaks to the other differences.

Using the NLSY97, we show in Table A1 of Appendix A.1.1 that those who take the AA-to-BA route spend fewer years enrolled in a BA, compared to enrollees who go straight to a BA. The difference is more than one year, and widens when we restrict attention to degree completers. Together, the lower number of years spent enrolled in a BA and the lower annual pecuniary cost of an AA documented in Table 1 indicate a lower overall cost of BA degree pursuit for AA-to-BA enrollees. Additionally, Table A2 of Appendix A.1.1 shows that taking the AA-to-BA route is associated with a lower annual likelihood of progressing towards BA completion, compared to going straight into BA enrollment (0.895 versus 0.934, respectively). This difference is driven by a lower annual continuation probability in the top skill tercile for those in the AA-to-BA group (the lower skill terciles continue at similar rates).

We also use the NLSY97 to document the impact of paths to BA degree attainment on earnings by computing the wage premium of each path. Table 5 reports the median wage and sample count within each skill tercile for those with only a high school degree, those with a BA degree who took the AA-to-BA route, and those with a BA degree who went straight to a BA. The last two columns report the BA wage premium for each path to BA degree attainment, computed as the ratio of median wages to those with only a high school degree. The BA wage premium is lower for those who took the AA-to-BA path in the top and bottom skill terciles and almost the same in the middle skill tercile. This indicates that, even controlling for skill and degree attainment, taking the AA-to-BA route results in a lower wage premium for many BA graduates.[11]

2.3 Paths to highest postsecondary enrollment

Postsecondary enrollment and demographic attributes How common are different paths to highest postsecondary enrollment, and what are the attributes of those who pursue various postsecondary education routes? Capitalizing on the more complete education profiles of the NLSY97, Table 6 reports descriptive statistics for three groups: all high school graduates, AA enrollees, and BA enrollees. AA enrollees include

[11] In Section 4.1, we show that the qualitative takeaway of an AA premium among those without a BA (shown in Table 4) and an AA-to-BA penalty among those with a BA (shown in Table 5) survives after accounting for age and skill along with other demographic attributes.

Table 5: Wage premiums of BA degree by skill tercile and path to BA enrollment

	High school		**AA to BA**		**Straight to BA**		**Wage premium**	
Skill	**Wage**	**Obs**	**Wage**	**Obs**	**Wage**	**Obs**	**AA to BA**	**Straight to BA**
1	13.269	6,020	17.573	334	18.447	768	1.324	1.390
2	15.414	3,662	21.170	602	21.175	2,434	1.373	1.374
3	15.535	1,440	22.154	507	24.037	5,676	1.426	1.547

Notes: The table tabulates the median wage within each skill tercile by highest degree attainment, specifically high school diploma, a BA having taken the AA-to-BA path to BA enrollment, and a BA having taken the straight-to-BA path to BA enrollment. The last two columns are the BA wage premiums for each skill bin, computed separately by path to BA enrollment. The sample is those not currently enrolled in postsecondary education who are aged 25-39. Observation counts are at the individual-year level. Source: NLSY97.

those who enroll in only an AA and those who go on to enroll in a BA; BA enrollees include those who go straight to a BA and those who take the AA-to-BA route. For each group, the rows of Panel A of the table report rates of postsecondary enrollment up to age 24, separated into three mutually exclusive paths to highest postsecondary enrollment: enrolling in only an AA, straight to BA (that is, no prior AA enrollment), or AA to BA. The last row of Panel A is the sum of the previous three, enrollment in any postsecondary education. Panel B reports demographic attributes: the average ASVAB score, real parental income in 1997, the share with at least one parent with a BA or higher in 1997, and the share who are female, black, or Hispanic.

The first column of Panel A of Table 6 reports enrollment statistics for the sample of all high school graduates. The most common path to highest postsecondary enrollment by age 24 is straight to BA (49 percent), followed by only AA (19 percent). At the same time, a quantitatively large share of high school graduates (9 percent) enroll in an AA and then a BA. Overall, more than 75 percent of high school graduates enroll in some form of PSE by age 24. Moving to the second column, the majority of those who enroll in an AA by age 24 enroll in only an AA (69 percent), although a sizable share go on to enroll in a BA (31 percent). The third column indicates that among those who enroll in a BA, 15 percent take the AA to BA route, while the remainder enroll directly in a BA program.

Comparing across columns in Panel B shows that there is sorting by several demographic attributes into AA and BA enrollment. In particular, compared to all high school graduates, those who enroll in an AA by age 24 have lower average values for skill, parental income, and parental education, while those who enroll in a BA have higher average values. The share who are female is slightly higher among AA and BA enrollees, while the share who are black or Hispanic is slightly higher for AA enrollees, and slightly lower for BA enrollees, compared to all high school graduates.

Overall, these statistics suggest that both AA and BA programs are popular choices for high school graduates and that many high school graduates take the AA-to-BA path to BA enrollment, representing a sizable share of both AA and BA enrollees. Additionally, there is sorting by skill and demographic attributes into each postsecondary program type.

Table 6: Postsecondary enrollment rates and demographic attributes

			Enrolled by age 24	
Panel	**Variable**	**All**	**AA**	**BA**
A: Enrollment by age 24	AA only	0.19	0.69	0.00
	Straight to BA	0.49	0.00	0.85
	AA then BA	0.09	0.31	0.15
	Enroll PSE	0.77	1.00	1.00
B: Demographic characteristics	Skill (ASVAB)	55.25	47.93	67.95
	Real parental income	79,949	66,004	97,013
	At least one parent BA+	0.34	0.21	0.49
	Female	0.55	0.57	0.58
	Black	0.21	0.22	0.18
	Hispanic	0.16	0.21	0.12
	Observations	2,257	631	1,307

Notes: The table reports enrollment rates by path to highest postsecondary enrollment and demographic attributes for three groups: high school graduates, AA enrollees by age 24, and BA enrollees by age 24. Note that a sample member can enroll in both an AA and a BA by age 24. Real parental income is in 2013 USD. Source: NLSY97.

Postsecondary enrollment outcomes, parental income, and skill Among high school graduates, PSE program enrollment is related to skill, parental income, parental education, and other demographic attributes. To examine the predictive power of each attribute holding other attributes fixed, Table 7 presents Average Marginal Effects (AMEs) for five probit regressions on various subsamples of high school graduates. In each regression, the dependent variable is an indicator for a specific postsecondary enrollment outcome and the controls are indicators for parental income terciles, skill terciles, having at least one parent with a BA or higher, and (although not shown in the main text) flags for being female, black, or Hispanic. Regression coefficients for all regressions in Table 7 are reported in Table A3 of Appendix A.1.2.

Model (1) regresses an indicator for enrolling in a BA program by age 24 on our set of controls for the sample of high school graduates. Consistent with the findings of Belley and Lochner (2007), the results indicate that, on average and holding other attributes fixed, being in a lower parental income tercile or a lower skill tercile negatively predicts enrolling in a BA, while having at least one parent with a BA or higher positively predicts this outcome. These AMEs are statistically significant at the one percent level.

In model (2), we regress an indicator for enrolling in an AA program by age 24 on our set of controls for the sample of high school graduates. Being in a lower parental income or skill tercile positively predicts enrolling in an AA, while having at least one parent with a BA or higher negatively predicts this outcome. All AMEs are statistically significant at the one percent level, except the parental income terciles which are significant at the five percent level. The comparison group in model (2) is mixed, in the sense that it contains both those who enroll in a BA and those not enrolling in any postsecondary education. In models (3) and (4), we show that model (2)'s AMEs represent the net effect of two opposing correlations: those with lower parental income, lower skill, or whose parents are not highly educated are less likely to enroll in an AA than to not enroll in any postsecondary education, but also more likely to enroll in an AA than a BA.

Model (3) reports how parental income, skill, and parental education predict enrolling in an AA conditional on either enrolling in an AA or not enrolling in any postsecondary education—that is, the sample is restricted

Table 7: Average Marginal Effects: Predicting postsecondary enrollment

	Dependent variable				
Control variable	**(1) Enroll BA**	**(2) Enroll AA**	**(3) Enroll AA**	**(4) Enroll AA**	**(5) AA to BA**
Flag: Parental income tercile 1	-0.16	0.06	-0.10	0.15	0.08
	(0.026)	(0.026)	(0.042)	(0.031)	(0.029)
Flag: Parental income tercile 2	-0.08	0.05	-0.06	0.08	0.05
	(0.023)	(0.023)	(0.040)	(0.025)	(0.022)
Flag: Skill tercile 1	-0.50	0.11	-0.31	0.40	0.17
	(0.026)	(0.026)	(0.038)	(0.033)	(0.040)
Flag: Skill tercile 2	-0.21	0.10	-0.11	0.17	0.05
	(0.022)	(0.023)	(0.038)	(0.025)	(0.022)
Flag: At least 1 parent BA+	0.23	-0.11	0.14	-0.19	-0.08
	(0.022)	(0.021)	(0.040)	(0.024)	(0.021)
Additional demographic flags	Y	Y	Y	Y	Y
Sample	HS grads	HS grads	Enr AA or no PSE	Enroll PSE	Enroll BA
Observations	2,257	2,257	1,144	1,744	1,307
pseudo- R^2	0.26	0.04	0.07	0.17	0.07

Notes: The table reports probit regression AMEs for regressions with the following dependent variables: (1) the likelihood of enrolling in a BA at some point up to age 24, conditional on completing high school (HS); (2) the likelihood of enrolling in an AA at some point up to age 24, conditional on completing high school; (3) the likelihood of enrolling in an AA conditional on either enrolling in an AA or no PSE at some point up to age 24; (4) the likelihood of enrolling in an AA conditional on enrolling in any PSE at some point up to age 24; (5) the likelihood of previously enrolling in an AA, conditional on enrolling in a BA at some point up to age 24. Standard errors in parentheses. Additional demographic flags included but not shown here are an indicator for being female, black, and Hispanic. Source: NLSY97.

to high school graduates who do not enroll directly in a BA. Therefore, in this model the comparison group is those who did not enroll in any postsecondary education. Being in a lower parental income tercile or a lower skill tercile negatively predicts enrolling in an AA, while having at least one parent with a BA positively predicts this outcome. These AMEs are statistically significant at the one percent level, except for the first parental income tercile (significant at the five percent level) and the second parental income tercile (not significant at the ten percent level).

Model (4) reports how parental income, skill, and parental education predict enrolling in an AA conditional on enrolling in any PSE (that is, either an AA or a BA). In model (4), the comparison group is BA enrollees. Being in a lower parental income tercile or a lower skill tercile positively predicts enrolling in an AA among all PSE enrollees, while having at least one parent with a BA negatively predicts this outcome. These AMEs are statistically significant at the one percent level.

Model (5) reports how the same set of controls predict taking the AA-to-BA path among BA enrollees. Being in a lower parental income or skill tercile positively predicts taking this route, while having at least one highly educated parent negatively predicts it. These AMEs are statistically significant at the one percent level, except for the second parental income and skill terciles, which are significant at the five percent level.

To summarize, having lower skill or poorer or less educated parents predicts a higher likelihood of enrolling in an AA program among both high school graduates and PSE enrollees, as well as a higher likelihood of taking the AA to BA route among BA enrollees. These same attributes negatively predict enrolling in a BA among all high school graduates and negatively predict enrolling in an AA among high school graduates who either enroll in an AA or do not enroll in postsecondary education.

In Table A8 of Appendix A.2.1, we estimate probit regressions analogous to those of Table 7 in the HSLS:09. The results are very similar to those of Table 7, although because of the HSLS:09's shorter panel dimension we do not observe medium-term education outcomes with the same accuracy as the NLSY97 and this lowers the statistical significance of some of the predictors in the HSLS:09.

In the next section, we build a quantitative model of college enrollment in which consumers choose between AA and BA programs. The AA program offers the option to transfer to the BA program after successfully completing one or two years and incorporates attributes documented in Section 2. After parameterizing the model, we validate our theory of postsecondary enrollment by comparing the results in Table 7 with those from analogous regressions run on model output from the model's baseline equilibrium.

3 Model

Guided by our empirical findings, we build an overlapping generations model with an AA program and a BA program. These two programs differ in both accessibility and quality. Program features related to accessibility are admission likelihoods, program durations, tuition and fees, effort costs, and financial aid. Program features related to quality are consumption values, the exogenous components of continuation likelihoods, and labor market returns. Additionally, AA enrollees may have the option to transfer to the BA program, but not vice versa. For those who transfer from AA to BA, the BA consumption value, continuation likelihoods, and labor market returns are distinct from those who directly enroll in a BA after high school.

Section 3.1 provides a model overview organized by agent type. Section 3.2 presents consumer value functions before and during enrollment in an AA or BA program, because these value functions capture most of the novel elements of the model. Appendix B presents the value functions after the highest level of degree attainment, functional forms, equilibrium definition, and the computational algorithm.

When we compute transition paths, the value functions, policy functions, prices, taxes, transfers, and equilibrium distributions will be time-dependent. We omit time subscripts for ease of exposition except for the equilibrium definition in Appendix B.

3.1 Overview

Time is discrete and runs forever; each period lasts one year. The economy contains a two-year postsecondary institution, a four-year postsecondary institution, heterogeneous overlapping generations of consumers, a government, and a final goods firm.

3.1.1 Two- and four-year postsecondary education institutions

The two-year institution offers an AA program, while the four-year institution offers a BA program. We organize differences between the two programs into features related to accessibility and features related to quality.

Program features related to accessibility The AA program follows an open admissions policy, which allows any high school graduate to enroll. In contrast, the BA program features a straight-to-BA enrollment option shock, $q(s)$, indexed by skill s. Only high school graduates who receive the favorable straight-to-BA enrollment option shock may enroll directly in the BA program; meanwhile, others can transfer to the BA program if they enroll in and successfully complete the first or second academic year in the AA program. The straight-to-BA enrollment option shock captures selective admissions due to unmet admissions requirements discussed in Section 2.1—as well as capacity constraints and other frictions—which result in the lower skill being less likely to be admitted directly into a BA program.

Apart from program duration and admissions policies, the two postsecondary institution types also feature distinct tuition and fees, $\kappa(d_e)$, and effort cost, $\lambda(d_e)$. The argument $d_e \in \{N, A, B\}$ records the postsecondary program enrollment status, where the elements of the set denote nonenrollment, AA enrollment, and BA enrollment, respectively. When $d_e = N$, the values for the corresponding variables are set to zero.

The extent of financial aid also varies between the two programs. The college room and board expenditure amount, $\bar{c}(d_e)$, is program-specific and is used only to determine the cost of attendance, $\kappa(d_e) + \bar{c}(d_e)$, which then determines an upper bound for Pell grants and subsidized student loans. Pell grants are denoted by $\theta^{Pell}(f, d_e)$, where f is the Expected Family Contribution (that is, they are need-based); the formula for Pell grants is provided in Section 3.1.3. Subsidized student loans are discussed in more detail in Appendix B.2.2. Additionally, grants from private sources and public sources other than Pell, $\theta^{other}(s, d_e)$, are skill- and program-specific (that is, they are merit-based). Net tuition, defined as tuition and fees net of all grants, is therefore given by

$$\tilde{\kappa}(d_e, s, f) = \kappa(d_e) - \theta^{Pell}(f, d_e) - \theta^{other}(s, d_e). \tag{1}$$

Program features related to quality Enrollment is associated with consumption value given by CV^A and $CV^B\left(j^A\right)$ for the AA and BA program, respectively. In $CV^B\left(j^A\right)$, the argument j^A records the number of years of AA completion for BA enrollees, which allows the model to reflect scarring effects for AA-to-BA enrollees in comparison to straight-to-BA enrollees. The exogenous component of continuation likelihoods is program- and skill-specific. The exogenous probabilities of continuing each academic year are given by $p^A(s)$ and $p^B\left(s, j^A\right)$ for the AA and BA program, respectively.

Completed education affects labor market returns via the deterministic component of life cycle earnings, $\epsilon(j, e, s)$, where j denotes age and $e \in \{\text{HS, AA, BA d/o, AA-to-BA, Straight-to-BA}\}$ denotes completed education. In the set of values for completed education, "HS" refers to a high school graduate with neither a PSE degree nor two or more years of BA enrollment, "AA" refers to an AA graduate without a BA degree (but potentially with one or more years of BA enrollment), "BA d/o" refers to a BA dropout with two or more years of BA enrollment who does not have an AA degree, "AA-to-BA" refers to a BA graduate with a history of an AA-to-BA transfer, and "Straight-to-BA" refers to a BA graduate who had enrolled directly in the BA program. These five groups are mutually exclusive and make up the set of working-age individuals. This grouping allows us to incorporate premiums for an AA degree and a BA degree, a premium for those who drop out of the BA program after two or more years of enrollment, and a scarring effect for BA graduates who have a history of AA enrollment.

In addition to the deterministic component of life cycle earnings, completed education affects the persistence, $\rho_\eta(e)$, innovation variance, $\sigma^2_\eta(e)$, and the initial draw variance, $\sigma^2_{\eta 0}(e)$, which are the parameters for the AR(1) process for earnings, and also the wage rate per efficiency unit, $w(e)$, Social Security transfers, $ss(e,s)$, and probability over child skill, $\pi_{s_c}(s_c|e)$. The function for Social Security transfers is presented in equation (21) of Appendix B.2.

3.1.2 Consumers

This section presents the consumers' preferences and an overview of their life cycle.

Preferences The consumer's utility function over consumption and leisure is given by, $U(c,\ell,j,d_e)$, in which the inputs are household consumption, c, hours worked, ℓ, adult age, j, which also records AA academic year when $j \in \{1,2\}$ and BA academic year when $j \in \{1,2,3,4\}$, and postsecondary education enrollment status, d_e. The functional form is

$$U(c,\ell,j,d_e) = \frac{\left[\left(\frac{c}{\zeta_j}\right)^{\upsilon}\left(1-\ell-\lambda(d_e)\right)^{1-\upsilon}\right]^{1-\sigma}}{1-\sigma} \tag{2}$$

where ζ_j is an adult equivalence parameter that reflects the presence of children in the household, υ is consumption share, and σ determines the consumer's relative risk aversion.

In addition to the objects that affect utility in equation (2), a consumer enrolled in a postsecondary program experiences the consumption value gains described in Section 3.1.1, scaled by their idiosyncratic weight for the consumption value of college, γ. Furthermore, a consumer who is delinquent on student loans experiences a stigma cost captured by a utility penalty, ξ_D.

Life cycle overview Consumers become an adult and start making decisions when they turn 18, which corresponds to model age $j=1$. Adults survive each period with probability ψ_j and live for a maximum of J periods.

When a consumer enters the economy as an adult, they are indexed by skill endowment, s, stochastic idiosyncratic earnings productivity, η, initial net assets, a, Expected Family Contribution (EFC), f, and weight for the consumption value of college, γ. The skill endowment of a new adult is drawn once from a distribution that depends on the parent's completed education. The stochastic component of earnings follows a log AR(1) process that depends on completed education. Net assets at the start of adulthood are determined by a one-time inter vivos transfer from the consumer's parent, and are recorded with $a \geq 0$. The EFC depends on parental income as explained in Section 3.1.3. The weight on the consumption value of college is drawn from a log-normal distribution such that $\log(\gamma)$ has a mean zero and variance σ^2_γ.

Consumers are subject to borrowing constraints, and make the postsecondary education enrollment decision after the realization of the straight-to-BA enrollment option shock. Expenses while enrolled in the AA or

BA program can be financed with federal student loans, inter vivos transfers from parents, earnings while in college, Pell grants, and other grants. Federal student loans are the only form of debt in the model and are recorded with $a < 0$.

Repayment of federal student loans begins when a consumer is no longer enrolled in any postsecondary education. At this stage, a consumer chooses between full repayment and delinquency. Upon paying off student loans, consumers solve a standard consumption-savings problem, which is affected by the presence of a child from age j_c until age $j_c + j_a$. At the beginning of age $j_c + j_a$, the family draws the child's skill endowment, s_c. The altruistic parent then makes an inter vivos asset transfer to their child that takes into account the child's skill.

Consumers retire at age j_r and subsequently receive Social Security transfers until they die. Assets of the deceased are used to fund private grants and then the remainder is redistributed as age-dependent accidental bequests, Tr_j. The function for the age-dependent accidental bequests is presented in equation (22) of Appendix B.2.

The consumer's income depends on their age, completed education, skill, AR(1) productivity, and net assets as well as hours worked. For ease of exposition, we summarize the consumer's pretax income as

$$y\left(j, e, s, \eta, a, \ell\right) = w\left(e\right) \epsilon\left(j, e, s\right) \eta \ell \mathbb{I}_{j<j_r} + ss\left(e, s\right) \mathbb{I}_{j \geq j_r} + r\left[a \mathbb{I}_{a>0} \mathbb{I}_{j>1} + Tr_j\right]. \tag{3}$$

3.1.3 Government

The government provides grants for college education, runs the federal student loan program, funds Social Security, and finances a required government consumption set as an exogenous fraction g of gross domestic product (GDP). Tax revenue is collected from a flat consumption tax, τ_c, and a progressive income tax and transfer function, $T\left(y\right)$, which is levied on pretax income, y. The income tax and transfer function follows the specification of Heathcote, Storesletten, and Violante (2017), and is given by

$$T\left(y\right) = y - \tau_l y^{1-\tau_p}, \tag{4}$$

where τ_p governs the tax progressivity and τ_l is the level parameter. In a stationary equilibrium, the level parameter is used to balance the government budget in every period. For the transition path, we consider both a balanced-budget case in which the government balances its budget in every period using the level parameter and a deficit-financed case in which the government issues debt to fund its expenditure for a fixed number of periods and thereafter repays its debt and uses the level parameter to balance the budget. See equations (23) and (24) of Appendix B.2 for additional details.

Pell grants The Pell grant function assigns an amount that is based on the EFC, f, and the cost of attendance, $\kappa\left(d_e\right) + \bar{c}\left(d_e\right)$. Specifically, the Pell grant has a maximum value of θ^{Pell}_{max}, and is decreasing in the EFC, with a value given by the function

$$\theta^{Pell}\left(f, d_e\right) = \max\left[\min\left[\kappa\left(d_e\right) + \bar{c}\left(d_e\right) - f, \theta^{Pell}_{max} - f\right], 0\right]. \tag{5}$$

Federal student loans The federal student loan program is characterized by a college-year specific student loan limit for subsidized and unsubsidized loans, $\bar{A}(j)$, a college-year specific student loan limit for subsidized loans, $\bar{A}_s(j)$, and a student loan interest rate, $r_{SL} = r + \tau_{SL}$, where r is the risk-free interest rate on savings and τ_{SL} is an add-on set by the government. The subsidized loan function, $a_s\left(f, j, d_e, a', j^A\right)$, is provided in equation (16) of Appendix B.2.

Expected Family Contribution The EFC, f, is an input into the Pell grant and subsidized loan functions and is determined as follows. If the consumer has parental income below the automatic zero EFC income threshold, $y_{f=0}$, the EFC is set to zero. Otherwise, parental income is used to determine adjusted available income, $y_{ad}(y)$, which is then used to calculate the EFC using a progressive step function, $G(y)$. The progressive EFC step function and the adjusted available income function are such that, among those who do not qualify for an automatic zero EFC, individuals with a lower disposable parental income will be assigned a lower EFC. The functions are provided in equations (17) and (18) of Appendix B.2.

3.1.4 Final goods firm

Output is produced by a final goods firm using a Cobb-Douglas production function that combines aggregate capital and aggregate labor. Aggregate labor combines efficiency units from workers with at most a HS degree excluding those with two or more years of BA enrollment, workers with at most an AA degree or dropouts with two or more years of BA enrollment, and workers with a BA degree via a constant elasticity of substitution aggregator with elasticity of substitution $1/(1-\iota)$ and share parameters υ_1 and υ_2.[12] Specifically, the final goods production function is given by

$$Y = ZK^{\alpha}\left[\left[(1-\upsilon_1-\upsilon_2)\, L^{\iota}_{(e=\text{HS})} + \upsilon_1 L^{\iota}_{(e\in\{\text{AA,BA d/o}\})} + \upsilon_2 L^{\iota}_{(e\in\{\text{AA-to-BA,Straight-to-BA}\})}\right]^{1/\iota}\right]^{1-\alpha}, \tag{6}$$

where K is aggregate capital, Z is total factor productivity, $L_{(e=\text{HS})}$ is total efficiency units of labor from workers with at most a HS degree who do not have two or more years of BA enrollment, $L_{(e\in\{\text{AA, BA d/o}\})}$ is total efficiency units of labor with at most an AA degree or dropouts with two or more years of BA enrollment, $L_{(e\in\{\text{AA-to-BA, Straight-to-BA}\})}$ is total efficiency units of labor with a BA degree, and α is the capital share. The capital stock depreciates at rate δ.

3.2 Consumer value functions before and during enrollment in college

Given their type, (s, η, a, f, γ), the lifetime expected value to an 18-year-old (model age $j = 1$) is given by

$$\begin{aligned} V^0(s,\eta,a,f,\gamma) = {}& (1-q(s)) \max\left[V(j,e,s,\eta,a), V^A(j,s,\eta,a,f,\gamma)\right] + \\ & q(s)\max\left[V(j,e,s,\eta,a), V^A(j,s,\eta,a,f,\gamma), V^B\left(j,s,\eta,a,f,\gamma,j^A\right)\right], \end{aligned} \tag{7}$$

[12]For labor without a BA degree, we assume perfect substitutability between those with an AA degree and those with two or more years of BA enrollment so that the wage rate for these two groups is the same: $w(e = \text{AA}) = w(e = \text{BA d/o})$. For labor with a BA degree, we assume perfect substitutability between those who transferred from the AA to the BA program and those who enrolled directly in the BA program so that the wage rate for these two groups is the same: $w(e = \text{AA-to-BA}) = w(e = \text{Straight-to-BA})$.

where $j = 1$, $e = \text{HS}$, and $j^A = 0$. In the equation, $V(j, e, s, \eta, a)$ is the value of not attending college, $V^A(j, s, \eta, a, f, \gamma)$ is the value of enrollment in the AA program, and $V^B\left(j, s, \eta, a, f, \gamma, j^A\right)$ is the value of enrollment in the BA program. For the value of not attending college, see equation (10) of Appendix B.1.

The value of enrollment in the AA program at age $j = 1$ is given by

$$\begin{aligned} V^A(j,s,\eta,a,f,\gamma) = & \max_{c\geq 0, \ell\in\mathscr{L}, a'\geq -\bar{A}(j)} U(c,\ell,j,d_e) + \beta\psi_j E_{\eta'|e,\eta}\Big[\left(1-p^A(s)\right)V\left(j+1,e,s,\eta',a'\right) \\ & + p^A(s)\max\left[V\left(j+1,e,s,\eta',a'\right), V^A\left(j+1,s,\eta',a',f,\gamma\right), V^B\left(j+1,s,\eta',a',f,\gamma,j^A\right)\right]\Big] + \gamma CV^A \\ & \text{s.t.} \\ & (1+\tau_c)c + a' + \tilde{\kappa}(d_e,s,f) = \tilde{y}(j,e,s,\eta,a,\ell) + a + Tr_j + r_{SL}\left(\mathbb{I}_{a<0}a - a_s\left(f,j,d_e,a',j^A\right)\right), \end{aligned} \tag{8}$$

where $d_e = A$, $e = \text{HS}$, and $j^A = 1$. In equation (8), $\mathscr{L}$ is the set of feasible work hours, β is the discount factor, a' is the stock of assets or federal student loans in the next period, and $\tilde{y} = y - T(y)$ is disposable income. The continuation value reflects that, if the consumer passes to the second year of the AA program, they may choose to drop out, continue in the AA program, or transfer into the second year of the BA program.

At age $j = 2$, the value of enrollment in the AA program is the same as in equation (8) with one difference: in the continuation value, if the consumer graduates from the AA program, there is no endogenous dropout decision and the consumer instead chooses between entering the labor market as an AA graduate and transferring into the third year of the BA program.

The value of enrollment in the BA program at ages $j \in \{1, 2, 3\}$ is given by

$$\begin{aligned} V^B\left(j,s,\eta,a,f,\gamma,j^A\right) = & \max_{c\geq 0, \ell\in\mathscr{L}, a'\geq -\bar{A}(j)} U(c,\ell,j,d_e) + \beta\psi_j E_{\eta'|e,\eta}\Big[\left(1-p^B\left(s,j^A\right)\right)V\left(j+1,e'\left(j,j^A\right),s,\eta',a'\right) \\ & + p^B\left(s,j^A\right)\times\max\left[V\left(j+1,e'\left(j,j^A\right),s,\eta',a'\right), V^B\left(j+1,s,\eta',a',f,\gamma,j^A\right)\right]\Big] + \gamma CV^B\left(j^A\right) \\ & \text{s.t.} \\ & (1+\tau_c)c + a' + \tilde{\kappa}(d_e,s,f) = \tilde{y}(j,e,s,\eta,a,\ell) + a + Tr_j + r_{SL}\left(\mathbb{I}_{a<0}a - a_s\left(f,j,d_e,a',j^A\right)\right) \\ & e'\left(j,j^A\right) = \begin{cases} \text{HS} & \text{if } j^A\in\{0,1\} \text{ and } j=1 \\ \text{AA} & \text{if } j^A = 2 \\ \text{BA d/o} & \text{if } j^A\in\{0,1\} \text{ and } j\in\{2,3,4\}, \end{cases} \end{aligned} \tag{9}$$

where $d_e = B$ and $e = \text{HS}$. In equation (9), those exiting college enter the labor market as a BA dropout. For these individuals, the function $e'\left(j, j^A\right)$ records whether the individual enters the labor market as a HS graduate who does not have two or more years of BA enrollment, an AA graduate, or a dropout with two or more years of BA enrollment.

At age $j = 4$, the value of enrollment in the BA program is the same as in equation (9) with modifications to the continuation value and the function $e'\left(j, j^A\right)$. Specifically, the continuation value does not contain a dropout decision and, for those who complete the BA program, the function $e'\left(j, j^A\right)$ is repurposed to record whether the graduate is an AA-to-BA ($j^A \in \{1,2\}$) or straight-to-BA ($j^A = 0$) enrollee. Furthermore, the idiosyncratic earnings productivity is drawn from the distribution for BA graduates. The function $e'\left(j, j^A\right)$ is unchanged for dropouts at age $j = 4$.

4 Parameterization

This section presents the externally estimated parameters related to the AA and BA program and all internally calibrated parameters in Sections 4.1 and 4.2, respectively. The remaining externally estimated parameters are presented in Appendix C. Note that, because the model is calibrated so that GDP per capita for those 18 and over is one in the initial equilibrium, this maps into empirical values normalized by GDP per capita for those 18 and over. To reflect this, we indicate that an external parameter or target moment is normalized in several rows of Tables 8 and 9 in this section as well as Table C1 of Appendix C. The value of the denominator in the empirical normalization is computed by combining information on GDP from the Bureau of Economic Analysis (BEA) from BEA Table 1.1.5 with population levels from the U.S. Census Bureau (Census Bureau of the United States, 2020) for 2013-2015.

4.1 Externally estimated parameters related to the AA and BA program

Table 8 reports the externally estimated parameters related to the AA and BA program organized into Panels A and B based on features related to accessibility and quality, respectively.

In Panel A, annual tuition and fees, $\kappa\left(d_e\right)$, is set to 0.057 and 0.258 for the AA and BA programs, respectively, using information from the HSLS:09. These estimates are drawn from Table 1 and then normalized by GDP per capita for those 18 and over in 2013. Similarly, the program-and-skill specific grants other than Pell, $\theta^{other}(s, d_e)$, are also set using estimates from the HSLS:09, drawing from Table A11 of Appendix A.2.3. Lastly, the college room and board amount, $\bar{c}\left(d_e\right)$, is set to 0.091 and 0.142 using data from NCES Table 330.10 (2019). Note that the college room and board amounts are only used to determine an upper bound for need-based aid (Pell grants and subsidized loans) in equations (5) and (16). The estimates in Panel A imply that the AA program is much cheaper than the BA program in terms of tuition and fees, but AA enrollees receive a lower amount of other grants and face a lower upper bound for need-based aid because of the lower value for college room and board.

In Panel B, the deterministic component of the life cycle earnings process, $\epsilon\left(j, e, s\right)$, is assumed to have the functional form given by

$$\epsilon\left(j, e, s\right) = \exp\left(\beta_0^{age}\left(e\right) + \beta_1^{age}\left(e\right)\left(17 + j\right) + \beta_2^{age}\left(e\right)\left(17 + j\right)^2 + \beta^s\left(e, s\right) + \beta^e\left(e\right)\right)$$

where $\beta_0^{age}\left(e\right)$, $\beta_1^{age}\left(e\right)$, and $\beta_2^{age}\left(e\right)$ are coefficients for the age-specific second-order polynomial for education group e; specifically, the coefficients of the age-specific polynomial and skill shifters are assumed to be the same for those without a BA, $e \in$ {HS, AA, BA d/o}, and the same for those with a BA, $e \in$ {AA-to-BA, Straight-to-BA}. The coefficient $\beta^s\left(e, s\right)$ is a skill tercile shifter, and $\beta^e\left(e\right)$ is a shifter

Table 8: Externally estimated parameters related to the AA and BA program

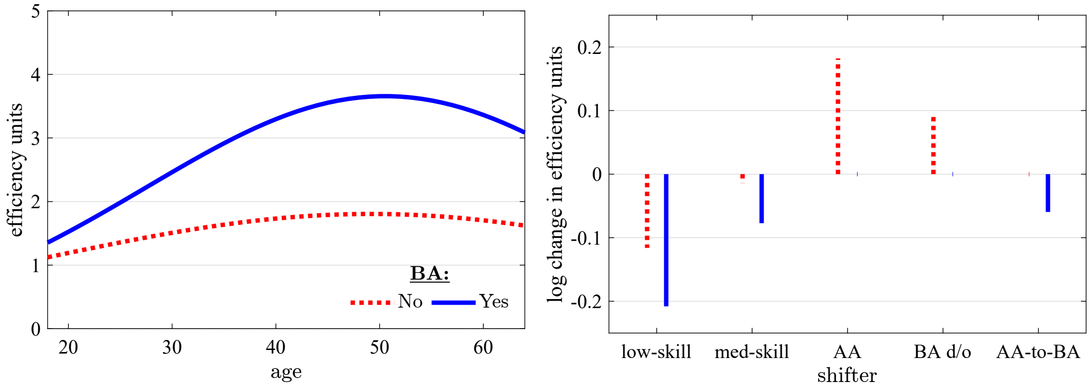
Parameter	Description	Data source	Value	
Panel A: Program parameters related to accessibility			$\boldsymbol{d_e = A}$	$\boldsymbol{d_e = B}$
$\kappa(d_e)$	Annual tuition, normalized	HSLS:09	0.057	0.258
$\theta^{other}(s, d_e)$	Grants net Pell by s, normalized		(0.007,0.016,0.019)	(0.083,0.087,0.124)
$\bar{c}(d_e)$	Room and board, normalized	NCES	0.091	0.142
Panel B: Program parameters related to quality: earnings process			$\boldsymbol{e \in}$ **{HS, AA, BA d/o}**	$\boldsymbol{e \in}$ **{AA-to-BA, Straight-to-BA}**
$\beta_1^{age}(e)$	Coefficients of age-specific	PSID + NLSY97	0.0480	0.0949
$\beta_2^{age}(e)$	polynomial		-0.000488	-0.000939
$\beta_0^{age}(e)$			1.825	1.314
$\beta^s(e, s = 1)$	Skill tercile shifters by s		-0.116	-0.208
$\beta^s(e, s = 2)$			-0.0141	-0.0772
$\beta^e(e = \text{BA d/o})$	BA d/o shifter		0.0928	n/a
$\beta^e(e = \text{AA})$	AA degree shifter		0.182	n/a
$\beta^e(e = \text{AA-to-BA})$	AA-to-BA shifter		n/a	-0.0595
$\rho_\eta(e)$	AR(1) persistence		0.921	0.922
$\sigma_\eta^2(e)$	AR(1) innovation variance		0.034	0.044
$\sigma_{\eta 0}^2(e)$	AR(1) initial draw variance		0.045	0.135

Notes: The table presents the externally estimated parameters related to the AA and BA program. Panel A reports moments indexed by PSE enrollment type (A or B) and Panel B reports moments indexed by completed education, e, further discretized into less-than-BA- or BA-degree groups. In the table, "n/a" refers to the not applicable cases. The remaining externally estimated parameters are presented in Table C1 of Appendix C.

that depends only on completed education that allows us to estimate, among those without a BA, separate premiums for AA graduates and for dropouts with two or more years of BA enrollment, and among those with a BA, a scarring effect for those with a history of AA enrollment. We estimate the second-order age polynomial using the Panel Study of Income Dynamics or PSID (Survey Research Center, Institute for Social Research, 2021) and then estimate the remaining shifters in the NLSY97 after cleaning out age effects on wages using our PSID estimates (see Appendix A.1.5 for further details).

Figure 1 provides a visual illustration of the deterministic component of the life cycle earnings process. Subfigure 1a depicts the efficiency units for individuals whose highest educational attainment is a high school degree excluding any dropouts with two or more years of BA enrollment (e = HS, dotted line) and for individuals whose highest degree attainment is a BA and had directly enrolled in the BA program (e = Straight-to-BA, solid line). The subfigure implies a large BA efficiency premium.

Subfigure 1b depicts the skill tercile and completed education shifters. A comparison across skill shifters within a BA completion status implies that lower skill is associated with a larger drop in efficiency units. A comparison across BA completion status within a skill tercile implies a smaller BA efficiency premium for those with lower skill. Among those without a BA, the BA dropout shifter implies a roughly 9 percent increase in efficiency units for dropouts with two or more years of BA enrollment and the AA degree shifter implies an approximately 18 percent increase in efficiency units for an AA degree. Lastly, among those with a BA, the AA-to-BA shifter implies a roughly 6 percent decrease in efficiency units for a history of AA enrollment.[13]



(a) High-skill by BA completion status

(b) Skill and completed education shifters

Figure 1: Deterministic component of life cycle productivity

Notes: Subfigure 1a depicts the deterministic component of life cycle earnings productivity, $\exp(\beta_0^{age}(e) + \beta_1^{age}(e)(17+j) + \beta_2^{age}(e)(17+j)^2)$, for two consumer types that represent the baseline case within BA-attainment categories. Specifically, for those without a BA, the red dotted line reports the age polynomial value for a consumer with high skill and without a PSE degree or two or more years of BA enrollment. For those with a BA, the solid blue line represents the age polynomial value for a consumer with high skill and a straight-to-BA enrollment history. Subfigure 1b depicts the skill and completed-education shifters ($\beta^s(e,s)$ and $\beta^e(e)$, respectively), which affect level shifts to the baseline type's earnings profile. For a description of completed-education types, see Section 3.1. Point estimates are drawn from Panel B of Table 8. The life cycle earnings productivity components are normalized so that $\epsilon(j=1, e=HS, s=1) = 1$.

Returning to Panel B of Table 8, the remaining parameters govern the stochastic component of the earnings process, which follows a log AR(1) process with persistence, $\rho_\eta(e)$, and an innovation term drawn from a Normal distribution with variance, $\sigma_\eta^2(e)$. When individuals enter the economy as a young adult without a BA at age $j=1$, they make their initial AR(1) productivity draw from a log-normal distribution; if individuals enter the labor market as a BA graduate at age $j=5$ they redraw their AR(1) productivity from a different log-normal distribution. These distributions have a zero mean and variance $\sigma_{\eta_0}^2(e)$.

4.2 Internally calibrated parameters

Table 9 reports the internally calibrated parameters organized into Panels A through D based on parameter type. Although parameters and moments are grouped using the most significant one-to-one relationship between each parameter and target moment, and are discussed accordingly, the parameters are calibrated jointly and each parameter can affect all target moments.

Panel A reports AA and BA parameters related to accessibility. The straight-to-BA enrollment option shock, $q(s)$, is parameterized to target AA enrollment rates by skill for those from the top parental income tercile, estimated using the NLSY97 and reported in Table A4 of Appendix A.1.3. Additional details on the estima-

[13] In Appendix C.2, we show that the model accounts for the empirical pattern that the AA wage premium is lower than the BA wage premium within each skill tercile. We also show that the model accounts for the lower BA wage premium for the many BA graduates who take the AA-to-BA route in comparison to straight-to-BA enrollees.

tion of the empirical moments including sample selection criteria are provided in Appendix A.1. Focusing on enrollment rates of individuals from high-income families minimizes the role of borrowing constraints in driving AA enrollment. The calibrated values imply that those with lower skill are academically disadvantaged in that they are less likely to receive the option to directly enroll in a BA. The program-specific college effort cost, $\lambda(d_e)$, is set to match the average weekly hours worked as a fraction of 40 hours (full-time work) for first-year students while in an AA and BA program, respectively. The target moments are estimated using the HSLS:09 and are drawn from Table 2 of Section 2.1. The calibrated effort cost for the AA program is smaller than that of the BA program.

Panel B reports AA and BA parameters related to quality. All target moments in this panel are based on the NLSY97 and are drawn from Tables 3 and A2 of Section 2 and Table A4 of Appendix A.1.3. The consumption value offered by the AA program, CV^A, is calibrated to match the AA enrollment rate. For the BA program, the consumption value for direct enrollees, $CV^B\left(j^A = 0\right)$, is calibrated to match the total BA enrollment rate, while the consumption value for AA-to-BA transfers, $CV^B\left(j^A \in \{1, 2\}\right)$, is calibrated to match the share of AA enrollees who transfer to a BA. A result of our calibration is that the AA program features the lowest consumption value, which is consistent with the discussion on consumption amenity spending from Section 2. Furthermore, within the BA program, those who enroll directly experience a higher consumption value than those who transfer from the AA program.

Table 9: Internally calibrated parameters

Parameter			**Target moment**		
Symbol	**Description**	**Value**	**Description**	**Data value**	**Model value**
Panel A: AA and BA parameters related to accessibility					
$q(s)$	Straight-to-BA enrollment option shock	(0.603,0.789,0.862)	AA enrollment | High income	*(0.404,0.230,0.141)*	*(0.405,0.230,0.141)*
$\lambda\left(d_e\right)$	Effort cost: $d_e \in \{A, B\}$	(0.516,0.540)	Average hours / Full time hours	*(0.425,0.225)*	*(0.422,0.226)*
Panel B: AA and BA parameters related to quality					
CV^A	Consumption value AA	6.996	AA enrollment	*0.280*	*0.278*
$CV^B\left(j^A = 0\right)$	Consumption value BA | Straight-to-BA enrollee	9.996	Total BA enrollment	*0.579*	*0.584*
$CV^B\left(j^A \in \{1, 2\}\right)$	Consumption value BA | AA-to-BA enrollee	8.513	Transfer rate to BA | AA	*0.310*	*0.312*
$p^A\left(s\right)$	AA continuation probability	(0.568,0.656,0.777)	Graduation or transfer to BA | AA enrollee	*(0.403,0.498,0.687)*	*(0.403,0.497,0.687)*
$p^B\left(s, j^A = 0\right)$	BA continuation probability | Straight-to-BA enrollee	(0.876,0.911,0.958)	BA graduation rate | Straight-to-BA enrollee	*(0.564,0.681,0.839)*	*(0.566,0.682,0.839)*
$p^B\left(s, j^A \in \{1, 2\}\right)$	BA continuation probability | AA-to-BA enrollee	(0.884,0.910,0.908)	BA graduation rate | AA-to-BA enrollee	*(0.638, 0.729, 0.753)*	*(0.639,0.730,0.753)*
Panel C: Preferences					
σ^2_γ	Variance consumption value weight	0.212	Enrollment response | $1,000 subsidy	*0.040*	*0.040*
β	Discount factor	0.990	Capital-to-output ratio	*3.000*	*3.000*
β_c	Parent altruism toward child	0.062	Average transfer, normalized	*0.609*	*0.603*
υ	Consumption share utility	0.402	Average worker is full time	*0.333*	*0.333*
ξ_D	Federal delinquency cost	0.027	Federal delinquency rate | BA	*0.088*	*0.079*
Panel D: Production technology and Social Security					
Z	Total factor productivity	1.771	GDP per capita 18+	*1.000*	*1.000*
ν_1	AA graduate or BA d/o labor share	0.282	AA wage premium | Middle skill	*1.184*	*1.183*
ν_2	BA graduate labor share	0.369	BA wage premium | Middle skill	*1.374*	*1.374*
χ	SS replacement rate	0.186	SS expenditure, fraction of GDP	*0.047*	*0.047*

Notes: The table presents the internally calibrated parameters and the associated target moments.

The remaining rows of Panel B contain parameters related to the exogenous likelihood of continuation in each program. The skill-specific annual probability of being allowed to continue the AA program, $p^A(s)$, is calibrated to target the skill-specific AA graduation or transfer to BA rates for AA enrollees. Similarly, the skill-specific annual probabilities of being allowed to continue the BA program for straight-to-BA enrollees, $p^B(s, j^A = 0)$, and for AA-to-BA enrollees, $p^B(s, j^A \in \{1, 2\})$, are calibrated to target skill-specific BA graduation rates for straight-to-BA enrollees and AA-to-BA enrollees, respectively. The calibrated values of $p^A(s)$ are less than $p^B(s, j^A)$ for all skill levels, reflecting the lower quality of the AA program. Additionally, the calibrated values of $p^B(s, j^A)$ are almost the same for the low and medium skill levels regardless of whether the individual enrolled directly in the BA program ($j^A = 0$) or transferred from the AA to the BA program ($j^A \in \{1, 2\}$). However, for the high skill, the same probability is lower for an AA-to-BA enrollee than a straight-to-BA enrollee. This captures a scarring effect for AA-to-BA enrollees with high skill in terms of their continuation likelihood once they enroll in a BA.[14]

Panel C reports preference parameters. The variance of the log-normal distribution for the weight on the consumption value of college, σ_γ^2, is calibrated to account for an estimate of the responsiveness of the enrollment rate from a quasi-natural experiment in which prospective students are given a $1,000 subsidy to attend postsecondary education (either AA or BA). Based on the preferred estimate of Deming and Dynarski (2009), the calibration targets a PSE enrollment rate response of 4.00 percentage points. In the model, the enrollment rate response is computed in a partial equilibrium in which the distribution of 18-year-olds and general equilibrium prices, taxes, and transfers are held fixed at their initial steady state values. Targeting the enrollment response using the variance of the weight on the consumption value of college places quantitative discipline on the results from the subsidy expansion experiments in Section 6: those who place a high weight on the consumption value of college are more likely to have selected into PSE enrollment in the status quo equilibrium while the remaining pool of nonenrollees are not as likely to value enrollment in a PSE.[15]

Returning to Panel C, the discount factor, β, is calibrated to target a capital-to-output ratio of 3, consistent with Jones (2016). The degree of parental altruism, β_c, is set so that the model matches average parent-to-child 6-year total transfers during the ages 18-23 in the NLSY97, reported in Table A7 of Appendix A.1.6. The consumption share in the utility function, υ, is calibrated so that the average non-retiree works full time, ft, which is parameterized in Panel A of Table C1 in Appendix C. The parameter governing the costs of being delinquent on student loans, ξ_D, targets the average cohort delinquency rate for BA programs for 2013-2015 from the Federal Student Aid (FSA) report FSA (2021b).

Panel D reports production technology and Social Security parameters. Total factor productivity, Z, is set so that GDP per capita for the population aged 18 and over is one in the model. The parameters that determine the labor share for AA graduates and BA dropouts with two or more years of BA enrollment, ν_1, and the BA

[14] In Appendix C.3, we show that the model accounts for the highest postsecondary degree attainment rates by age 30 observed in the data. That is, the model accounts for the shares of high school graduates with neither a PSE degree nor two or more years of BA enrollment, AA graduates without a BA degree, BA dropouts with two or more years of BA enrollment who do not have an AA degree, BA graduates with a history of an AA-to-BA transfer, and BA graduate who had enrolled directly in the BA program.

[15] If we consider an alternative model specification in which we do not allow for any variation in the weight on the consumption value and re-calibrate the alternative model to target the same set of moments except the enrollment rate response, the resulting enrollment rate response is more than twice as high.

graduate labor share, ν_2, are set so that the model matches the AA and BA wage premiums for the middle skill tercile reported in Table 4. The Social Security replacement rate, χ, targets the average ratio of total Social Security expenditure to GDP, estimated using 2013-2015 data from BEA Table 1.1.5 and BEA Table 2.1.

To summarize, our model specification and parameterization together reflect the fact that the AA program is more accessible than the BA program. Any high school graduate can enroll in the AA program (by assumption), whereas the BA program is more selective in admissions, especially for the lower skill. Additionally, the AA program is cheaper in the sense that it lasts for two years instead of four years, the annual tuition is lower, and the required effort is lower (which allows individuals to earn more from employment). However, the AA program is of lower quality than the BA program in the sense that it features a lower consumption value, lower continuation likelihood for enrollees, and it leads to a lower wage premium for graduates. Individuals who are successful in the AA program are allowed to transfer to the BA program, but they experience a scarring effect in terms of the BA consumption value and labor market returns for all skill levels and the BA continuation likelihood for the high skill level.

5 College Enrollment Decision and Model Validation

In Section 5.1, we analyze the college enrollment policy function. Subsequently, in Section 5.2, we validate the model by showing that it accounts for empirical patterns related to college enrollment and the level of net tuition in each program.

5.1 College enrollment decision

Figure 2 plots the college enrollment decision of high school graduates in the baseline calibration as a function of the initial characteristics: skill, net assets, EFC, consumption value weight, and AR(1) productivity. We restrict attention to high school graduates who receive the favorable straight-to-BA enrollment option shock, so that each individual faces the full three-way choice between not enrolling in any PSE, enrolling in the AA program, and enrolling in the BA program. In each subfigure, the vertical axis records this choice, ordered by education level from non-enrollment ("NE") at the bottom, AA in the middle, and BA at the top. In the figure, the three columns use net assets, consumption value weight, and AR(1) productivity component along the x-axis, respectively; the top and bottom rows fix EFC at its lowest ($f = 0$) and highest ($f = \kappa(B) + \bar{c}(B)$) values; within each subfigure, the decision rule is depicted separately for the low skill (solid line) and high skill (dashed line).

Comparing the two lines within each subfigure, higher skill shifts the decision rule up: those with high skill reach the AA and BA thresholds at lower values of net assets/EFC/AR(1) productivity, and hence, enroll in (weakly) a higher level of education than those with low skill. This is because the expected return to education is increasing in skill.[16]

[16] Although differences in the return to education across skill levels contribute to AA enrollment even among those who could enroll directly in the BA, the main reason high school graduates with low skill are more likely than those with high skill to enroll in an

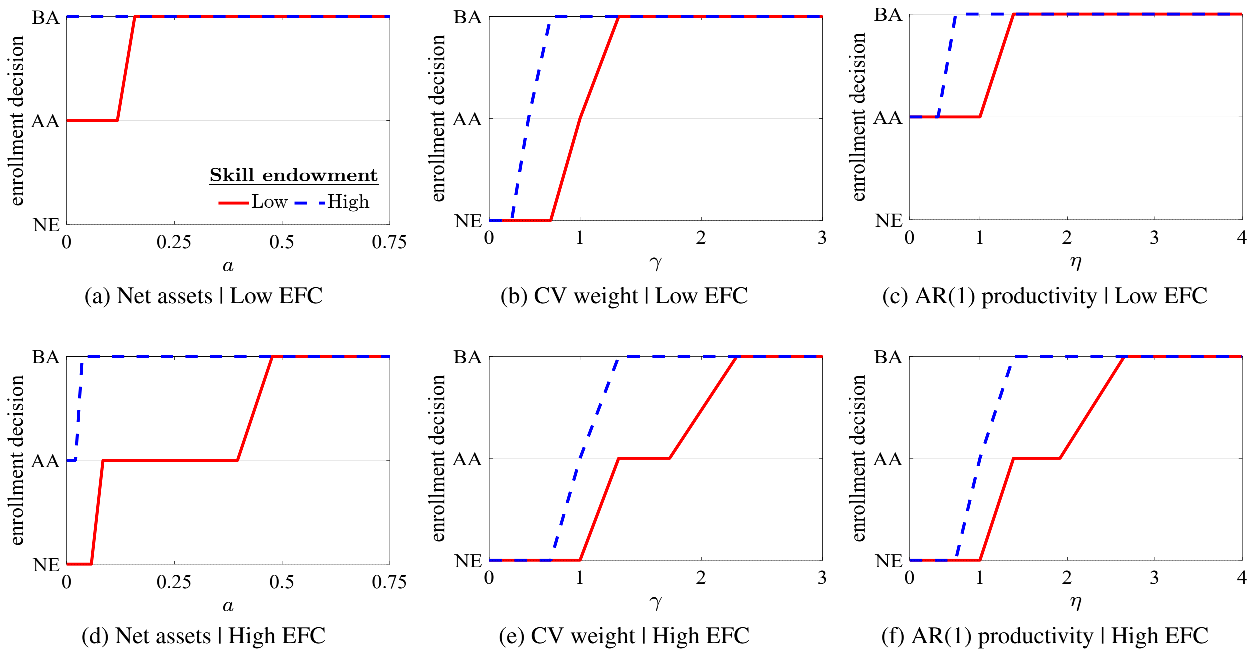


Figure 2: College enrollment decision by initial characteristics

Notes: The figure depicts the enrollment decision in the baseline calibration by initial characteristics for high school graduates who receive the favorable straight-to-BA enrollment option shock. Enrollment decision options are enrolling in a BA, enrolling in an AA, or non-enrollment ("NE"). Within each subfigure, the decision is depicted separately for low- and high-skill high school graduates; the top and bottom rows fix EFC at its lowest ($f = 0$) and highest ($f = \kappa(B) + \bar{c}(B)$) values, respectively; and the three columns vary net assets, the consumption value weight, and AR(1) productivity along the x-axis, respectively. For each subfigure, the remaining states are held at their medians, except initial net assets, which is set to zero in the second and third columns. In all subfigures, the x-axis is restricted to lower values to highlight the thresholds at which individuals change their enrollment decision.

Comparing the two rows, a lower EFC (top row) shifts the decision rule up: a graduate with a lower EFC reaches the AA and BA thresholds at lower values of net assets/EFC/AR(1) productivity. This is because a lower EFC qualifies the individual for more need-based aid, which lowers the pecuniary cost of enrollment.

Turning our attention to each of the columns of the figure, in the first column, the decision is increasing in net assets: the very poor choose non-enrollment, the slightly richer choose the AA, and the even richer choose the BA. In the second column, the decision is increasing in the consumption value weight: those who place little weight on the consumption value of college choose non-enrollment, those with intermediate weights choose the AA, and those with high weights choose the BA. The third column shows the same ordering in the AR(1) productivity component.

5.2 Model Validation

In Section 5.1, we analyzed the college enrollment policy function in the baseline equilibrium. Next, we show that the equilibrium of the baseline model also accounts for patterns of sorting into AA and BA enrollment discussed in Section 2.3. Table 10 reports AMEs from five probit regressions that predict the

AA rather than a BA program is selective admissions, captured by the skill-specific straight-to-BA enrollment option shock.

likelihood of BA enrollment among high school graduates, AA enrollment among high school graduates, AA enrollment among high school graduates who do not directly enroll in a BA, AA enrollment among PSE enrollees, and AA to BA enrollment among high school graduates who ever enroll in a BA, respectively. In all five regressions, the controls in both the data and the model include indicators for parental income terciles, skill terciles, and parental education; additional controls included in the data regression are flags for being female, black, or Hispanic. The empirical estimates are drawn from Table 7.

Table 10: Average Marginal Effects: predicting PSE enrollment in data and model

		Dependent variable				
Panel	**Control variable**	**(1) Enroll BA**	**(2) Enroll AA**	**(3) Enroll AA**	**(4) Enroll AA**	**(5) AA to BA**
A: Data	Parental income tercile 1	−0.16	0.06	−0.10	0.15	0.08
	Parental income tercile 2	−0.08	0.05	−0.06	0.08	0.05
	Skill tercile 1	−0.50	0.11	−0.31	0.40	0.17
	Skill tercile 2	−0.21	0.10	−0.11	0.17	0.05
	At least one parent BA+	0.23	−0.11	0.14	−0.19	−0.08
B: Model	Parental income tercile 1	−0.36	0.06	−0.42	0.21	0.08
	Parental income tercile 2	−0.31	0.04	−0.40	0.16	0.06
	Skill tercile 1	−0.36	0.19	−0.11	0.37	0.22
	Skill tercile 2	−0.20	0.14	−0.02	0.19	0.09
	At least one parent BA+	0.08	−0.01	0.09	−0.04	−0.02
	Sample	HS grads	HS grads	Enroll AA or no PSE	Enroll PSE	Enroll BA

Notes: The table reports probit regression AMEs with the following dependent variables: (1) the likelihood of enrolling in a BA at some point up to age 24, conditional on completing high school; (2) the likelihood of enrolling in an AA at some point up to age 24, conditional on completing high school; (3) the likelihood of enrolling in an AA conditional on not directly enrolling in a BA at some point up to age 24; (4) the likelihood of enrolling in an AA conditional on enrolling in any PSE at some point up to age 24; (5) the likelihood of previously enrolling in an AA, conditional on enrolling in a BA at some point up to age 24. The control variables are flags for parental income tercile 1, parental income tercile 2, skill tercile 1, skill tercile 2, and at least one parent with a BA or higher. Controls included in the data regression but not reported here are indicators for the sample member being female, black, and Hispanic. Data source: NLSY97.

In all five regressions, the model's baseline equilibrium produces AMEs with signs that are consistent with those in the data. In particular, the model reproduces the empirical pattern that those with lower skill or poorer or less educated parents, are less likely to enroll in a BA among all high school graduates and in an AA among high school graduates that excludes straight-to-BA enrollees (first and third columns), but more likely to enroll in an AA program among both high school graduates and PSE enrollees (second and fourth columns), as well as more likely to take the AA-to-BA route among BA enrollees (fifth column).

In one of our main experiments in Section 6, we analyze the implications of making the AA program tuition-free. The implications of this experiment critically depend on the AA enrollment rate response from that policy. In Table 11, we report the rise in the AA enrollment rate following a $1,000 subsidy for enrollment in the AA program. The empirical estimate is from Denning (2017), based on a quasi-natural experiment of tuition discounts for community colleges in Texas. The model enrollment response compares well with the data.

Lastly, for the credibility of our results from the experiment in which we make the AA program tuition-free, it is important that our model's baseline successfully reproduces the average net tuition observed in

Table 11: Change in AA enrollment rate given $1,000 AA subsidy in data and model

Quasi-natural experiment	Data	Model
$1,000 AA subsidy	5.1	5.1

Notes: The table presents estimates for the change in the AA enrollment rate given a $1,000 subsidy for enrollment in the AA program. In the model, the change in the AA enrollment rate is computed in a partial equilibrium in which the distribution of 18-year-olds and general equilibrium prices, the tax rate, and transfers are held fixed at their initial steady state values. Data source: Denning (2017).

the data.[17] As such, Table 12 reports program-specific average annual net tuition and fees after total grant aid is taken into account for first-year enrollees. The empirical estimates are drawn from Table 1 and then normalized by GDP per capita for those 18 and over. The model does reasonably well in producing estimates close to those observed in the data.

Table 12: AA and BA net tuition in data and model

Program	Data	Model
AA	0.012	0.022
BA	0.129	0.139

Notes: The table reports the average net tuition and fees after total grant aid is taken into account for first-year enrollees in AA and BA programs, respectively. The estimates are normalized by GDP per capita for those 18 and over. Data source: HSLS:09.

6 Main Experiments: Subsidizing AA or BA Programs

We perform two main counterfactual experiments. First, we examine the aggregate and welfare implications of making the AA program tuition-free. Second, we examine the implications of a spending-equivalent tuition subsidy expansion for the BA program.

In Section 6.1, we describe the two counterfactual experiments and then compare the model's prediction for the fiscal cost of making the AA program tuition-free to a budget-proposal estimate from the U.S. Department of Education. In Section 6.2, we analyze the aggregate implications of the two subsidy expansions across steady states in general equilibrium. In Section 6.3, we analyze the transition dynamics of key variables that matter for welfare. In Section 6.4, we report the resulting welfare changes in both partial and general equilibrium, taking transition dynamics into account.

[17] While we incorporated the rich college financial aid system in the US into our model and parameterized it to match program-specific sticker prices for tuition, we did not target average net tuition. Net tuition incorporates grant aid, which is a function of youth skill and Expected Family Contribution.

6.1 Experiment description

Let $\bar{\theta}^A$ denote the additional annual tuition subsidy for an enrollee in the AA program in the experiment in which the AA program is made tuition-free (the "AA subsidy expansion"). The AA subsidy is given by

$$\bar{\theta}^A = \max_{s,f} \tilde{\kappa}\left(A, s, f\right),$$

where $\tilde{\kappa}\left(A, s, f\right)$ is net tuition defined in equation (1). The AA subsidy is set to the maximum value of net tuition across all skill and EFC levels, ensuring any individual enrolled in the AA program does not pay tuition. This subsidy is added to the right hand side of the budget constraints of AA enrollees after the policy change; the Pell and the other grants that an individual was already eligible for in the initial equilibrium are unaffected. Our modeling of this policy is consistent with the proposed *Free Community College program*, which was structured as a "first-dollar program." In a first-dollar program, the new financial aid is applied to college costs before other sources of aid, such as Pell and other grants, which can instead be used to cover expenses beyond tuition and fees such as room and board (U.S. Department of Education, 2023).

Let $\bar{\theta}^B$ denote the additional annual tuition subsidy for an enrollee in the BA program in the experiment in which a spending-equivalent subsidy is provided for the BA program (the "BA subsidy expansion"). The BA subsidy is calibrated so that the resulting increase in the ratio of total government spending to output is the same across steady states in general equilibrium in both experiments.[18] A visual illustration of net tuition in the AA and BA programs and the levels of the AA and BA tuition subsidies are depicted in Figure D1 of Appendix D.1.1.

Table 13 reports the resulting changes in fiscal aggregates and the associated subsidies from the two experiments in Panels A and B, respectively. As indicated by Panel A, in both experiments, the ratio of total government spending to output increases by 0.038 percentage points. We also note that our chosen spending-equivalence criterion results in essentially the same increase in the ratios of public education grant and federal student loan spending to output in the two experiments, making them even more comparable.

Panel B of Table 13 indicates that the annual per-enrollee subsidy for the AA program is approximately \$3,500, whereas the spending-equivalent subsidy for the BA program is \$600. The BA tuition subsidy is significantly smaller than the AA tuition subsidy because the BA enrollment rate is higher than the AA enrollment rate (extensive margin) and the BA program lasts for four years instead of two years and features a higher annual continuation probability (intensive margin).

Table 14 compares the predicted annual increase in public education grants in the AA subsidy expansion with an estimate from the U.S. Department of Education (ED) based on the FY 2024 Congressional Justification (U.S. Department of Education, 2023). The ten-year ED budget request of \$90 billion for the *Free Community College* proposal is converted to an annual estimate of \$9 billion and then reported as a percentage of GDP in 2023 from BEA Table 1.1.5. The model estimate aligns well with the ED estimate.

[18] Total government spending in our model is the sum of expenditure on public education grants, the federal student loan program, government consumption, and Social Security. Expressions for these variables are provided in Appendix B.2.1.

Table 13: Changes in fiscal aggregates after subsidy expansions and associated subsidies

Panel		Subsidy expansion AA	Subsidy expansion BA
A: Changes in fiscal aggregates	Public education grants	0.030	0.028
Units: percentage of output	Federal student loans	0.008	0.006
	Government consumption	0	0
	Social Security	0	0.004
	Total government spending	0.038	*0.038*
B: Annual per-enrollee subsidy	$\bar{\theta}^A$ and $\bar{\theta}^B$	0.050	0.009
Units: see table notes	In 2013 dollars	3,481	600

Notes: The table reports general-equilibrium changes in fiscal aggregates across steady states after each subsidy expansion (Panel A) and the associated AA or BA subsidies (Panel B). The subsidy units are per enrollee per year, and are reported both as parameter values and in 2013 dollars. The total government spending change in the second column is in italics because it is targeted in the calibration.

Table 14: Annual increase in total public education grants

Units	Department of Education	AA subsidy expansion
Percentage of initial output	0.032	0.028

Notes: The table compares the estimated cost of the *Free Community College* proposal from the U.S. Department of Education (ED) with the annual increase in public education grant spending from the model experiment in which the AA program is made tuition-free. Both statistics are reported as a percentage of initial output (Panel A of Table 13's similar statistic differs slightly because it uses contemporaneous output). The details of the ED estimate are provided in the main text.

6.2 Aggregate implications across steady states in general equilibrium

Table 15 reports aggregate changes across steady states in general equilibrium following the AA and BA subsidy expansions. The aggregates are organized into Panels A through C based on statistic type.

Panel A reports changes in education statistics. The subsidy expansion for the AA program leads to a large increase in PSE enrollment, the result of a large increase in AA enrollment. The decline in straight-to-BA enrollment explains why the increase in PSE enrollment is smaller than the increase in AA enrollment: some of the new AA enrollees substituted out of straight-to-BA enrollment as opposed to non-enrollment. These changes in the enrollment rates imply that the democratization effect in which non-enrollees transition to AA enrollment dominates the diversion effect in which those who would have enrolled in the BA program now enroll in the AA program instead.[19] A byproduct of increased AA enrollment is an increase in the mass of transfers to a BA. As a result of these enrollment changes, the population share of PSE graduates increases by 2.16 percentage points—this increase is completely driven by AA degree attainment. Hence, the AA subsidy expansion is effective in subsidizing accessibility (that is, increasing PSE enrollment) and in achieving the proposed policy goal of increasing the share of PSE graduates.

[19] The patterns of enrollment changes from the AA subsidy expansion in our model are qualitatively consistent with the findings of Mountjoy (2022), although we change a different cost margin in our analysis. Specifically, Mountjoy (2022) develops a novel instrumental variable approach and applies it to a reduced-form decomposition of two-year enrollment changes after a decrease in distance to an AA program. He finds that for enrollment rates the democratization effect is larger than the diversion effect.

Table 15: Steady state changes after subsidy expansions in general equilibrium

Panel	Variable	Description	Subsidy expansion AA	Subsidy expansion BA
A: Education statistics	Enrollment rate:	PSE	7.21	0.85
Units: percentage point change		AA	11.27	−1.46
		Straight-to-BA	−4.06	2.31
		AA-to-BA	1.78	−0.05
	Educational attainment:	PSE	2.16	0.83
		AA	3.25	−0.71
		BA	−1.09	1.54
B: Macroeconomic aggregates	Labor productivity		−0.41	0.95
Units: percentage change	Output		−0.46	0.75
	Consumption		−0.43	0.71
	Capital		−0.49	0.70
	Labor (efficiency units):		−0.45	0.78
	Hours worked		−0.19	−0.03
C: Tax rate, accidental bequests, and prices	Income tax rate:	At baseline mean income	0.09	−0.05
Units: percentage point/percentage change	Accidental bequests		−0.74	0.59
	Risk-free rate		0	0.01
	Wage rate:	HS	0.44	0.96
		AA or BA d/o	−2.93	0.61
		BA	0.53	−0.71

Notes: The table reports general-equilibrium changes from the initial baseline equilibrium to the final steady state in the AA and BA subsidy expansions. Panels A, B, and C report changes in education statistics, macroeconomic aggregates, and income tax rate, accidental bequests, and prices, respectively. Labor productivity is computed for the working-age population as the weighted sum of individual productivities, $\epsilon(j, e, s)\,\eta$, divided by the mass of individuals.

In contrast to the AA subsidy expansion, the BA subsidy expansion leads to a smaller rise in PSE enrollment because the spending-equivalent BA tuition subsidy is smaller per recipient than the analogous AA tuition subsidy (see Table 13). The second and third rows of Panel A indicate that the BA subsidy expansion increases straight-to-BA enrollment primarily due to students substituting out of the AA program. Fewer high school graduates transfer from the AA to the BA program due to the decline in AA enrollment. The population share of PSE graduates increases by 0.83 percentage points, driven completely by BA degree attainment. Hence, the BA subsidy expansion is less effective than the AA subsidy at subsidizing accessibility and increasing the share of PSE graduates.

Panel B reports changes in macroeconomic aggregates. The first row of the panel reports the change in average labor productivity: this is a measure we construct, which helps to provide intuition for the behavior of the several macroeconomic aggregates and many of our results from the two subsidy expansions. Specifically, average labor productivity is computed for the working-age population, as the weighted sum of individual productivities, $\epsilon(j, e, s)\,\eta$, divided by the mass of individuals. Labor productivity decreases in the AA subsidy expansion and increases in the BA subsidy expansion (see Subfigure D2a of Appendix D.1.2 for a depiction of these changes by age). In the AA subsidy expansion, the decrease in labor productivity leads to the decrease in output, consumption, capital, and labor; in the BA subsidy expansion, labor productivity increases, which leads to an increase in the same four variables. Hence, the BA subsidy is more effective than the AA subsidy in subsidizing quality, as reflected by the increase in labor productivity and aggregates such as output.

The primary reason that labor productivity decreases in the AA subsidy expansion is the diversion effect discussed above (those who would have enrolled in the BA program now enroll in the AA program). In Appendix D.1.3, we show that, in partial equilibrium, labor productivity decreases with the diversion effect but increases without the diversion effect.[20] This exercise cleanly quantifies a key mechanism of our structural analysis and highlights that when the AA program is made tuition-free the diversion effect dominates the democratization effect in regard to labor productivity. This result holds true despite the fact that the democratization effect dominates when it comes to enrollment, a popular outcome measure in reduced-form work examining related subsidies to AA enrollment. In contrast, the BA subsidy expansion increases labor productivity by boosting BA attainment among those who previously enrolled in an AA or did not enroll in postsecondary education.

Additionally, in our framework, the AA program features an option to transfer to the BA program: although the AA subsidy expansion diverts some individuals from straight-to-BA enrollment to AA enrollment, these individuals, later on, may transfer to the BA program. In Appendix D.1.4, we show that this transfer option acts to mitigate the fall in labor productivity after the AA subsidy expansion.

Returning to the remaining macroeconomic aggregate in Panel B, the last row indicates that hours worked decreases more in the AA subsidy expansion than in the BA subsidy expansion. This is because the AA subsidy expansion leads to a larger rise in PSE enrollment, which shifts time use from labor to college effort in a quantitatively meaningful way. The effects of both subsidy expansions on hours worked by age is depicted by Subfigure D2b of Appendix D.1.2. The large fall in hours worked from the AA subsidy expansion also contributes to the fall in output.

In Panel C, the first row indicates that the income tax rate increases with the AA subsidy and decreases with the BA subsidy. This implies that the AA subsidy expansion does not pay for itself in the long run, whereas the BA subsidy expansion does. Public grant expenditure increases in both subsidy expansions, which puts upward pressure on the income tax rate. Additionally, the fall in labor productivity in the AA subsidy expansion compounds the increase in the income tax rate. In contrast, the rise in labor productivity in the BA subsidy expansion more than offsets the increase in public grant expenditure, leading to a fall in the income tax rate.[21]

The second row of Panel C shows that accidental bequests decrease with the AA subsidy and increase with the BA subsidy. In our framework, most of the increase in grant expenditure is funded by the government (almost seventy percent), and the remainder is funded by private sources. This is done by deducting the private grant expenditure from the assets of the deceased; subsequently, the remainder of those assets are transferred to older working-age consumers as accidental bequests (see equation (22) of Appendix B.2.3). As a result, the increase in private grant expenditure that follows from both subsidy expansions puts down-

[20] In this partial equilibrium, the distribution of 18-year-olds and general equilibrium prices, the tax rate, and transfers are held fixed at their initial steady state values.

[21] Progressive income taxation is key for the fall in the income tax rate under the BA subsidy expansion. In the BA subsidy expansion, when labor productivity increases, income tax revenue increases; however, government consumption also increases because output increases. With flat income taxation, the rise in tax revenue is largely offset by the rise in government consumption. However, with progressive income taxation, the increase in labor productivity increases income, which then increases income tax revenue more than government consumption; this allows the government to reduce the income tax rate. In an alternative re-calibrated model with flat income taxation, we verify that both subsidy expansions lead to an increase in the income tax rate.

ward pressure on accidental bequests.[22] Additionally, in the AA subsidy expansion, the decrease in labor productivity makes the economy poorer, which further contributes to the fall in accidental bequests. In contrast, in the BA subsidy expansion, the increase in the labor productivity makes the economy richer, which more than offsets the increase in private grant expenditure, resulting in an increase in accidental bequests.[23]

The third row of Panel C indicates that the change in the interest rate is small in both experiments. The last three rows report changes in the wage rates. In the AA subsidy expansion, the increase in the AA labor supply outweighs the decrease in the BA dropout labor supply resulting in a fall in the wage rate for AA graduates and BA dropouts; the fall in the HS and BA labor supplies leads to an increase in the HS and BA wage rates (see Panel A of Table D2 of Appendix D.1.5 for changes in labor by type). In the BA subsidy expansion, the increase in the BA labor supply leads to a decrease in the BA wage rate; the fall in the HS and AA labor supplies drive the increase in the HS and AA or BA dropout wage rates.

6.3 Aggregate implications across the transition path in general equilibrium

In the context of expanding postsecondary education subsidies, grant expenditure increases immediately but some of the potential benefits are delayed until cohorts that benefit directly from the tuition subsidy enter the labor market. Therefore, taking the transition dynamics into account is important for a complete welfare analysis. In this section, we analyze the transition path for key variables that matter for the welfare analysis in Section 6.4.

For our analysis of transition dynamics, we compare the impact of the two policy changes on a balanced-budget transition in which the income tax rate level parameter is adjusted to balance the government budget in every period. The analysis of a balanced-budget transition is motivated by the fact that this is a common assumption used by the literature and it serves as a benchmark. When we turn to welfare implications in Section 6.4, we also analyze a deficit-financed transition in which the government borrows in the early periods of the transition to finance the increase in government spending.

Figure 3 depicts the transition dynamics for the key variables. Specifically, Subfigure 3a depicts the changes in the income tax rate at initial steady state mean income. Across steady states, the AA subsidy expansion leads to an increase in the income tax rate and the BA subsidy expansion leads to a decrease in the income tax rate. However, in the early periods of the transition, both subsidy expansions lead to higher income tax rates relative to the initial level. Subfigure 3b shows that, across steady states, the AA subsidy expansion decreases accidental bequests and the BA subsidy expansion increases it. However, in the early periods of the transition, both subsidy expansions result in lower accidental bequests relative to their initial level. These patterns in the two subfigures are explained by the immediate increase in public and private grant expenditure following the policy change and the subsequent gradual adjustment in labor productivity (depicted in

[22] Our main takeaways comparing the subsidy expansions are not sensitive to whether or not grant subsidies are partially financed using the assets of the deceased (that is, private funds). In Appendix D.3, we consider a sensitivity analysis in which the public share of total grant expenditure is one and the private share is zero. In this sensitivity analysis, the subsidy expansions do not directly impose a downward pressure on accidental bequests.

[23] The other form of transfers in general equilibrium is Social Security. The changes in Social Security transfers are reported in Panel B of Table D2 of Appendix D.1.5.

Subfigure 3d).

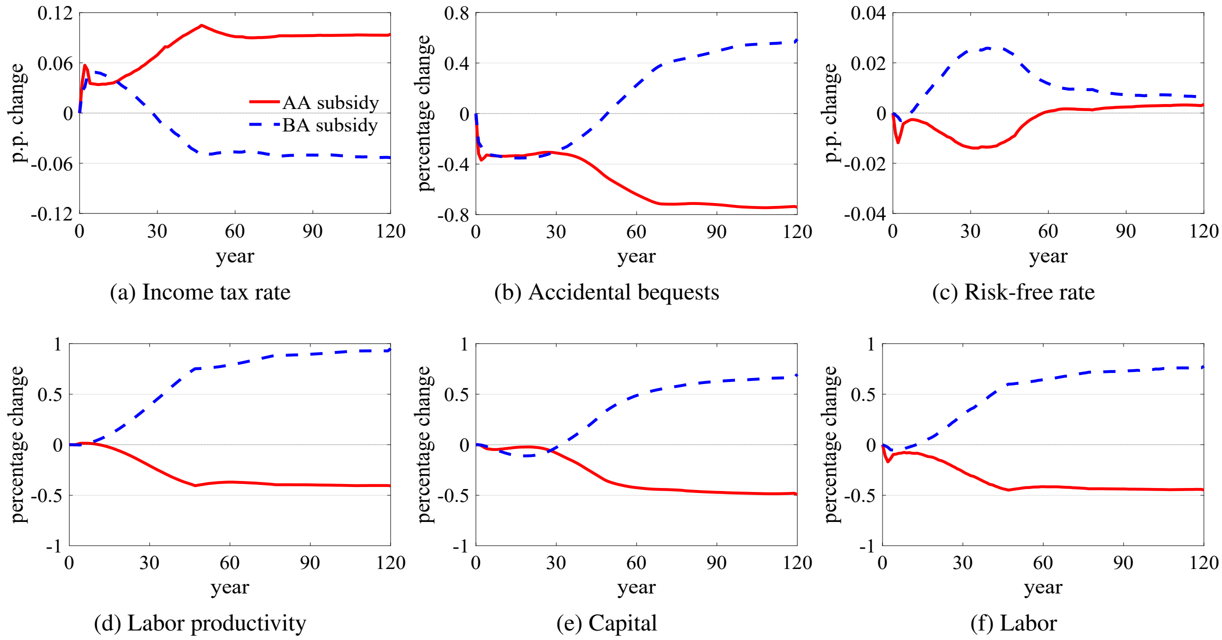


Figure 3: Changes in key macroeconomic aggregates under a balanced-budget transition

Notes: The figure depicts the changes in the income tax rate (at initial steady state mean income), accidental bequests, risk-free rate, labor productivity, capital, and labor from the two subsidy expansions under a balanced-budget transition.

Subfigure 3c depicts the transition dynamics for the risk-free rate. Although changes in the risk-free rate are small across steady states, we observe larger movements in the early periods of the transition. In the AA subsidy expansion, the risk-free rate falls below its initial level, whereas in the BA subsidy expansion, the risk-free rate rises above its initial level. Subfigures 3d, 3e, and 3f provide intuition for the contrasting behavior of the risk-free rate in the early periods of the transition.

Specifically, Subfigure 3d depicts the fall in labor productivity from the AA subsidy expansion and the rise in labor productivity from the BA subsidy expansion. These productivity changes lead to qualitatively similar changes in both aggregate capital and aggregate labor, as depicted by Subfigures 3e and 3f. However, as productivity changes, the change in aggregate labor outpaces the change in capital in the early periods of the transition. Consequently, in the AA subsidy expansion, aggregate labor falls at a faster rate than the capital stock, which makes capital relatively more abundant and drives down the risk-free rate. In the BA subsidy expansion, aggregate labor rises at a faster rate than the capital stock, which makes capital relatively more scarce and drives up the risk-free rate.

To summarize, both subsidy expansions increase grant expenditure, which puts upward pressure on the income tax rate and downward pressure on accidental bequests. At the same time, the AA subsidy expansion leads to a fall in labor productivity, which compounds the rise in the income tax rate and the fall in accidental bequests; it also decreases the risk-free rate in the early periods of the transition. The same mechanisms work in the opposite direction for the BA subsidy expansion. In the next section, we examine how these general equilibrium effects and transitions dynamics matter for welfare changes.

6.4 Welfare implications

Our welfare analysis focuses on current and future generations of consumers who are aged 18 and over. To measure welfare changes, we compute one-time compensation transfers for each individual.[24] The timing of the transfer differs depending on the age of the consumer in the period when the transition begins. If the consumer is aged 18 and over, we compute a one-time transfer made in the first period of the transition that makes the consumer indifferent between the value of experiencing the transition and remaining in the initial steady state. If the consumer is younger than 18 or not yet born, we compute the one-time transfer made in the period in which the individual enters the economy as an adult so that, behind the veil of ignorance, the consumer is indifferent between entering the economy as an 18-year-old during the transition and entering the economy as an 18-year-old in the initial steady state. The transfers estimated in a future period are discounted to the present value using the risk-free rate to make them comparable with the transfers computed for the generations of adults alive in the first period of the transition.

We assume that the economy is in its initial steady state in period zero, and that the transition is announced unexpectedly at the beginning of period one. For the generation that is exactly 18 years old in the period when the transition begins, we assume that the transition is announced after the realization of the 18-year-old's skill but before the parents make the inter vivos transfer decision. This timing assumption implies that a consumer who is 18 years old in period one knows what inter vivos transfer they would have received in the initial steady state and what inter vivos transfer they will receive in the transition. Similarly, the 18-year-old knows what their EFC would have been in the initial steady state and what it will be in the transition. Therefore, the one-time compensation transfer computed for such a consumer takes into account the effects of adjustments in the inter vivos transfer and EFC due to the policy change. Similarly, for future generations of 18-year-olds, effects on initial characteristics such as net assets, EFC, and skill are taken into account in the welfare computation. Additional details are provided in Appendix B.4.

Table 16 reports welfare changes from the AA and BA subsidy expansions under three cases: (1) a partial equilibrium case ("PE") in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) the balanced-budget transition; and (3) the deficit-financed transition (described in more detail below). Panel A reports the total of one-time transfers as a percentage of initial steady state GDP for all consumers: this includes current and future generations of consumers who are aged 18 and over. Panel B reports averages of one-time transfers as a percentage of initial steady state GDP per capita 18+ for mutually exclusive consumer groups. The first seven rows are restricted to consumers who are aged 18 and over and are alive in the first period of the transition ($t = 1$): they include (1) high school graduates ("HS grads"), (2) sophomores in the AA program ("AA enrollees"), (3) sophomores, juniors, and seniors in the BA program ("BA enrollees"), (4) those of working-age after the college phase ($j > 5$) but before the inter vivos transfer age ("Pre-IVT age"), (5) those at the inter vivos transfer age ("At IVT age"), (6) those of working-age after the inter vivos transfer age ("Post-IVT age"), and (7) retirees. The last row is restricted to future generations

[24] In Table D4 of Appendix D.1.8, we analyze the share of population that is better off.

of 18-year-olds.

Table 16: Welfare changes in one-time transfers: total and by mutually exclusive consumer groups

			AA subsidy expansion			BA subsidy expansion		
Panel	**Consumer group**	**Period**	**PE**	**Balanced-budget**	**Deficit-financed**	**PE**	**Balanced-budget**	**Deficit-financed**
A: Total	All	$t \geq 1$	1.03	-0.71	-0.03	1.39	1.15	1.61
B: Average	HS grads	$t = 1$	0.95	0.32	0.53	1.41	1.20	1.30
	AA enrollees	$t = 1$	5.17	3.52	3.72	0.50	0.93	1.02
	BA enrollees	$t = 1$	0.15	-0.35	-0.17	1.81	1.36	1.44
	Pre-IVT age	$t = 1$	1.25	-0.78	0.14	1.13	1.12	1.72
	At IVT age	$t = 1$	1.99	-0.49	0.72	1.98	1.47	2.22
	Post-IVT age	$t = 1$	0	-1.44	-0.57	0	-0.35	0.20
	Retirees	$t = 1$	0	-0.40	-0.07	0	-0.29	0
	Future generations	$t > 1$	1.16	0.25	0.12	2.19	2.05	1.96

Notes: The table reports welfare changes from the AA and BA subsidy expansions under three cases: (1) "PE" is a partial equilibrium case in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) "Balanced-budget" is a general equilibrium case in which the income tax rate level parameter is adjusted to balance the government budget in every period; and (3) "Deficit-financed" is a general equilibrium case in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate. Panel A reports total one-time transfers as a percentage of initial GDP. Panel B reports average one-time transfers as a percentage of initial GDP per capita 18+ for mutually exclusive consumer groups. "IVT" refers to inter vivos transfer. "Period" refers to the transition period.

In partial equilibrium, Panel A shows that both subsidy expansions lead to total gains amounting to a one-time transfer worth more than 1 percent of initial GDP. Panel B provides a breakdown of the gains by the mutually exclusive consumer groups described in the previous paragraph. We observe strictly positive average gains for all consumer groups except the groups in the period of the transition who are older than the inter vivos transfer age ("Post-IVT age" and "Retirees"). In partial equilibrium, these consumers who are older than the inter vivos transfer age experience a welfare change of zero because there are no altruistic gains or losses (their children have already moved out) and there are no general equilibrium effects. Panel B also indicates that in both subsidy expansions, parents at the inter vivos transfer age ("At IVT age") experience larger gains than the HS graduates ("HS grads") although the latter group is the direct beneficiary of the tuition subsidy. The parents benefit more because they can reduce their inter vivos transfer in response to the tuition subsidy (see Table D2 of Appendix D.1.5 for the decline in inter vivos transfers); incorporating this feedback effect shifts some of the gains from the high school graduate to the parent.[25]

In the balanced-budget case, Panel A shows that the AA subsidy expansion leads to total losses that amount to 0.71 percent of initial GDP in contrast to the gains above 1 percent observed in partial equilibrium; the BA subsidy expansion leads to total gains that are somewhat lower than they were in partial equilibrium but still above 1 percent of initial GDP. Panel B shows that, especially in the AA subsidy expansion, the gains turn to losses for various consumer groups that are alive in the first period of the transition ("pre-IVT age," "at IVT age," "post-IVT age"). Therefore, once we take general equilibrium effects and the transition dynamics into account, the AA subsidy expansion leads to losses and the BA subsidy expansion leads to substantial gains.

[25] In the AA subsidy expansion, when this feedback effect is not taken into account, the gains to the HS graduates increase from 0.95 to 2.53 percent of initial GDP per capita 18+; similarly, in the BA subsidy expansion, the gains increase from 1.41 to 2.07 percent of initial GDP per capita 18+.

From partial to general equilibrium, why are gains flipped to losses in the AA subsidy expansion, but only dampened slightly in the BA subsidy expansion? In what follows, we elaborate on how the fall in labor productivity in the AA subsidy expansion and the rise in labor productivity in the BA subsidy expansion, depicted in Subfigure 3d, explain these implications for welfare.

As discussed in Sections 6.2 and 6.3, the fall in labor productivity in the AA subsidy expansion compounds the rise in the income tax rate and the fall in accidental bequests, and also drives the fall in the risk-free rate in the early periods of the transition. In Appendix D.1.6, we perform a decomposition analysis and show that the main contributor to the welfare losses is the increase in the income tax rate, although the decrease in accidental bequests and risk-free rate also contribute to the welfare losses.[26] The rise in labor productivity in the BA subsidy expansion leads to the opposite effect. The decomposition analysis that we perform in Appendix D.1.6 reveals that although the changes in the income tax rate and accidental bequests under the BA subsidy expansion act to dampen welfare gains, the extent of this dampening is smaller in comparison to that of the AA subsidy expansion. Additionally, the increase in the risk-free rate in the early periods of the transition under the BA subsidy expansion, which is beneficial to savers, contributes significantly to the welfare gains.

Thus far, we have focused on a balanced-budget transition. Next, we consider the welfare implications under a deficit-financed transition. In this transition, the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the government budget in every period while the government pays down its debt at a constant decay rate of 0.93.[27] The analysis of the deficit-financed transition is motivated by two factors: first, deficit-financing is a common approach used by governments in policy implementation; and, second, it allows the government to delay the tax burden of financing the tuition subsidy until potential recipients of the subsidy have directly benefited from the policy change. Because future welfare changes are discounted, this type of transition allows the government to shift some of the welfare losses to the future and to some extent reduce the total welfare losses or increase the total welfare gains.[28]

In the deficit-financed case, the gains observed in partial equilibrium under the AA subsidy expansion are dampened enough to turn to losses; however, the losses are smaller in magnitude compared to the balanced-budget case. In the BA subsidy expansion, the gains remain above one percent. Therefore, after taking general-equilibrium effects and transition dynamics into account, in the deficit-financed transition the AA subsidy expansion leads to no gains and the BA subsidy expansion leads to substantial gains.

In our main experiments, we analyzed subsidy expansions in which either the AA program is subsidized exclusively, or the BA program is subsidized exclusively. In Appendix D.2, we analyze experiments in which tuition subsidies are weakly expanded to both the AA and BA programs jointly, subject to the spending-equivalence criterion.[29] We find that the exclusive spending-equivalent BA subsidy expansion (that is, the

[26] Although those with student loans benefit from a lower risk-free rate, savers are hurt.

[27] The constant decay rate is within the range of 0.9 and 0.95 used in Auclert, Bardóczy, and Rognlie (2021).

[28] Appendix D.1.7 compares the transition dynamics of the tax rate and prices in the balanced-budget and deficit-financed transitions.

[29] Specifically, in each case of the joint tuition subsidy expansion, the tuition subsidy expansion for the AA program is varied between zero and its tuition-free AA expansion value, and then the BA subsidy expansion is chosen to make total government

BA subsidy expansion analyzed in the main text) maximizes total welfare.

In summary, we argue that in terms of welfare, the total benefits from subsidizing accessibility (AA subsidy expansion) are negative or close to zero, whereas the total benefits of subsidizing quality (BA subsidy expansion) are positive and substantial. Our results also highlight the importance of taking general-equilibrium effects and transition dynamics into account for a complete welfare analysis.

7 Conclusion

Several recent policy proposals have sought to make AA programs tuition-free at the national level. However, the existing structural macroeconomic literature on postsecondary financial aid has focused predominantly on BA programs. In this paper, we document tradeoffs between AA and BA programs related to accessibility and quality, as well as correlations of AA, BA, and AA-to-BA enrollment with observable characteristics such as youth skill, parental income, and parental education. Next, we build a quantitative model with an AA and BA program, which reflects the key tradeoffs. The quantitative model accounts for the correlations of youth skill, parental income, and parental education with AA, BA, and AA-to-BA enrollment. We then use our framework to analyze the aggregate and welfare implications of making two-year programs tuition-free, and compare it with an alternative spending-equivalent BA subsidy.

Our results show that making the AA program tuition-free is a cost-effective way to increase the share of postsecondary graduates, consistent with the goal of this proposed policy. However, the AA subsidy expansion leads to lower labor productivity and output. Furthermore, once we take general-equilibrium effects and transition dynamics into account, the welfare gains observed in partial equilibrium are essentially eliminated or turn to losses. In contrast, the spending-equivalent BA tuition subsidy leads to higher labor productivity and output, as well as substantial welfare gains. In fact, with a limited government budget, we show that it is optimal to exclusively expand a tuition subsidy for the BA program.

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Online Appendix for: "Two-year Programs and College Financial Aid"

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Contents

A Data Appendix

A.1 The 1997 National Longitudinal Survey of Youth

We use our NLSY97 postsecondary outcome estimation sample for results regarding postsecondary enrollment patterns and related model parameterization moments in the main text (Bureau of Labor Statistics, U.S. Department of Labor, 2019). We also use the NLSY97 in our estimation of the life cycle earnings process, which is estimated using the earnings profile estimation sample, and in our estimation of inter vivos transfers, which is based on a third sample. When we convert nominal dollar values to real values, we have converted to 2013 dollars using the Consumer Price Index or CPI (Bureau of Labor Statistics, U.S. Department of Labor, 2024). Our three estimation sample cleaning procedures are described next, along with descriptions of how several key variables are constructed.

Estimation samples Our postsecondary outcomes estimation sample is at the individual level, and consists of respondents who we observe until age 30 (i.e., up to and including age 30), and whose education records indicate that they completed high school at the age of 17, 18, 19, or 20. We also require that we observe the ASVAB tercile, parental income tercile and parental education in 1997 for each respondent and that the variable reporting enrollment and graduation status from high school and two- or four-year college degree programs in each wave not imply inconsistent information.[30]

Throughout our analyses using the postsecondary outcomes estimation sample, we interpret two-year program as synonymous with an AA and four-year program as synonymous with a BA. When we define AA or BA enrollment, we refer to being enrolled in a two-year or four-year program, respectively, at some point by age 24. When we define graduation rates for AA or BA programs, we refer to being a graduate of a two-year or four-year program, respectively, by age 30. We define someone who takes the AA-to-BA enrollment path as someone who we observe being enrolled in a two-year program before being enrolled in a four-year program; we do not require that the respondent graduate from the two-year program before starting the four-year program in this definition. When we assign observations to terciles of skill (i.e., the ASVAB score) and parental income in 1997, we do so using the distribution of valid observations for skill or parental income among survey respondents who have completed high school at the ages of 17, 18, 19, or 20, and whose parental education in 1997 is observed, but do not impose other requirements on the sample.

We apply two methods to identify observations with inconsistent enrollment/graduation status histories that are to be dropped: a general approach and an age-threshold approach. With the general approach, we use the interview year in which a respondent first reports enrolling in a two-year, four-year, or graduate program, and the year in which they first report having some college, a two-year degree, or a four-year degree.[31]

[30] In our NLSY97 analyses, for variables where we condition on an outcome occurring up to and including age 24 or 30 we account for the fact that the NLSY97 switched to collecting data every other year instead of every year after 2011 in the following way. Given the age of the respondent in 1997, we identify the year in which they are 24 or 30. If that year is not an interview year, then we use enrollment or graduation status reported when they are 25 or 31, respectively. Additionally, for all of our samples, we do not condition the sample on observing gender, race, and ethnicity because these variables have valid values for everyone in the NLSY97.

[31] Enrolling in a graduate program or earning a graduate degree implies the respondent earned a four-year degree before that. This

We then drop observations who claim to have attained "some college" without any record of a previous postsecondary enrollment spell, as well as observations who report obtaining a four-year (two-year) degree without reporting four-year (two-year) enrollment in a previous year. When we additionally apply the age threshold method to the sample, we perform a similar routine but collapse education histories into flags for AA or BA educational enrollment by age 24 and degree attainment by age 30, or some college by age 30. We drop observations where there is a record of achieving some college by age 30 but no record of whether it was enrollment in a two-year or a four-year program by age 24, and where the respondent reports being a two-year program graduate by age 30 but there is no record of two-year program enrollment by age 24, or where they report being a four-year program graduate by age 30 but there is no record of four-year program enrollment by age 24. This leaves us with a total of 2,257 individuals.

The earnings profile estimation sample reshapes the data to be at the individual-year level (an unbalanced panel), where the year is the year of the interview. Observations in this sample are selected in a similar way as the postsecondary outcomes estimation sample, in that we impose a baseline set of conditions and then clean out inconsistent education histories. For the baseline set of conditions, we require that the respondent have completed high school while aged 17, 18, 19, or 20, and that we observed the skill tercile of the respondent. In a given survey wave, we also require that they be aged from 25 to 39 and not enrolled in postsecondary education as well as that we observe their marital status, and number of children (topcoded at 4). We also restrict attention to observations where there is a valid value for the hourly real wage at the respondent's main job in that wave, after dropping observations where real wages are above 400 dollars or below 2 dollars in 1993 USD converted to real values using the CPI (these nominal wage level cutoffs are from Guvenen (2009)). Our sample selection procedure yields 4,130 individuals (25,680 individual-year observations) in the earnings profile estimation sample.

For the inter vivos transfers estimation sample, which also uses panel data, our baseline requirements are that the survey year be 1997-2003, that the respondent be aged 18 to 23 at the time of the interview and have completed high school, that they be counted as independents by the NLSY97, and have valid values for current enrollment and parental income tercile. We then clean out observations with inconsistent education histories. This yields 3,942 individuals (12,582 individual-year observations) in the inter vivos transfer sample.

A.1.1 Tradeoffs of AA-to-BA versus straight-to-BA: supporting tabulations

Differences in time spent enrolled in a BA To examine the impact of the AA-to-BA route on the total cost of pursuing a BA degree, using the NLSY97 we show in Table A1 both the average (with its standard error) and median years spent enrolled in a BA up to age 30, conditional on enrolling in a BA by age 24. The values are reported in separate rows for those who take the straight-to-BA route and those who take the

captures those who complete the BA degree and start graduate school immediately, and so never report not being enrolled and having a four-year degree. Additionally, if the respondent enrolled in a four-year program before a two-year program, we set the first year of two-year program enrollment back to none. This is consistent with our concept of path to highest enrollment and is also implemented when we construct the age threshold education flags. We also do this for two-year degree attainment for such cases, to avoid those who earn an AA degree being dropped from the estimation sample because their education histories now seem inconsistent.

AA-to-BA route. In addition, this breakdown is reported for all BA enrollees (first two columns) and for those who earn a BA by age 30 (last two columns).

Table A1: Years spent enrolled in BA up to age 30

	BA enrollees		BA graduates	
Path to BA enrollment	**Mean (SE)**	**Median**	**Mean (SE)**	**Median**
Straight to BA	4.23 (0.047)	4.00	4.61 (0.039)	4.00
AA to BA	2.94 (0.104)	3.00	3.19 (0.112)	3.00
Observations	1,307		985	

Notes: The table reports the mean (standard error) and median of years spent enrolled in a BA up to age 30. The statistic is reported separately for those who enroll straight in a BA and for those who take the AA to BA path. The "BA enrollees" columns condition on enrolling in a BA by age 24, and the "BA graduates" columns additionally condition on earning a BA by age 30. Source: NLSY97.

The first column indicates that taking the AA-to-BA route reduces years spent enrolled in a BA by at least one year on average, compared to going straight to a BA. The third column implies that this difference in averages is not driven by BA dropouts: it is slightly higher among those who go on to earn the BA degree. The lower average duration of BA enrollment for AA-to-BA enrollees is also not driven by outliers: comparing median values in the second and fourth columns reveals a similar one-year difference. Together, the lower number of years spent enrolled in a BA and the lower annual cost of an AA documented in Table 1 indicate a lower overall cost of BA degree pursuit for AA-to-BA enrollees.

Differences in continuation likelihoods Among those who enroll in a BA by age 24, on average taking the AA-to-BA route is associated with a lower likelihood of progressing towards BA completion each year compared to going straight into BA enrollment. This pattern is shown in Table A2. The first two columns report the share who earn a BA by age 30 within each path to BA enrollment, broken down by skill tercile. These overall BA completion rates are increasing in skill for both paths to BA enrollment. However, those who take the AA to BA route arrive in their BA program with one or two academic years under their belt, unlike their straight-to-BA peers. To address this, we report annualized continuation rates in the last two columns. Consistent with the findings of Table A1, these annualized probabilities assume that those who take the AA-to-BA path need to successfully pass three years enrolled in a BA to graduate, while those who take the straight-to-BA route need four years. Once annualized, we observe a lower continuation probability for those in the AA-to-BA group in the top skill tercile (the lower two skill terciles have more similar continuation probabilities for the two BA enrollment paths).

A.1.2 Predicting PSE enrollment outcomes: regression coefficients

Table A3 reports probit regression coefficients for the AMEs reported in Table 7 of the main text.

A.1.3 Target moments for internally calibrated parameters

The first three columns of Table A4 report various education-related rates used in the internal calibration of the model, by skill tercile and overall. In the table's first column we report the AA enrollment rate by skill

Table A2: BA degree attainment rates by path to BA enrollment

	Overall share		Annualized probability	
Skill	**AA to BA**	**Straight to BA**	**AA to BA**	**Straight to BA**
1	0.638	0.564	0.861	0.867
2	0.729	0.681	0.900	0.908
3	0.753	0.839	0.910	0.957
Total	0.716	0.760	0.895	0.934
Observations	194	1,113	194	1,113

Notes: The table reports BA attainment rates by path to BA enrollment, within each skill tercile and in total (first and second columns) as well as the annualized continuation probability (third and fourth columns). The annualized probabilities assume AA-to-BA enrollees need three years of BA enrollment to graduate, while straight-to-BA enrollees need four years. Source: NLSY97.

Table A3: Regression coefficients: Predicting PSE enrollment outcomes

	Dependent variable				
Control variable	**(1) Enroll BA**	**(2) Enroll AA**	**(3) Enroll AA**	**(4) Enroll AA**	**(5) AA to BA**
Flag: Parental income tercile 1	-0.53	0.20	-0.28	0.47	0.35
	(0.084)	(0.081)	(0.115)	(0.093)	(0.126)
Flag: Parental income tercile 2	-0.27	0.16	-0.17	0.26	0.25
	(0.077)	(0.074)	(0.110)	(0.081)	(0.105)
Flag: Skill tercile 1	-1.52	0.34	-0.84	1.17	0.66
	(0.087)	(0.081)	(0.115)	(0.100)	(0.139)
Flag: Skill tercile 2	-0.68	0.33	-0.32	0.52	0.25
	(0.075)	(0.072)	(0.115)	(0.078)	(0.102)
Flag: At least 1 parent BA+	0.78	-0.36	0.40	-0.60	-0.37
	(0.074)	(0.071)	(0.116)	(0.076)	(0.097)
Flag: Female	0.30	0.03	0.27	-0.11	0.06
	(0.061)	(0.058)	(0.078)	(0.067)	(0.090)
Flag: Black	0.58	-0.16	0.31	-0.54	-0.45
	(0.086)	(0.080)	(0.106)	(0.098)	(0.134)
Flag: Hispanic	0.19	0.10	0.32	-0.05	-0.04
	(0.089)	(0.083)	(0.106)	(0.100)	(0.137)
Constant	0.63	-0.82	0.49	-0.62	-1.20
	(0.082)	(0.081)	(0.126)	(0.087)	(0.114)
Sample	HS grads	HS grads	Enr AA or No PSE	Enroll PSE	Enroll BA
Observations	2,257	2,257	1,144	1,744	1,307
pseudo- R^2	0.26	0.04	0.07	0.17	0.07

Notes: The table reports probit regression coefficients for regressions with the following dependent variables: (1) the likelihood of enrolling in a BA at some point up to age 24, conditional on completing high school; (2) the likelihood of enrolling in an AA at some point up to age 24, conditional on completing high school; (3) the likelihood of enrolling in an AA conditional on either enrolling in an AA or no PSE some point up to age 24; (4) the likelihood of enrolling in an AA conditional on enrolling in any PSE at some point up to age 24; (5) the likelihood of previously enrolling in an AA, conditional on enrolling in a BA at some point up to age 24. Standard errors in parentheses. Source: NLSY97.

tercile for those with parents in the top parental income tercile. The second and third columns report AA and BA enrollment rates by skill tercile, respectively, among high school graduates.

Table A5 reports another set of calibration targets, the distribution across skill terciles for children from families with highest parental education in the following three categories: No college, some college, or a bachelor's degree. Parental education is reported as number of years completed, so it is mapped into these education outcome categories using the thresholds given in the column labels. Unlike other tables using the NLSY97, the sample for this table conditions only on parental education group.

Table A4: Internal calibration targets: moments by skill tercile from the NLSY97

	Enrollment type		
Skill	**AA | High income**	**AA**	**BA**
1	0.404	0.353	0.245
2	0.230	0.320	0.589
3	0.141	0.190	0.827
Total	0.202	0.280	0.579
Observations	778	2,257	2,257

Notes: The table reports rates within each skill tercile of AA enrollment rates by age 24 for those with parents in the top parental income tercile as well as AA and BA enrollment rates by age 24. Source: NLSY97.

Table A5: Child skill distribution by parental education

	Highest parental educational attainment (years of education)		
Skill	**No college:** ≤ 13	**Some college:** $14-15$	**BA:** $16+$
1	0.476	0.287	0.137
2	0.328	0.380	0.312
3	0.196	0.333	0.551
Observations	2,512	1,013	1,582

Notes: The table reports the conditional distribution of child skill across terciles within each parental education grouping among high school graduates. Source: NLSY97.

A.1.4 Validation moments for parameterized model

Table A6 reports highest postsecondary degree attainment rates by age 30 in the NLSY97 postsecondary attainment estimation sample. Types of attainment are mutually exclusive; shares in the table sum to one.

Table A6: Highest PSE degree attainment rates among high school graduates

Category	**Type**	**Share**
No BA degree	HS	0.397
	BA d/o	0.100
	AA degree	0.066
BA degree	AA-to-BA	0.062
	Straight-to-BA	0.375
Observations	2,257	

Notes: The table reports highest postsecondary degree attainment rates by age 30 among high school graduates. HS includes those with no postsecondary enrollment, AA dropouts, and BA dropouts with at most one year of enrollment. Shares sum to one. Source: NLSY97.

A.1.5 Life cycle earnings profile estimation

To estimate the parameters of the earnings profile we use the Panel Study of Income Dynamics or PSID (Survey Research Center, Institute for Social Research, 2021), in combination with the NLSY97, earnings profile estimation sample described at the start of this section. We use both the PSID and NLSY97 because the longer panel dimension of the PSID allows us to observe more ages than the NLSY97 and to estimate a more complete life cycle age profile, while variables in the NLSY97 allow us to incorporate the joint effects of skill, AA degree attainment, BA degree attainment with or without the AA-to-BA enrollment path, or BA dropout after 2 or more years on labor earnings. We estimate the stochastic component of the earnings process on the residuals from the NLSY97 skill-shifter and education-subtype-shifter estimation, which uses age-free wages cleaned by applying the age profiles estimated using the PSID.

The PSID sample uses data from 1968-2019. We only use observations from the SRC core sample. Our sample cleaning adapts that of Guvenen (2009) and creates five flags for different sets of criteria that observations must satisfy to be in our sample. First, we require that the respondent be a household head or their spouse/cohabiting partner, and that the respondent have nonmissing values for the number of children in their family unit under 18 years of age, the sex and marital status of the household head, state of residence (territory residence is not admissible), and labor force status, and for whom hours worked are below 5110 hours per year. Second, we flag observations for whom we observe education and who are aged between 18 and 64. Third, we flag observations where the education history is consistent in the sense that educational progress is monotonic as the respondent ages. Fourth, we flag observations where real wage level is equal to or above 2 and equal to or below 400 in 1993 dollars, converted to 2013 dollars using the CPI. Fifth, we flag those who are not self employed. We require that an individual satisfy the requirements of these five flags for at least 20 potentially non-consecutive years in order to be in our sample. The PSID estimation sample contains 1,491 individuals for the no-BA group (33,292 individual-year observations) and 1,044 individuals (25,502 individual-year observations) in the BA group. The NLSY97 earnings profile estimation sample has 2,529 individuals in the no-BA group (15,359 individual-year observations) and 1,734 individuals (10,321 individual-year observations) in the BA group. Note that, in the PSID or the NLSY97 sample, a single individual can appear in both education groups over the course of their working life, but in only one group within a given survey year.

In both the PSID and NLSY97 estimations, besides the controls of interest we also control for survey year, marital status, number of children (top-coded at 4), and gender. The PSID estimation additionally controls for state of residence, and the NLSY97 estimation additionally controls for being the supplemental sample. Panel B in Table 8 of Section 4 reports the life cycle earnings process estimation results.

A.1.6 Inter vivos transfers

We estimate average inter vivos transfers to college-aged children from their parents in the NLSY97, reported in Table A7, using the inter vivos estimation sample described at the start of this section.

We compute yearly nominal transfers from parents to their child as follows. We account for the implicit transfer from parents to their children who live with them and do not pay rent by flagging those living with their parents and paying no rent. We then impute value of the unpaid rent using average rent paid by sample members with the same family income tercile, college enrollment status, and interview year who are not living with their parents after dropping the top and bottom percentile of the rent distribution. Next, we annualize rent and add it to yearly net income received from parents. Finally, we add any yearly allowances received.

The transfer ratio reported in the first row of Table A7 is computed as follows. Within each year, we multiply the quantity by six, and then find the mean within the enrollment, parental income tercile, and year grouping of the individual. We then divide the mean by that year's nominal GDP per capita for those over 18. The result is a unitless ratio of implied total transfers received to per capita income for each individual while they are young adults of college age. We then average this ratio across individuals and years to compute the transfer ratio while sample members are young adults of college age. GDP per capita for those 18 and over in each year is computed as described in Section 4.

Table A7: Inter-vivos transfers: average transfers + sample summary statistics

Variable	**Mean**
Transfer ratio	0.609
Transfers	6,699
Observations	12,582
Individuals	3,942

Notes: The table reports statistics related to inter vivos transfers in the NLSY97. Sample: independents aged 18-23 from 1997 to 2003. Dollars are in real values. Observations are at individual-year level. Source: NLSY97.

A.2 The High School Longitudinal Study of 2009

The High School Longitudinal Study of 2009 (HSLS:09) is a panel survey conducted by the United States Department of Education that is representative of 9th graders in the US in 2009 (National Center for Education Statistics, U.S. Department of Education, 2020). The panel collects information in the base year of 2009 (BY), in a first follow-up in 2011 (F1), an update in the summer of 2013 (2013 Update), in a second follow-up in 2016 (F2), and via postsecondary transcripts and student records (PETS-SR). The BY and F1 rounds of data collection implement several questionnaires to sample members, their parents, and school administrators. In the 2013 Update either the student or the parent (on behalf of the student) completes a questionnaire; in F2, only the student completes a questionnaire. PETS-SR contains information on grants and tuition and fees collected directly from postsecondary institutions and administrative records.

Our HSLS:09 estimation sample requires that students earn a high school diploma in 2013 and have valid observations for honors-weighted high school GPA, race, gender, parental education, and parental income. GPA is our measure of skill in the HSLS:09 because there is no standardized test like the ASVAB of the NLSY97. A valid parental income observation means we can see nonimputed discretized parent income at least once before high school completion; if we observe it twice, we use the average real value after assigning a dollar value to each income bin, as described in the discussion surrounding Table A12 of Appendix A.3.1. Dollar values are converted to 2013 dollars using the CPI. We also require that the respondent live with at least one parent at some point in the first two waves of data collection. In addition, we require that we have sufficient information to construct the respondent's postsecondary education outcome and that this education outcome satisfy consistency conditions. Analogous to the NLSY97 sample cleaning, we reset AA enrollment records to null for those who enroll in a BA program, then an AA program (we do not measure degree attainment in the HSLS:09). The rest of the sample cleaning is very similar to Moschini and Raveendranathan (2026).

A.2.1 Predicting postsecondary enrollment outcomes

Table A8 reports Average Marginal Effects (AMEs) from a set of probit regressions run on the HSLS:09 estimation sample. These probits are similar to that of Table 7 of the main text, which are run on the NLSY97 postsecondary outcomes sample. Coefficients for Table A8 are presented in Table A9.

Using the sample of high school graduates, model (1) regresses an indicator for BA enrollment within three years of high school completion on indicators for parental income tercile, skill tercile, parental educational attainment, and (although their AMEs are not reported) demographic flags for being female, black, or Hispanic. Those with poorer parents or lower skill are less likely to enroll in a BA, while those with at least one parent with a BA or higher are more likely to enroll in a BA. All reported AMEs are significant at the one percent level. Model (2) predicts AA enrollment within three years of high school completion on the same set of controls and using the same sample as model (1). Lower parental income or skill positively predicts AA enrollment, while at least one highly educated parent negatively predicts it; reported AMEs are significant at the one percent level except for parental education, which is significant at the five percent level, and the first parental income tercile, which is not significant at the ten percent level. In models (3), (4), and (5), we use the same controls to predict enrolling in an AA among those who either enroll in an AA or no PSE, PSE enrollees, and taking the AA-to-BA route among BA enrollees, respectively. Poorer parents and lower skill negatively predict AA enrollment among those who either enroll in an AA or no PSE; these controls positively predict AA enrollment among PSE enrollees and AA-to-BA enrollment among BA enrollees, while parental education negatively predicts those outcomes. In model (3), the reported AMEs are significant at the one percent level except for the first and second parental income terciles, which are significant at the five and ten percent level, respectively. In model (4), all of the AMEs are significant at the one percent level. In model (5), the first parental income tercile is not significant at the ten percent level, the second parental income tercile is significant at the five percent level, and parental education is significant at the ten percent level. The other reported AMEs are significant at the one percent level. The shorter panel dimension of the HSLS:09 compared to the NLSY97 affords a relatively narrow temporal window to observe AA-to-BA transfers, which may play a role in the lower significance of the AMEs in model (5) compared to the analogous exercise using the NLSY97.

A.2.2 Postsecondary application and acceptance rates

Table A10 reports application and acceptance rates for the subset of high school graduates for whom valid and internally consistent values for this set of variables are reported in the 2013 Update wave of the HSLS:09. These moments reflect application and acceptance outcomes in the fall of the 2013-2014 academic year. Specifically, the first two columns report the share of high school graduates with valid values who apply to an AA program and who are accepted to an AA program conditional on applying. The third and fourth columns report the same information for BA programs.

Table A8: Average Marginal Effects: Predicting PSE enrollment outcomes

	Dependent variable				
Control variable	**(1) Enroll BA**	**(2) Enroll AA**	**(3) Enroll AA**	**(4) Enroll AA**	**(5) AA to BA**
Parental income tercile 1	-0.20	0.05	-0.12	0.16	0.03
	(0.026)	(0.030)	(0.049)	(0.034)	(0.021)
Parental income tercile 2	-0.12	0.06	-0.07	0.11	0.04
	(0.021)	(0.024)	(0.039)	(0.026)	(0.017)
Skill tercile 1	-0.57	0.13	-0.32	0.46	0.10
	(0.021)	(0.028)	(0.038)	(0.032)	(0.030)
Skill tercile 2	-0.28	0.13	-0.15	0.20	0.04
	(0.021)	(0.022)	(0.040)	(0.024)	(0.015)
Flag: At least 1 parent BA+	0.16	-0.05	0.12	-0.12	-0.03
	(0.021)	(0.019)	(0.029)	(0.024)	(0.017)
Additional demographic flags	Y	Y	Y	Y	Y
Sample	HS grads	HS grads	Enroll AA or no PSE	Enroll PSE	Enroll BA
Observations	7,214	7,214	3,158	5,570	4,382

Notes: The table reports probit regression AMEs for: (1) the likelihood of enrolling in a BA at some point up to 3 years after high school completion; (2) the likelihood of enrolling in an AA at some point up to 3 years after high school completion; (3) the likelihood of enrolling in an AA at some point up to 3 years after high school completion, conditional on either enrolling in an AA or no PSE; (4) the likelihood of enrolling in an AA at some point up to 3 years after high school completion, conditional on enrolling in any PSE; and (5) the likelihood of taking the AA to BA path at some point up to 3 years after high school completion, conditional on enrolling in a BA. Standard errors in parentheses. Additional demographic flags included but not shown here are an indicator for being female, black, and Hispanic. Weights are Second follow-up student longitudinal weights. Source: HSLS:09.

Table A9: Regression coefficients: Predicting PSE enrollment outcomes

	Dependent variable				
Control variable	**(1) Enroll BA**	**(2) Enroll AA**	**(3) Enroll AA**	**(4) Enroll AA**	**(5) AA to BA**
Parental income tercile 1	-0.70	0.16	-0.33	0.53	0.20
	(0.088)	(0.099)	(0.134)	(0.108)	(0.146)
Parental income tercile 2	-0.41	0.21	-0.18	0.38	0.28
	(0.073)	(0.080)	(0.106)	(0.089)	(0.108)
Skill tercile 1	-1.75	0.45	-0.86	1.37	0.61
	(0.073)	(0.093)	(0.110)	(0.096)	(0.141)
Skill tercile 2	-0.87	0.46	-0.42	0.67	0.32
	(0.066)	(0.075)	(0.110)	(0.076)	(0.099)
Flag: At least 1 parent BA+	0.55	-0.16	0.33	-0.41	-0.21
	(0.070)	(0.065)	(0.078)	(0.076)	(0.111)
Flag: Female	0.08	0.04	0.13	0.00	0.05
	(0.056)	(0.058)	(0.072)	(0.067)	(0.100)
Flag: Black	0.37	0.05	0.37	-0.26	-0.51
	(0.120)	(0.124)	(0.162)	(0.138)	(0.224)
Flag: Hispanic	-0.04	0.27	0.35	0.22	0.10
	(0.096)	(0.107)	(0.127)	(0.116)	(0.127)
Constant	0.97	-1.14	0.47	-1.05	-1.62
	(0.090)	(0.088)	(0.120)	(0.103)	(0.140)
Sample	HS grads	HS grads	Enroll AA or no PSE	Enroll PSE	Enroll BA
Observations	7,214	7,214	3,158	5,570	4,382

Notes: The table reports probit regression coefficients for the probit regressions of Table A8. Standard errors in parentheses. Weights are second follow up student longitudinal weights. Source: HSLS:09.

A.2.3 Target moments for internally calibrated parameters

Table A11 reports moments used as targets in the internal calibration of the model framework, specifically grants that are not Pell grants received for AA and BA enrollees. Within each education group, normaliza-

Table A10: Postsecondary application and acceptance rates by skill tercile

	AA		BA	
Skill	**Applied**	**Accepted | applied**	**Applied**	**Accepted | applied**
1	0.466	0.893	0.264	0.739
2	0.328	0.936	0.592	0.916
3	0.120	0.917	0.892	0.986
Total	0.300	0.911	0.591	0.928
Observations	5,956			

Notes: The table reports statistics related to application and acceptance in AA or BA PSE programs in the 2013-2014 academic year. Rates are reported within each skill tercile and for all high school graduates with valid application and acceptance info in the 2013 Update wave. The sample count reflects the number of high school graduate observations that satisfy this criterion. Weights: Second follow-up student longitudinal weights. Source: HSLS:09.

tions are reported as a share of average tuition and fees and as a share of GDP per person 18+. Dollar values are for the enrollees' first academic year (2013-2014).

Table A11: Grants net of Pell subsidy rate by skill

	AA enrollees		BA enrollees	
Skill tercile	**TF**	**GDP**	**TF**	**GDP**
1	0.122	0.007	0.321	0.083
2	0.275	0.016	0.339	0.087
3	0.330	0.019	0.480	0.124
Observations	844		2,652	

Notes: The table reports average grant subsidy rates for grants net of Pell grants, by skill tercile, among first-year 2013-2014 AY AA and BA enrollees. First two columns normalize by unconditional tuition and fees (TF), last two columns normalize by GDP per capita 18+ (GDP). Weights: PETS-SR longitudinal weights. Source: HSLS:09.

A.3 Other sources

A.3.1 CPS ASEC

We use the Current Population Survey's Annual Social and Economic Supplement (CPS ASEC) in 2009 and 2012 to compute within-bin median incomes that we impute to income bins in the BY and F1 waves of the HSLS:09 to assign parental income a real dollar value. Table A12 reports these dollar values for each income bin. Income bin thresholds are drawn from the HSLS:09. These moments are computed for total income at the tax unit level. For the 2008 income values, the sample is tax units with an adult between 35 and 69 and a child aged 13 or 14. For the 2011 income values, the sample is tax units with an adult between 35 and 69 and a child aged 16 or 17. The ages of the child in these sample selection criteria reflect the ages of sample members in the BY and F1 waves of the HSLS:09. Income values are in 2013 dollars.

A.3.2 NCES Tabulations

Spending on consumption amenities We use NCES tabulations of average institutional spending on various spending categories to compute the average number of dollars per student spent on consumption ameni-

Table A12: CPS ASEC median income by income bin

Income bin	**Income (2008)**	**Income (2011)**
1	8,658	8,203
2	27,060	26,310
3	48,700	46,357
4	70,344	67,320
5	90,962	85,963
6	111,303	107,694
7	132,921	128,118
8	156,976	150,541
9	177,665	169,560
10	197,163	191,629
11	218,172	210,798
12	238,983	233,506
13	431,366	341,898
Observations	5,695	4,850

Notes: The table reports the within-bin median income used to impute dollar values to the HSLS:09 family income variable. These imputation moments from the CPS ASEC are in real dollars, rounded to the nearest dollar and converted to 2013 dollars using the CPI. Source: 2009 and 2012 CPS ASEC.

ties at two-year programs and four-year programs. Following Jacob, McCall, and Stange (2018) we measure spending on consumption amenities as the sum of spending on student services and auxiliary enterprises. In NCES tabulations, within two- or four-year programs, average spending on student services and auxiliary enterprises are reported separately for institutional "classifications," i.e. public, private not-for-profit, and private for-profit. Specifically, average spending on student services and auxiliary enterprises by public and private not-for-profit institutions in academic years 2013, 2014, and 2015 are found in NCES Table 334.10 (2024) and NCES Table 334.30 (2024), respectively. Private for-profit institutions have student services and auxiliary enterprise spending reported separately for the 2014 and 2015 academic years in NCES Table 334.60 (2015b) and NCES Table 334.60 (2018), respectively. For the 2013 academic year, NCES Table 334.60 (2015a) instead reports spending on student services combined with academic and institutional support. To address the fact that student services is not reported separately for the 2013 academic year, we multiply spending on student services plus academic and institutional support by the ratio of spending on student services to the sum of student services, academic support, and institutional support in 2014 and 2015 from NCES Table 334.60 (2015b) and NCES Table 334.60 (2018) to find an imputed value for spending on student services in 2013. Dollars are converted to 2013 dollars using the CPI.

We aggregate across institutional classifications in each academic year by taking a weighted sum, where weights are shares of enrollment totals in public, private not-for-profit, and private for-profit institutions among two- and four-year programs reported in NCES Table 303.25 (2023). Table A13 reports the components of each academic year's mean as well as the total mean across academic years. For each program type, the mean across years is reported in Section 2.1 in the main text.

Spending on instruction To measure spending on instruction, for public and private not-for-profit institutional classifications we use the same sources as Table A13. However, for instructional spending there is no need to impute for the 2013 value because it is reported as a distinct category in NCES Table 334.60

Table A13: Spending on consumption amenities by program type

		Spending				Enrollment share		
Program type	**Year**	**Average**	**Public**	**Private NFP**	**Private FP**	**Public**	**Private NFP**	**Private FP**
Two-year (AA)	2013	2,092	2,029	3,325	2,619	0.950	0.010	0.050
	2014	2,068	2,058	4,509	2,821	0.950	0.000	0.040
	2015	2,200	2,183	4,947	3,146	0.950	0.000	0.040
	mean	2,120						
Four-year (BA)	2013	6,244	5,629	8,626	3,317	0.600	0.290	0.110
	2014	6,611	5,809	8,908	4,839	0.610	0.290	0.100
	2015	6,560	5,919	9,087	3,488	0.610	0.290	0.090
	mean	6,471						

Notes: For the 2013, 2014, and 2015 academic year, the table reports the weighted average of spending on consumption amenities using enrollment shares as weights. The 'mean' row is the average across academic years. The table also reports the components of the weighted average, specifically the total spending on consumption amenities and the enrollment share for postsecondary institutions who are public, private not-for-profit, and private for-profit. Spending is in 2013 dollars. Sources: NCES tabulations cited in text.

(2015a). Table A14 reports average spending on instruction for two-year and four-year programs, average spending within each institutional classification, and enrollment share weights. On average during the 2013-2015 time period, spending on instruction is more than twice as high among four-year institutions compared to two-year institutions.

Table A14: Spending on instruction by program type

		Spending				Enrollment share		
Program type	**Year**	**Average**	**Public**	**Private NFP**	**Private FP**	**Public**	**Private NFP**	**Private FP**
Two-year (AA)	2013	5,689	5,640	8,370	4,954	0.950	0.010	0.050
	2014	5,683	5,765	9,391	5,148	0.950	0.000	0.040
	2015	5,966	6,065	7,667	5,095	0.950	0.000	0.040
	mean	5,779						
Four-year (BA)	2013	11,637	10,957	16,125	3,518	0.600	0.290	0.110
	2014	12,167	11,226	16,552	5,186	0.610	0.290	0.100
	2015	12,247	11,498	16,848	3,854	0.610	0.290	0.090
	mean	12,017						

Notes: For the 2013, 2014, and 2015 academic year, the table reports the weighted average of spending on instruction using enrollment shares as weights. The 'mean' row is the average across academic years. The table also reports the components of the weighted average, specifically the total spending on instruction and the enrollment share for postsecondary institutions who are public, private not-for-profit, and private for-profit. Spending is in 2013 dollars. Sources: NCES tabulations cited in text.

B Model Appendix

B.1 Consumer value functions after highest degree attainment

This section presents the value functions after the highest level of degree attainment is permanently determined.

Consumers are required to begin student loan payments the year after they are no longer enrolled in college, regardless of whether or not they completed college. The idiosyncratic state of a consumer when $j \neq j_c + j_a$ is given by the tuple (j, e, s, η, a). The consumer's value function is given by

$$V(j, e, s, \eta, a) = \max_{d_f \in \{0,1\}} (1 - d_f) V^R(j, e, s, \eta, a) + d_f V^D(j, e, s, \eta, a), \tag{10}$$

where $d_f \in \{0, 1\}$, $V^R(\cdot)$, and $V^D(\cdot)$ denote the student loan delinquency/default decision, the value of repayment, and the value of delinquency, respectively. The value of repayment for $j \neq j_c + j_a$ is given by

$$\begin{aligned} V^R(j, e, s, \eta, a) &= \max_{c \geq 0, \ell \in \mathscr{L}, a'} U(c, \ell, j, d_e) + \beta \psi_j E_{\eta'|e,\eta} V\left(j + 1, e, s, \eta', a'\right) \\ &\text{s.t.} \\ &(1 + \tau_c) c + a' = \tilde{y}(j, e, s, \eta, a, \ell) + a + Tr_j + \mathbb{I}_{\{a<0\}} r_{SL} a \\ &a' \geq \min\left[(1 + r_{SL}) a + \rho_R(j, a), 0\right], \end{aligned} \tag{11}$$

where $d_e = N$ and $\rho_R(j, a)$ is the full payment function (see equation (19)). The constraint on a' is the loan repayment constraint, which requires that if the consumer has outstanding federal loans, then the consumer must repay at least $\rho_R(j, a)$ of the outstanding principal plus interest.

Alternatively, these consumers can choose delinquency. The value function for $j \neq j_c + j_a$ is given by

$$\begin{aligned} V^D(j, e, s, \eta, a) &= \max_{c \geq 0, \ell \in \mathscr{L}} U(c, \ell, j, d_e) - \xi_D + \beta \psi_j E_{\eta'|e,\eta} V\left(j + 1, e, s, \eta', a'\right) \\ &\text{s.t.} \\ &(1 + \tau_c) c = \tilde{y}(j, e, s, \eta, a, \ell) + Tr_j - \rho_D(j, a, y(j, e, s, \eta, a, \ell)) \\ &a' = (1 + r_{SL}) a + \rho_D(j, a, y(j, e, s, \eta, a, \ell)) - \phi_D\left[\rho_R(j, a) - \rho_D(j, a, y(j, e, s, \eta, a, \ell))\right], \end{aligned} \tag{12}$$

where $d_e = N$ and $\rho_D(j, a, y)$ is the partial payment function (see equation (20)). In the case of delinquency, consumers are not allowed to save, and additionally have their disposable income above the amount $\bar{y}$ garnished at a rate τ_g, resulting in the partial payment. Therefore, they consume whatever remains from their disposable income, plus accidental bequests, after making the partial payment. The parameter ϕ_D is the fraction of the missed payment, the difference between full payment and partial payment, that is charged as a collection fee. The outstanding principal plus interest is then augmented by the missed payment plus the collection fee net of any partial payment. During delinquency the consumer also faces a stigma cost, ξ_D.

When $j = j_c + j_a$, in addition to the choices described above, the parent chooses an inter vivos transfer to their child, who will become an independent agent in that period. At the start of age $j_c + j_a$, the parent draws their child's skill type and then chooses whether or not to be delinquent on any student debt payments; the EFC of the child, f, is determined based on parental income. The value function before the draw of child skill type is given by

$$V(j, e, s, \eta, a) = \textstyle\sum_{s_c} \pi_{s_c}(s_c|e) \left[\max_{d_f \in \{0,1\}} (1 - d_f) V^R(j, e, s, \eta, a, s_c) + d_f V^D(j, e, s, \eta, a, s_c) \right]. \tag{13}$$

The value of repayment for $j = j_c + j_a$ is given by

$$\begin{aligned} V^R(j, e, s, \eta, a, s_c) = &\max_{c \geq 0, \ell \in \mathscr{L}, a', a_c \geq 0} U(c, \ell, j, d_e) + \beta \psi_j E_{\eta'|e,\eta} V\left(j+1, e, s, \eta', a'\right) \\ &+ \beta_c E_{\eta_c|e_c} E_{\gamma_c} V^0\left(s_c, \eta_c, a_c, \mathbb{I}_{\hat{y} > y_{f=0}} G(\hat{y}), \gamma_c\right) \end{aligned} \tag{14}$$

s.t.

$$(1+\tau_c) c + a' + a_c = \tilde{y}(j, e, s, \eta, a, \ell) + a + Tr_j + r_{SL} a \mathbb{I}_{a<0}$$

$$a' \geq \min\left[(1 + r_{SL}) a + \rho_R(j, a), 0\right]$$

$$\hat{y} = y(j, e, s, \eta, a, \ell = ft),$$

where $d_e = 0$ and $e_c =$ HS. In the equation, a_c is the inter vivos transfer to the child, $V^0(\cdot)$ is the child's value function, β_c disciplines the intensity of parental altruism toward the child, and $\hat{y}$ is income at full time work hours, $\ell = ft$. When computing the EFC, parental income assuming full time work hours is used to avoid moral-hazard incentives with respect to hours worked; otherwise, the parent may have an incentive to work fewer hours to lower the EFC so that the child qualifies for more need-based aid.[32]

When $j = j_c + j_a$ and the consumer chooses delinquency, we assume for simplicity that those consumers cannot make an inter vivos transfer to their child. Therefore, the value functions for delinquency are the same as in equation (12), with the difference that the parent has a term reflecting altruistic utility toward their child in their objective function.

B.2 Additional notation and functional forms and equilibrium definition

This section presents additional notation required to define the Social Security transfer function and the equilibrium, the functional forms not provided in the main text in Section 3.1, and the equilibrium definition.

B.2.1 Additional notation

We summarize the idiosyncratic state of a consumer with $\vec{\omega}$. This state depends on age and enrollment status in the following way:

$$\vec{\omega} = \begin{cases} (s, \eta, a, f, \gamma) & \text{for 18-year-olds, before realization of } q(s) \\ (s, \eta, a, f, \gamma, Q) & \text{for 18-year-olds, after realization of } q(s) \\ (j, s, \eta, a, f, \gamma) & \text{for consumers enrolled in the AA program} \\ \left(j, s, \eta, a, f, \gamma, j^A\right) & \text{for consumers enrolled in the BA program} \\ (j, e, s, \eta, a) & \text{for consumers not enrolled in PSE, if } j \neq j_c + j_a \\ (j, e, s, \eta, a, s_c) & \text{if } j = j_c + j_a, \end{cases} \tag{15}$$

[32] In reality this moral hazard incentive is most likely not large, because the EFC is updated for every academic year of college. For the purposes of tractability, in our model the EFC is determined in the year that the child leaves the household and stays constant thereafter.

where $Q \in \{0,1\}$ is an indicator for whether an individual received the option to directly enroll in a BA program.

Equilibrium distributions are summarized by $\Omega^0(\cdot)$, $\Omega^A(\cdot)$, $\Omega^B(\cdot)$, and $\Omega(\cdot)$, which denote the distributions of 18-year-olds before the realization of the straight-to-BA enrollment option shock, AA enrollees, BA enrollees, and consumers not enrolled in an AA or BA, respectively.

The policy functions for the dropout decisions that solve the endogenous discrete dropout problems in the continuation values of equations (8) and (9) are given by $d_d^A(\vec{\omega})$ and $d_d^B(\vec{\omega})$.

Lastly, G_c, E, D, and SS denote government consumption, total education grants, federal student loan program expenditure, and Social Security expenditure. They are computed as follows:

$$
\begin{aligned}
G_c =& gY \\
E =& \int \left[\theta^{Pell}(f,A) + \theta^{other}(s,A)\right] \Omega^A(\vec{\omega})\, d\vec{\omega} + \int \left[\theta^{Pell}(f,B) + \theta^{other}(s,B)\right] \Omega^B(\vec{\omega})\, d\vec{\omega} \\
D =& \int \left[\min[a,0] - \min\left[a'(\vec{\omega}),0\right] + r_{SL}\left(\mathbb{I}_{a<0}a - a_s\left(f,j,A,a',j^A\right)\right)\right] \Omega^A(\vec{\omega})\, d\vec{\omega} + \\
& \int \left[\min[a,0] - \min\left[a'(\vec{\omega}),0\right] + r_{SL}\left(\mathbb{I}_{a<0}a - a_s\left(f,j,B,a',j^A\right)\right)\right] \Omega^B(\vec{\omega})\, d\vec{\omega} + \\
& \int \Bigg[(1 - d_f(\vec{\omega}))\left[\min[a,0](1+r_{SL}) - \min\left[a'(\vec{\omega}),0\right]\right] + \\
& d_f(\vec{\omega})\big[-\rho_D(j,a,y(j,e,s,\eta,a,\ell)) + \\
& \phi_D \max\left[\rho_R(j,a) - \rho_D(j,a,y(j,e,s,\eta,a,\ell)),0\right]\big]\Bigg] \Omega(\vec{\omega})\, d\vec{\omega} \\
SS =& \int \mathbb{I}_{j \geq j_r} ss(e,s)\, \Omega(\vec{\omega})\, d\vec{\omega}.
\end{aligned}
$$

B.2.2 Additional functional forms

The subsidized loan function is given by

$$
a_s\left(f,j,d_e,a',j^A\right) =
\begin{cases}
-\mathbb{I}_{a'<0} \min\left[-a', \bar{A}_s(j), j\max\left[\tilde{COA}(f,A),0\right]\right] & \text{if } d_e = A \\
-\mathbb{I}_{a'<0} \min\left[-a', \bar{A}_s(j), j^A\max\left[\tilde{COA}(f,A),0\right] + (j-j^A)\max\left[\tilde{COA}(f,B),0\right]\right] & \text{if } d_e = B,
\end{cases}
\tag{16}
$$

where $\tilde{COA}(f,d_e) = \kappa(d_e) + \bar{c}(d_e) - f - \theta^{Pell}(f,d_e)$ is cost of attendance net of the EFC and Pell grant. The subsidized loan function computes the amount of federal loans taken out by the student that are treated as subsidized. The function implies that, subject to the limit $\bar{A}_s(j)$, either the cumulative cost of attendance net of the EFC and Pell grant or the total amount of student debt (whichever is smaller) can be treated as a subsidized loan. For an AA enrollee (case 1), the cumulative cost of attendance net of the EFC and Pell grant is computed by multiplying $\tilde{COA}(f,d_e)$ by the college year. For a BA enrollee (case 2), the cumulative cost of attendance net of the EFC and Pell grant depends on the potential cost of having enrolled in the AA program previously and the cost of enrollment in the BA program.

The adjusted available income function is given by

$$y_{ad}(y) = y - T(y) - y_{PA}, \tag{17}$$

where y_{PA} denotes the income protection allowance.

The progressive EFC step function is given by

$$G(y) = \begin{cases} \max\left[\tau_{f,1} y_{ad}(y), 0\right] & \text{if } \; y_{ad}(y) \le Y_{f,1} \\ \tau_{f,1} Y_{f,1} + \tau_{f,2}\left[y_{ad}(y) - Y_{f,1}\right] & \text{if } \; Y_{f,1} < y_{ad}(y) \le Y_{f,2} \\ \ldots & \\ \tau_{f,1} Y_{f,1} + \ldots + \tau_{f,6}\left[y_{ad}(y) - Y_{f,5}\right] & \text{if } \; Y_{f,5} < y_{ad}(y), \end{cases} \tag{18}$$

where y is pretax parental income, the set $\{Y_{f,i}\}_{i=1}^{5}$ determines the intervals of the step function, and $\tau_{f,1} < \tau_{f,2} < \ldots < \tau_{f,6}$ are the marginal rates within the intervals.

Federal student loan repayment leads to a full payment given by the function

$$\rho_R(j, a) = \begin{cases} -\left[\dfrac{r_{SL}}{1 - (1 + r_{SL})^{-(T_{SL} + 1 - j)}} \mathbb{I}_{j \le T_{SL}} + (1 + r_{SL}) \mathbb{I}_{j > T_{SL}}\right] a & \text{if } a < 0 \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

If there is an outstanding balance and j is still within T_{SL} periods of repayment, then the loan is amortized with an interest rate of r_{SL}. If j is not within T_{SL} periods of repayment, then the outstanding principal plus interest is due. If there is no outstanding loan balance, the payment amount is zero.

Federal student loan delinquency leads to a partial payment given by the function

$$\rho_D(j, a, y) = \min\left[\tau_g \max\left[y - T(y) - \bar{y}, 0\right], \rho_R(j, a)\right] \tag{20}$$

Social Security transfers are set equal to a fraction χ of the average labor earnings for the 30 years before retirement (conditional on the history of postsecondary education and skill), plus the average unconditional labor earnings for the 30 years before retirement, divided by two. The transfer function is given by

$$\begin{aligned} ss(e, s) = & \frac{\chi}{2}\left[\frac{\int w(e)\, \eta \epsilon(j, e, s)\, \ell(\vec{\omega})\, \Omega(\vec{\omega} | j_r - 30 \le j < j_r, e, s)\, d\vec{\omega}}{\int \Omega(\vec{\omega} | j_r - 30 \le j < j_r, e, s)\, d\vec{\omega}} + \right. \\ & \left. \frac{\int w(e)\, \eta \epsilon(j, e, s)\, \ell(\vec{\omega})\, \Omega(\vec{\omega} | j_r - 30 \le j < j_r)\, d\vec{\omega}}{\int \Omega(\vec{\omega} | j_r - 30 \le j < j_r)\, d\vec{\omega}}\right]. \end{aligned} \tag{21}$$

B.2.3 Equilibrium definition

Although we compute the transition path in our analysis, thus far we have omitted time subscripts for the ease of exposition. For the definition of equilibrium, we include a time subscript, t, to indicate which variables may change along a transition path. The equilibrium for the balanced-budget case is defined as follows.

Given an initial level of capital stock K_0 and initial distributions over idiosyncratic states $\{\Omega_0^0(\vec{\omega}), \Omega_0^A(\vec{\omega}), \Omega_0^B(\vec{\omega}), \Omega_0(\vec{\omega})\}$, a competitive equilibrium consists of sequences of household value functions $\{V_t^0(\vec{\omega}), V_t^B(\vec{\omega}), V_t^A(\vec{\omega}), V_t(\vec{\omega}), V_t^R(\vec{\omega}), V_t^D(\vec{\omega})\}$, household college entrance and dropout policy functions $\{d_{e,t}(\vec{\omega}), d_{d,t}^A(\vec{\omega}), d_{d,t}^B(\vec{\omega})\}$, household consumption, hours worked, and next period asset policy functions $\{c_t(\vec{\omega}), \ell_t(\vec{\omega}), a_t'(\vec{\omega})\}$, household delinquency policy functions $\{d_{f,t}(\vec{\omega})\}$, household inter vivos transfer policy function $\{a_{c,t}(\vec{\omega})\}$, production plans $\{Y_t, K_t, L_{(t,e)}\}$, tax policies $\{\tau_{l,t}\}$, prices $\{r_t, w_t(e)\}$, Social Security transfers $\{ss_t(e,s)\}$, accidental bequests $\{Tr_{t,j}\}$, and distributions $\{\Omega_t^0(\vec{\omega}), \Omega_t^A(\vec{\omega}), \Omega_t^B(\vec{\omega}), \Omega_t(\vec{\omega})\}$ such that:

(i) Given prices, transfers, and policies, the value functions and household policy functions solve the consumer problems in equations (7)-(14);
(ii) Given prices and the production technology, the firm's factor input choices solve its profit maximization problem;
(iii) Social Security transfers satisfy equation (21);
(iv) Accidental bequests are transferred to households between and including ages beq_{min} and beq_{max} after deducting expenditure on private education subsidies

$$Tr_{t+1,j} = \frac{\int (1-\psi_j)\, a_t'(\vec{\omega})\, \Omega_t(\vec{\omega})\, d\vec{\omega} - (1-pub_{share})E_t}{\sum_{j=beq_{min}}^{beq_{max}} N_{t+1,j}}, \tag{22}$$

where $N_{t,j}$ denotes the mass of population of age j at time t and pub_{share} denotes the share of total education grant funded publicly;[33]
(v) The parameter τ_l in the income tax and transfer function $T_t(y_t(j,e,s,\eta,a,\ell))$ balances the government's budget constraint in every period by solving:

$$\begin{aligned} G_{c,t} + pub_{share}E_t + D_t + SS_t = &\int [\tau_c c_t(\vec{\omega}) + T_t(y_t(j,e,s,\eta,a,\ell))]\, \Omega_t^A(\vec{\omega})\, d\vec{\omega} + \\ &\int [\tau_c c_t(\vec{\omega}) + T_t(y_t(j,e,s,\eta,a,\ell))]\, \Omega_t^B(\vec{\omega})\, d\vec{\omega} + \\ &\int [\tau_c c_t(\vec{\omega}) + T_t(y_t(j,e,s,\eta,a,\ell))]\, \Omega_t(\vec{\omega})\, d\vec{\omega}, \end{aligned} \tag{23}$$

[33] In our baseline calibration and in all of the counterfactual exercises, accidental bequests transferred to the households are always positive because the assets of those who die exceed the expenditure on private subsidies to education costs. If they did not exceed private subsidies, then transferred bequests would be negative, which is equivalent to a lump-sum tax. Additionally, in our baseline calibration and in all of our counterfactual exercises, the total public grants are always greater than total Pell grants.

(vi) Labor, capital, and goods markets clear in every period t; and

(vii) $\Omega_{t+1} = \Pi_t(\Omega_t)$, where Π_t is the law of motion that is consistent with consumer household policy functions and the exogenous processes for population, labor productivities, skill, and the probabilities of being allowed to continue college.

Note that in the stationary equilibrium, the equilibrium distribution will be stationary, all aggregates, prices, taxes, and transfers will be constant, and all value functions and policy functions will be time invariant. Furthermore, there is no government debt.

In the deficit-financed case, the government issues debt B_{t+1} during the periods of deficit-financing while fixing the level parameter τ_l at its initial steady state level so that equation (23) is modified to be:

$$\begin{aligned} G_{c,t} + pub_{share}E_t + D_t + SS_t + (1+r_t)B_t = &\int \left[\tau_c c_t(\vec{\omega}) + T_0\left(y_t(j,e,s,\eta,a,\ell)\right)\right]\Omega_t^A(\vec{\omega})\,d\vec{\omega} + \\ &\int \left[\tau_c c_t(\vec{\omega}) + T_0\left(y_t(j,e,s,\eta,a,\ell)\right)\right]\Omega_t^B(\vec{\omega})\,d\vec{\omega} + \\ &\int \left[\tau_c c_t(\vec{\omega}) + T_0\left(y_t(j,e,s,\eta,a,\ell)\right)\right]\Omega_t(\vec{\omega})\,d\vec{\omega} + \\ &B_{t+1}, \end{aligned} \tag{24}$$

where $B_0 = 0$ and $T_0(y_t)$ captures the feature that the level parameter τ_l is fixed at its initial steady state level. Subsequently, during the periods in which the government pays down its debt, $B_{t+1} = \rho_B B_t$, where ρ_B is the constant decay rate, and $T_0(y_t)$ is replaced with $T_t(y_t)$ in equation (24) to balance the budget in every period.

B.3 Computational algorithm

This section presents the computational algorithm to solve for the stationary equilibrium. The algorithm for the transition path is analogous except that the value functions, policy functions, prices, taxes, transfers, and equilibrium distributions are indexed by a time subscript.

1. Guess interest rate, r, wage rates, $w(e)$, the level parameter for the income tax rate, τ_l, Social Security transfers, $ss(e,s)$, and accidental bequests, Tr_j
2. Use backward induction to solve the consumer problems when $j = j_c + j_a + 1, \ldots, J$ (equations (10)-(12))
3. Guess value function before college, $V^0(s,\eta,a,f,\gamma)$ (equation (7))
4. Use backward induction to solve consumer problem when $j = 1, \ldots, j_c + j_a$ (equations (7)-(14)). In solving the consumer problem at $j = j_c + j_a$, use $V^0(s,\eta,a,f,\gamma)$ for the altruism term.
5. Use new value before the realization of the straight-to-BA enrollment option shock to update $V^0(s,\eta,a,f,\gamma)$; repeat 4.-5. until convergence
6. Guess initial distribution of 18-year-old consumers Ω^0
7. Solve for distributions of $\{\Omega^A, \Omega^B, \Omega\}$

8. Use distribution of Ω for $j = j_c + j_a$, exogenous processes for child skill and productivity, and the policy function for the inter vivos transfers to compute new estimates for distribution of initial 18-year-old consumers
9. Update Ω^0 and repeat 7.-9. until convergence
10. Given the stationary distributions of $\{\Omega^A, \Omega^B, \Omega\}$, solve for new guesses:
    - Compute interest and wage rates from the firm's first order conditions
    - Compute the level parameter for the income tax rate using the government budget constraint (equation (23))
    - Compute Social Security transfers and accidental bequests (equations (21) and (22))
11. Update guesses in 1., and repeat steps 2.-11. until convergence

B.4 Welfare computation

For consumers who are aged 18 and over when the transition begins, the one-time transfer is subtracted from earnings in the age-one budget constraint so that a positive transfer amount indicates welfare gains and a negative transfer amount indicates welfare losses. The specific budget constraint and state space that the transfer interacts with differs depending on the age and other attributes of the consumer. For example, for the consumer whose education is already permanently determined and who is not at the inter vivos transfer age $j_c + j_a$, the one-time compensation transfer solves

$$V_1\left(j, e, s, \eta, a; \Delta_y\right) = V_0\left(j, e, s, \eta, a\right), \tag{25}$$

where $V_1(\cdot)$ is the value the consumer would receive in $t = 1$ with the transition and a transfer amount of Δ_y and $V_0(\cdot)$ is the value to the consumer in the initial steady state. If the consumer chooses repayment, the one-time transfer enters the period-one budget constraint as follows

$$\left(1 + \tau_c\right) c + a' = \tilde{y}\left(j, e, s, \eta, a, \ell\right) + a + Tr_j + \mathbb{I}_{\{a<0\}} r_{SL} a - \Delta_y, \tag{26}$$

whereas if the consumer chooses delinquency, the analogous expression is instead

$$\left(1 + \tau_c\right) c = \tilde{y}\left(j, e, s, \eta, a, \ell\right) + Tr_j - \rho_D\left(j, a, y\left(j, e, s, \eta, a, \ell\right)\right) - \Delta_y. \tag{27}$$

Besides the example group discussed above, consumers who are adults in the period in which the transition begins includes 18-year-olds, AA enrollees aged $j = 2$, BA enrollees aged $j = \{2, 3, 4\}$, and parents aged $j = j_c + j_a$. The calculation of the one-time compensation transfer for these consumers is analogous to the example discussed here.

For consumers who turn 18 in period $t = 2, 3, ...$, the one-time compensation transfer is computed behind the veil of ignorance as follows

$$\int V_{\mathrm{t}}^{0}\left(s,\eta,a,f,\gamma;\Delta_{y}\right)\Omega_{\mathrm{t}}^{0}\left(\vec{\omega}\right)d\vec{\omega}=\int V_{0}^{0}\left(s,\eta,a,f,\gamma\right)\Omega_{0}^{0}\left(\vec{\omega}\right)d\vec{\omega}. \tag{28}$$

To make these transfers comparable with the transfers computed for the generations of adults alive in $t = 1$, the transfers estimated in a future period $t > 1$ are discounted to the present value using $\prod_{i=2}^{t}[1/(1 + r_i)]$.

C Parameterization Appendix

C.1 Remaining externally estimated parameters

Table C1 reports the remaining externally estimated parameters that are not reported in Table 8. These parameters are organized into Panels A through E based on type. In several rows in Table C1, an external parameter value is normalized by GDP per capita for those 18 and over and the parameter's row is flagged accordingly. GDP per capita for those 18 and over is computed the same way as for Tables 8 and 9, as described at the start of Section 4.

Panel A reports parameters related to demographics and hours, which we set by assumption unless stated otherwise. The period in which the child is born, j_c, is set to 13 so that consumers have a child when they turn 30; the age adulthood begins, j_a, is set to 18; j_r is chosen so that the retirement age is 65; and, finally, J sets maximum life span to 100 years. For $j < j_c + j_a$, we set survival probabilities ψ_j to one to rule out children without parents; ages $j \geq j_c + j_a$ use estimates from the 2010 Social Security Administration Life Tables presented in Bell and Miller (2020). The probabilities over child skill conditional on the parent's postsecondary education history, $\pi_{s_c}(s_c|e)$, are based on NLSY97 estimates and are drawn from Table A5 of Appendix A.1.3. These probabilities are estimated conditional on whether the parent's highest degree attainment is HS and they do not have two or more years of BA enrollment, $e =$ HS, whether the parent has either an AA degree or two or more years of BA enrollment but no BA degree, $e \in \{$AA,BA d/o$\}$, and whether the parent is a BA graduate, $e \in \{$AA-to-BA,Straight-to-BA$\}$. The minimum and maximum model ages at which accidental bequests are transferred to consumers are set to 33 and 43, respectively. This choice is based on Conesa et al. (2018) and implies that accidental bequests are received by consumers later in their working life who are between the ages of 50 and 60. We set full time hours, ft, to 1/3, which is equivalent to 8 hours of work per day, five days a week. The set of hours a consumer could work, $\mathscr{L}$, consists of full time hours scaled by 0, 0.75, 1.00, 1.25, and 1.50.

Panel B reports parameters that govern preferences and the goods production technology. The parameter that governs the relative risk aversion, σ, is set to $1/\upsilon + 1$ so that relative risk aversion is equal to 2 based on Chetty (2006). The adult equivalence scale, ζ_j, is set so that the child increment is 0.3 following the Organization for Economic Co-operation and Development (OECD) modified scale. The capital share parameter, α, is set to 0.36 following Kydland and Prescott (1982). The depreciation rate of capital, δ, is set to 0.076, as in Krueger and Ludwig (2016). The parameter that dictates the elasticity of substitution between labor without and with a BA, ι, is set to 0.75, which implies an elasticity of substitution of 4, based on Bils,

Table C1: Externally estimated parameters

Parameter	Description	Data target	Value
Panel A: Demographics and hours			
j_c	Child bearing age	30 years	13
j_a	Years for child to move out	18 years	18
j_r	Retirement age	65 years	48
J	Maximum life span	100 years	83
ψ_j	Survival probability	2010 SSA Life Tables	-
beq_{min}	Minimum age to receive accidental bequest	Conesa et al. (2018)	33
beq_{max}	Maximum age to receive accidental bequest		43
$\pi_{s_c}(s_c\|e = \text{HS})$	Probability over child skill	NLSY97	$(0.476, 0.328, 0.196)$
$\pi_{s_c}(s_c\|e \in \{\text{AA,BA d/o}\})$			$(0.287, 0.380, 0.333)$
$\pi_{s_c}(s_c\|e \in \{\text{AA-to-BA,Straight-to-BA}\})$			$(0.137, 0.312, 0.551)$
ft	Full time hours	8 hours per day	$1/3$
$\mathscr{L}$	Set of feasible work hours	Scale full time hours	$(0, 0.75, 1, 1.25, 1.5)ft$
Panel B: Preferences and production technology			
σ	Risk aversion	Risk aversion = 2, Chetty (2006)	$\frac{1}{v} + 1$
ζ_j	Adult equivalence scale	OECD modified scale	$1 + 0.3\mathbb{I}_{j_c \leq j < j_c + j_a}$
α	Capital share	Kydland and Prescott (1982)	0.360
δ	Depreciation rate	Krueger and Ludwig (2016)	0.076
ι	Elasticity of substitution	Bils, Kaymak, and Wu (2024)	0.750
Panel C: Grants and federal student loans			
θ^{Pell}_{max}	Pell maximum, normalized	Statutory	0.081
pub_{share}	Public share education grants	OECD	0.697
$\bar{A}(j)$	Subsidized and unsubsidized loan limit, normalized		(0.077,0.167,0.271,0.376)
$\bar{A}_s(j)$	Subsidized loan limit, normalized		(0.049,0.111,0.188,0.264)
τ_{SL}	Interest rate add-on		0.021
T_{SL}	Maximum years to repay		14
τ_g	Student loan garnishment rate		0.150
$\bar{y}$	Garnishment-exempt income, normalized		0.151
ϕ_D	Student loan collection fee	Luo and Mongey (2024)	0.185
Panel D: EFC			
$y_{f=0}$	Automatic zero EFC income threshold, normalized	Statutory	0.345
$Y_{f,i}$	EFC formula intervals, normalized		(0.220,0.276,0.332,0.388,0.444)
$\tau_{f,i}$	EFC formula marginal rates		(0.220,0.250,0.290,0.340,0.400,0.470)
y_{PA}	Income protection allowance, normalized		0.306
Panel E: Government spending and tax policy			
g	Government consumption	BEA	0.146
τ_p	Income tax progressivity	CBO	0.177
τ_c	Consumption tax rate	OECD	0.044

Notes: The table presents the externally estimated parameters that are not presented in Table 8 of Section 4.1.

Kaymak, and Wu (2024).

Panel C reports parameters related to grants and federal student loans. Statutory values are used for all parameters except for the public share of total education grants and the student loan collection fee. The maximum Pell amount, θ^{Pell}_{max}, is set to 0.081 based on Office of Postsecondary Education, Federal Student Aid (2013). The public share of total education grants, pub_{share}, is set to 0.697 based on the OECD's Education at a Glance report (OECD, 2012). The college-year specific loan limits for any federal student loans, $\bar{A}(j)$, and for subsidized student loans, $\bar{A}_s(j)$, are set using the yearly limits reported by the Congressional Research Service (CRS) in Smole (2019). The values imply that individuals can borrow more in the third and fourth years in comparison to the first two years resulting in a lower average annual loan limit for AA programs compared to BA programs. The interest rate add-on, τ_{SL}, is set to 0.021 as reported by the Chief Operating Officer for Federal Student Aid (FSA) in Chief Operating Officer for FSA (2021). The repayment period for student loans, T_{SL}, is set to 14. This value implies that for a BA enrollee who graduates the maximum repayment period is 10 years, which is equal to the period for the Standard Repayment Plan (10 years) outlined in Smole (2019). The garnishment rate in the event of student loan delinquency, τ_g, is set following the rate established by the 2005 Deficit Reduction Act (109th Congress of the United States of America, 2006). The income exempt from garnishment in delinquency, $\bar{y}$, is set to 0.151 based on our calculations using results from Yannelis (2020). The last parameter in this panel, the student loan collection fee, ϕ_D, is set to 0.185 following Luo and Mongey (2024).

Panel D reports parameters related to the EFC, which are based on statutory values. These parameters are drawn from Tables A3 and A6 of the EFC formula guide for 2013-2014 prepared by the Federal Student Aid Office, U.S. Department of Education (2014). The income threshold below which applicants are assigned an automatic zero EFC, $y_{f=0}$, is set to 0.345. The parameters that determine the intervals and the marginal rates of the EFC step function, $\{Y_{f,i}\}_{i=1}^{5}$ and $\{\tau_{f,i}\}_{i=1}^{6}$, are reported in the next two rows of the panel. For the income protection allowance reported in the last row, y_{PA}, we use the allowance for a household with three members including the student and one student in college to set the parameter to 0.306.

Panel E reports parameters related to government spending and tax policy. Government consumption as a share of GDP, g, is set to 0.146 using estimates of the numerator and denominator from BEA Table 1.1.5 and BEA Table 3.1. The income tax progressivity, τ_p, and consumption tax rate, τ_c, are set to 0.177 and 0.044, respectively. These values are estimated in Moschini and Raveendranathan (2026) using data from the Congressional Budget Office (CBO) and OECD.

C.2 Skill-specific wage premiums in data and model

Table C2 reports the AA and BA wage premium by skill tercile. The empirical estimates are drawn from Table 4. In both the model and the data, the AA and BA wage premium is computed as the median hourly earnings of individuals with an AA or BA degree divided by the median hourly earnings of those with only a high school degree, for workers aged 25 to 39, within a given skill tercile. The age range is chosen to match the available NLSY97 sample used for the empirical estimates. The model captures the empirical pattern that the AA wage premium is lower than the BA wage premium within each skill tercile.

Table C2: AA and BA wage premium by skill tercile in data and model

	Data		**Model**	
Skill	**AA**	**BA**	**AA**	**BA**
Low	1.167	1.365	1.163	1.312
Medium	*1.184*	*1.374*	*1.183*	*1.374*
High	1.204	1.537	1.183	1.479

Notes: The table reports the AA and BA wage premium by skill tercile. The middle row estimates are reported in italics because they are targeted in the calibration. Data source: NLSY97.

Table C3 reports the BA wage premium for those who transfer from the AA to the BA program and for those who enroll directly in the BA program within each skill tercile. The empirical estimates are drawn from Table 5. In both the data and the model, the BA wage premium for each path to BA degree attainment is computed as the median hourly earnings of individuals with a BA degree given their path divided by the median hourly earnings of those with only a high school degree, for workers aged 25 to 39, within a given skill tercile. Again, the age range is chosen to match the available NLSY97 sample used for the empirical estimates. The model accounts for the lower BA wage premium for the many BA graduates taking the AA-to-BA route in comparison to straight-to-BA enrollees.

Table C3: BA wage premium for AA to BA and straight to BA in data and model

	Data		Model	
Skill	**AA to BA**	**Straight to BA**	**AA to BA**	**Straight to BA**
Low	1.324	1.390	1.282	1.360
Medium	1.373	1.374	1.342	1.424
High	1.426	1.547	1.429	1.517

Notes: The table reports the BA wage premium by skill tercile and path to BA degree attainment. Data source: NLSY97.

C.3 Highest postsecondary degree attainment rates in data and model

Table C4 reports the highest postsecondary degree attainment rates by age 30 broken down by the five types of completed education levels considered in our model. The types of attainment are mutually exclusive and the percentages sum to 100. The empirical estimates are drawn from Table A6 of Appendix A.1.4. The model accounts for the attainment shares observed in the data.

Table C4: Highest PSE degree attainment rates among high school graduates by age 30

Category	**Type**	**Data**	**Model**
No BA degree	HS	39.7	39.7
	BA d/o	10.0	10.8
	AA degree	6.6	6.1
BA degree	AA-to-BA	6.2	6.2
	Straight-to-BA	37.5	37.3

Notes: The table reports highest postsecondary degree attainment rates by age 30 among a cohort of high school graduates. The percentages sum to 100. Data source: NLSY97.

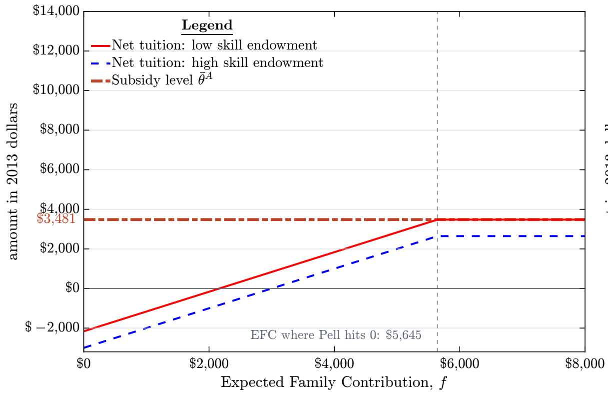
D Results Appendix

D.1 Additional results from the main experiments

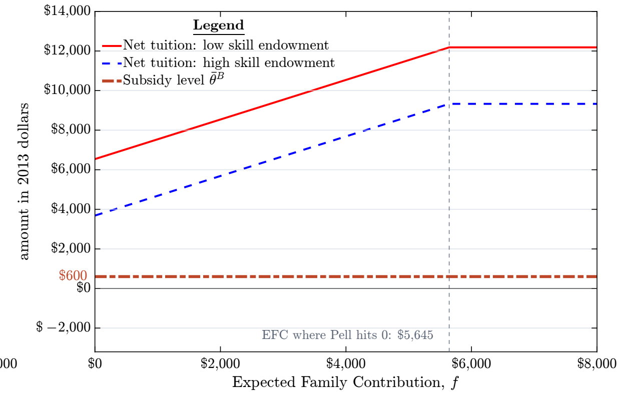
This section presents additional results from the main experiments in Section 6.

D.1.1 Visual illustration of AA and BA net tuition and subsidy expansions

Figure D1 provides a visual illustration of net tuition from equation (1) given our parameter values as well as the level of AA and BA tuition subsidies from our main experiments in Section 6.



(a) AA net tuition and subsidy expansion



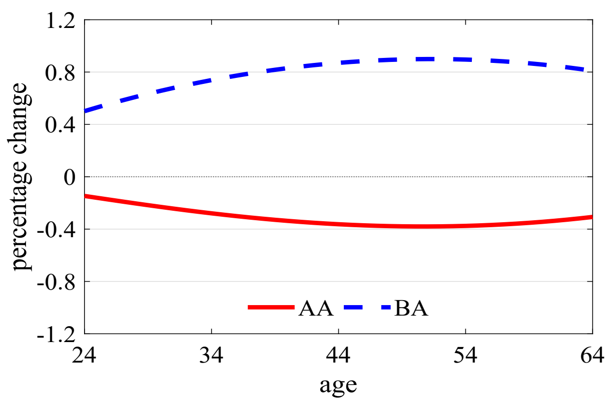
(b) BA net tuition and subsidy expansion

Figure D1: Net tuition and AA and BA subsidy

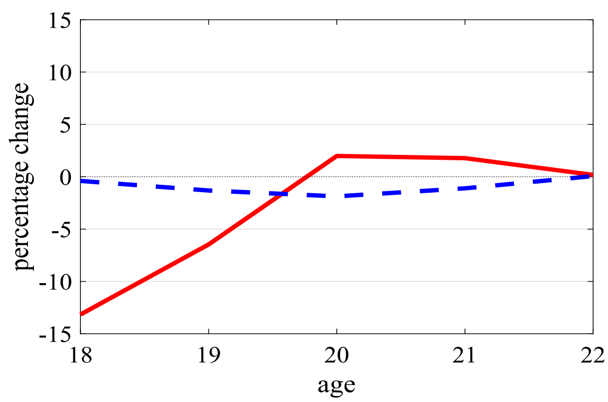
Notes: Subfigure D1a depicts net tuition by Expected Family Contribution for AA enrollees with low and high skill and the AA subsidy for the experiment in which the AA program is made tuition-free. Subfigure D1b depicts net tuition by Expected Family Contribution for BA enrollees with low and high skill and the BA subsidy for the experiment in which a spending-equivalent subsidy is provided for the BA program. All amounts are in 2013 dollars. Net tuition for those with medium skill (not depicted in these subfigures) lies between the low- and high-skill lines.

D.1.2 Changes in hours worked and productivity by age across steady states

Following the AA and BA subsidy expansions, Figure D2 depicts changes across steady states by age in labor productivity for working-age consumers after the college phase and in labor hours worked during the college phase. Subfigure D2a indicates that the AA subsidy expansion lowers the labor productivity of the working-age population, whereas the BA subsidy expansion raises the labor productivity. Subfigure D2b indicates that the AA subsidy expansion results in a significant decline in hours worked for ages 18 and 19 in comparison to the BA subsidy expansion.



(a) Productivity by age: after college phase



(b) Hours worked by age: college phase

Figure D2: Steady state changes after subsidy expansions in general equilibrium

Notes: The figure depicts the percentage change across steady states by age in labor productivity for working-age consumers after the college phase and in labor hours worked during the college phase in Subfigures D2a and D2b, respectively. Labor productivity by age is computed as the weighted sum of individual productivity, $\epsilon(j, e, s)\,\eta$, divided by the mass of individuals at a given age.

D.1.3 The impact of the diversion effect on labor productivity in partial equilibrium

Recall that in the AA subsidy expansion, the diversion effect on enrollment is one in which those who would have enrolled in the BA program now enroll in the AA program instead. To cleanly isolate the impact of the diversion effect on labor productivity, we analyze the AA subsidy expansion in a partial equilibrium in which the distribution of 18-year-olds and general equilibrium prices, the tax rate, and transfers are held fixed at their initial steady state values.

The results are presented in Figure D3. In the line labeled "without diversion effect," we control for the diversion effect in the computation of the equilibrium by assuming that those who would optimally choose to enroll in the AA program post-policy but would have enrolled in the BA program pre-policy, still (sub-optimally) enroll in the BA program. The line labeled "with diversion effect" serves as the benchmark as it does not do this adjustment to the policy functions. The figure shows that labor productivity increases without the diversion effect but decreases once the diversion effect is incorporated.

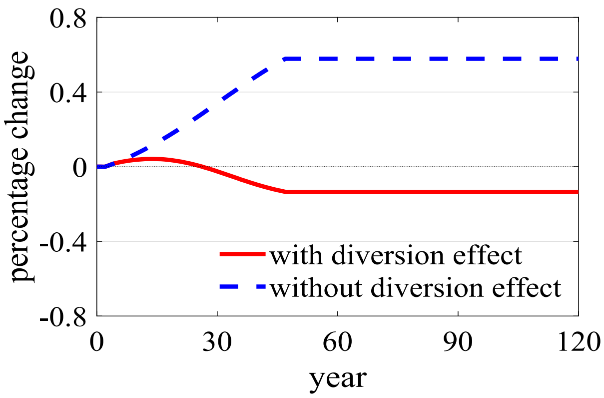


Figure D3: AA subsidy expansion: labor productivity in partial equilibrium

Notes: The figure depicts the percentage change in labor productivity from the AA subsidy expansion in partial equilibrium with and without the diversion effect. The partial equilibrium is one in which the distribution of 18-year-olds and general equilibrium prices, the tax rate, and transfers are held fixed at their initial steady state values. Labor productivity is computed for the working-age population as the weighted sum of individual productivities, $\epsilon(j, e, s)\,\eta$, divided by the mass of individuals.

D.1.4 The impact of the AA-to-BA transfer option on labor productivity in partial equilibrium

One of the features that distinguishes our analyses from existing structural studies with heterogeneous college quality is the AA-to-BA transfer option. We use the following exercise to quantify the impact of the transfer option on average labor productivity in the baseline equilibrium: in a partial equilibrium in which the initial distribution of 18-year-olds, prices, the tax rate, and transfers (accidental bequests and Social Security) are all held fixed at their initial steady-state values, we shut off the AA-to-BA transfer outcome when computing equilibrium. This is done by assuming that those who would choose to transfer to the BA program at the end of the first year continue the AA program and those who would choose to transfer to the BA program at the end of the second year enter the labor market directly as AA graduates. The result is reported in the first row of Table D1: labor productivity decreases by 2.39 percent. This result indicates that the AA-to-BA transfer option acts to raise labor productivity in equilibrium in a quantitatively meaningful way.

Table D1: Steady state change in labor productivity in partial equilibrium

Policy change	AA-to-BA transfer option in: Initial steady state	Final steady state	Labor productivity
None	Yes	No	−2.39
AA subsidy expansion	Yes	Yes	−0.13
AA subsidy expansion	No	No	−0.50

Notes: The table reports the change in average labor productivity under three cases described in the main text. The analysis is performed in a partial equilibrium in which the distribution of 18-year-olds and general equilibrium prices, the tax rate, and transfers are held fixed at the values of the initial steady state of the baseline equilibrium.

In the next two rows of Table D1, we consider how the AA-to-BA transfer option affects the results of the AA subsidy expansion. In a partial equilibrium in which the initial distribution of 18-year-olds and prices, the tax rate, and transfers are fixed at the initial steady state values of the baseline equilibrium, we find that with the AA-to-BA transfer option, the AA subsidy expansion decreases the average labor productivity by 0.13 percent; however, without the transfer option, the AA subsidy expansion decreases the average labor productivity by 0.50 percent. Therefore, the AA-to-BA transfer option in our model mitigates the extent to which labor productivity falls. This is because although some individuals who were straight-to-BA enrollees in the initial steady state choose to enroll in the AA program after the policy change, some of them will still transfer from the AA to the BA program after a year or two. This result indicates the importance of taking the AA-to-BA transfer option into account in our analysis.

D.1.5 Changes in other aggregates across steady states

Table D2 reports changes in aggregates across steady states in general equilibrium that were not reported in the main text in Table 15. The aggregates are organized into Panels A through C based on statistic type.

Panel A reports changes in labor supplies by type. These changes provide context for the changes in wage rates reported in Panel C of Table 15.

Panel B reports changes in Social Security transfers. In both subsidy expansions, the signs of the change in Social Security transfers primarily reflect the signs of the changes in the wage rates for each group shown in Panel C of Table 15.[34]

Panel C reports changes in the averages and shares of initial characteristics of high school graduates whose distribution can change across equilibria, specifically net assets, EFC, and skill.[35] The qualitative changes in net assets are the same in both experiments: initial net assets fall. In both subsidy expansions, the tuition subsidy crowds out inter vivos transfers from parents. Additionally, in the AA subsidy expansion, the inter vivos transfers fall more as the economy becomes poorer; the BA subsidy expansion leads to an opposing

[34] The only exception is the HS group under the AA subsidy expansion. While this group experiences a rise in the wage rate, high school graduates with higher AR(1) productivity transition from non-enrollment to AA enrollment, contributing to a decline in earnings, and hence, the group's Social Security transfer.

[35] Changes in AR(1) productivity and the weight for the consumption value of college are not reported because their distributions remain the same across equilibria.

effect on transfers as the economy becomes richer. Similarly, the AA subsidy expansion leads to a fall in the average EFC as the economy becomes poorer while the BA subsidy expansion leads to the opposite effect. The AA subsidy expansion leads to more parents with an AA degree resulting in a slight increase in the share of children with medium skill. In contrast, the BA subsidy leads to more parents with a BA degree, resulting in more children with high skill. This effect on the skill distribution also captures a better quality outcome of the BA subsidy expansion.

Table D2: Steady state changes in other aggregates after subsidy expansions in general equilibrium

		Subsidy expansion	
Panel	**Description**	**AA**	**BA**
A: Labor input (efficiency units)	HS	-2.22	-3.29
Units: percentage change	AA	51.49	-12.10
	BA d/o	-12.42	5.26
	BA	-2.59	3.57
B: Social Security by skill	HS	(-0.07,-0.08,-0.06)	(0.88, 0.87, 0.91)
	AA	(-1.78,-1.76,-1.77)	(0.73, 0.75, 0.79)
	BA d/o	(-1.65,-1.70,-1.69)	(0.76, 0.75, 0.71)
	BA | AA to BA	(0.05, 0.09, 0.11)	(-0.01,-0.14,-0.14)
	BA | Straight to BA	(0.08, 0.11, 0.13)	(-0.06,-0.11,-0.13)
C: Average of initial characteristics	Net assets	-4.51	-0.82
Units: percentage or percentage point change	EFC	-0.83	1.32
	Low skill share	0	-0.49
	Medium skill share	0.12	-0.03
	High skill share	-0.12	0.53

Notes: The table reports general-equilibrium changes from the initial baseline equilibrium to the final steady state in the AA and BA subsidy expansions. Panels A, B, and C report changes in labor supplies by type, Social Security transfers, and average of initial characteristics, respectively

D.1.6 Balanced-budget general equilibrium decomposition for welfare changes

In this section, we perform analyses to isolate the effect of changes in the tax rate, prices, and transfers from the two subsidy expansions on welfare under the balanced-budget transition. Table D3 reports the total welfare changes in one-time transfers as a percentage of initial GDP for the AA and BA subsidy expansions in Panels A and B, respectively. Within each panel, the first eight rows report welfare changes for mutually exclusive consumer groups and the last row reports welfare changes for all consumer groups.[36] Columns "(1)" and "(2)" report the welfare changes in partial equilibrium and in general equilibrium for comparison purposes; in columns "(3)" through "(7)," each identified general equilibrium object is held fixed at its initial steady state value while the remaining general equilibrium objects change as they do in the general equilibrium case of column (2). For example, in the column labeled "(3)," the income tax rate is held fixed at its initial steady state level while the prices (risk-free rate and wage rates) and transfers (accidental bequests and Social Security) adjust they do in the general equilibrium case.

Panel A indicates that, in the AA subsidy expansion, the key change that flips the gains observed in partial

[36] Unlike in the main text in which averages were reported for mutually exclusive consumer groups, in Table D3 we report the total welfare change within each group because our primary focus in this section is the welfare change for the whole economy.

equilibrium to losses in general equilibrium is the increase in the income tax rate. As column (3) indicates, when the income tax rate does not change but all other general equilibrium objects change, we observe gains; however, as columns (4) through (7) indicate, when each of the other general equilibrium objects is held fixed, we still observe losses. Columns (2), (4), and (5) show that both the fall in the risk-free rate and accidental bequests in the AA subsidy expansion act to increase total welfare losses in general equilibrium: when these objects are held fixed, in comparison to the general equilibrium case, the total welfare losses are reduced.

Table D3: Total welfare changes in one-time transfers: by mutually exclusive and all consumer groups

			Equilibrium concept						
Panel	**Consumer group**	**Period**	**(1) PE**	**(2) GE**	**(3) fix τ_l**	**(4) fix r**	**(5) fix Tr_j**	**(6) fix $w(e)$**	**(7) fix $SS(e,s)$**
A: AA subsidy	HS grads	$t=1$	0.02	0.01	0.01	0.01	0.01	0.01	0.01
	AA enrollees	$t=1$	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	BA enrollees	$t=1$	0	-0.01	0	-0.01	-0.01	-0.01	-0.01
	Pre-IVT age	$t=1$	0.55	-0.34	0.14	-0.14	-0.15	-0.42	-0.26
	At IVT age	$t=1$	0.03	-0.01	0.01	0	0	-0.01	-0.01
	Post-IVT	$t=1$	0	-0.35	-0.19	-0.24	-0.26	-0.38	-0.34
	Retirees	$t=1$	0	-0.10	-0.04	-0.06	-0.10	-0.10	-0.11
	Future generations	$t>1$	0.42	0.09	0.22	0.10	0.13	0.24	0.11
	All	$t\geq 1$	1.03	-0.71	0.16	-0.33	-0.37	-0.66	-0.60
B: BA subsidy	HS grads	$t=1$	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	AA enrollees	$t=1$	0	0	0	0	0	0	0
	BA enrollees	$t=1$	0.04	0.03	0.03	0.03	0.03	0.04	0.03
	Pre-IVT age	$t=1$	0.49	0.49	0.68	0.15	0.67	0.56	0.41
	At IVT age	$t=1$	0.03	0.02	0.03	0.01	0.03	0.03	0.02
	Post-IVT	$t=1$	0	-0.09	0.02	-0.21	-0.01	-0.09	-0.07
	Retirees	$t=1$	0	-0.08	-0.01	-0.10	-0.08	-0.08	-0.04
	Future generations	$t>1$	0.80	0.74	0.73	0.71	0.74	0.87	0.73
	All	$t\geq 1$	1.39	1.15	1.51	0.61	1.42	1.36	1.11

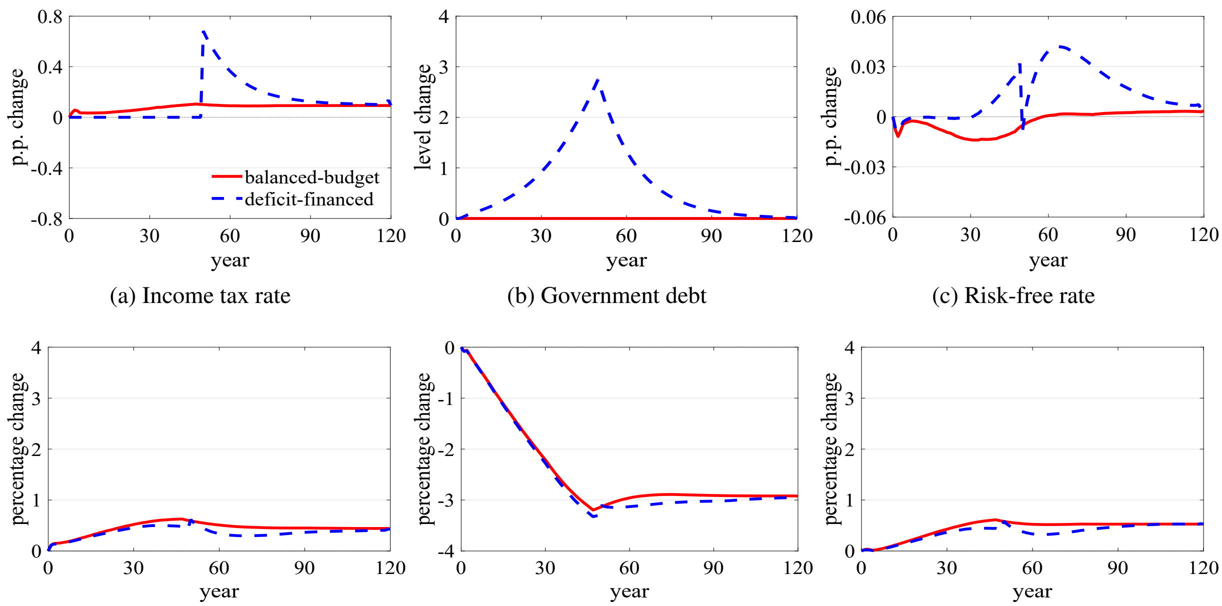
Notes: The table reports total welfare changes (total one-time transfers as a percentage of initial GDP) from the AA and BA subsidy expansions in each panel. The seven columns are as follows: (1) "PE" is a partial equilibrium case in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) "GE" is the balanced-budget general equilibrium case; (3) "fix τ_l" is a partial equilibrium case in which the income tax rate level parameter τ_l is held fixed at its initial steady state value while the remaining GE objects adjust as they do in GE; (4) "fix r" is analogous to (3) except that the interest rate is held fixed instead; (5) "fix Tr_j" is analogous to (3) except that accidental bequests are held fixed instead; (6) "fix $w(e)$" is analogous to (3) except that the wage rates are held fixed instead; (7) "fix $SS(e,s)$" is analogous to (3) except that the Social Security transfers are held fixed instead. "IVT" refers to inter vivos transfer.

Panel B indicates that, in the BA subsidy expansion, the increase in the risk-free rate in the early periods of the transition acts to increase total welfare gains in general equilibrium: column (4) shows that when the risk-free rate does not change but all other general equilibrium objects change, we observe gains that amount to a one-time transfer worth 0.61 percent of initial GDP in comparison to the general equilibrium gains of 1.15 percent in column (2). Columns (3) and (5) indicate that changes in the income tax rate and accidental bequests act to dampen total welfare gains in general equilibrium. Additionally, a comparison of Panel A with Panel B shows that the extent to which these changes dampen total welfare is smaller in the BA subsidy expansion. To illustrate, the change in the income tax rate in AA subsidy expansion contributes to total welfare losses of 0.87 percent of initial GDP (0.16+0.71=0.87); the change in the income tax rate in the BA subsidy expansion contributes to total welfare losses of 0.36 percent (1.51-1.15=0.36). Similarly, the change in accidental bequests in the AA subsidy expansion contributes to total welfare losses of 0.34 percent of initial GDP (-0.37+0.71=0.34); the change in accidental bequests in the BA subsidy expansion contributes to total welfare losses of 0.27 percent (1.42-1.15=0.27).

For both subsidy expansions, a comparison of the last two columns with columns (3) through (5) indicates that the net effect of changes in the wage rates and Social Security transfers on welfare changes are smaller compared to the net effects of the income tax rate, risk-free rate, and accidental bequests.

D.1.7 A comparison of the balanced-budget and deficit-financed transition dynamics

This section compares the transition dynamics for the income tax rate, government debt, and prices between the balanced-budget and deficit-financed cases from both the AA subsidy expansion and the BA subsidy expansion.



(a) Income tax rate (b) Government debt (c) Risk-free rate

(d) Wage rate: HS (e) Wage rate: AA or BA d/o (f) Wage rate: BA

Figure D4: AA subsidy expansion: Changes in tax rate, government debt, and prices by transition type

Notes: The figure depicts the changes in the income tax rate (at initial steady state mean income), government debt, risk-free rate, and wage rates from the AA subsidy expansion under two transition types: (1) a "balanced-budget" transition in which the income tax rate level parameter is adjusted to balance the government budget in every period and (2) a "deficit-financed" transition in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate.

Figure D4 depicts the changes in the income tax rate (at initial steady state mean income), government debt, risk-free rate, and wage rates from the AA subsidy expansion under the two transition types. In the balanced-budget transition, the income tax rate increases in every period (Subfigure D4a). In the deficit-financed transition, the government increases its debt to finance the higher expenditure in the first 50 years after the policy is implemented (Subfigure D4b). Because higher government debt lowers productive capital in the economy, the risk-free rate is higher (Subfigure D4c) and the wage rates are lower (Subfigures D4d, D4e, and D4f) under the deficit-financed transition in comparison to the budget-balanced transition. These results imply that, in the deficit-financed transition, while shifting the tax burden to the future is better for the generation of consumers who are alive in the first period of the transition because the tax rate does not have to increase as much, there is a tradeoff for this generation in that the wage rates are lower.[37]

Figure D5 depicts the changes in the income tax rate (at initial steady state mean income), government debt, risk-free rate, and wage rates from the BA subsidy expansion under the two transition types. The takeaways from the AA subsidy expansion in Figure D4 also hold when comparing the two types of transitions under the BA subsidy expansion.

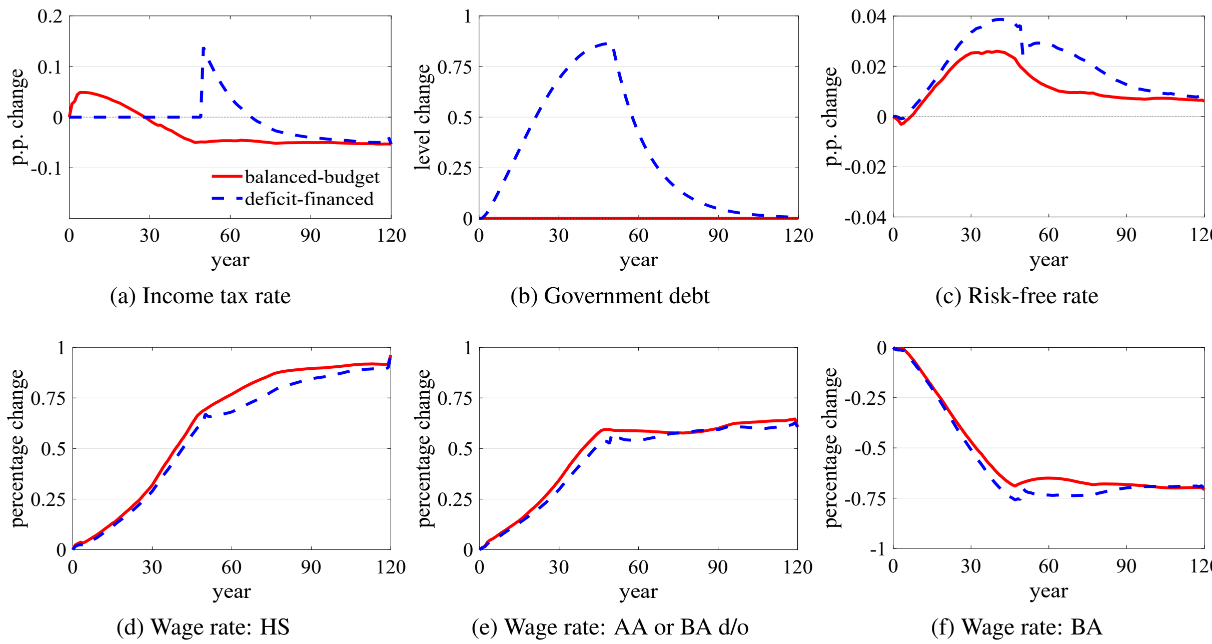


Figure D5: BA subsidy expansion: Changes in tax rate, government debt, and prices by transition type

Notes: The figure depicts the changes in the income tax rate (at initial steady state mean income), government debt, risk-free rate, and wage rates from the BA subsidy expansion under two transition types: (1) a "balanced-budget" transition in which the income tax rate level parameter is adjusted to balance the government budget in every period and (2) a "deficit-financed" transition in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate.

D.1.8 Percentage of population strictly better off from the subsidy expansions

Table D4 reports the percentage of population that is strictly better off from the AA and BA subsidy expansions under three cases: (1) a partial equilibrium case ("PE") in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) the balanced-budget transition; and (3) the deficit-financed transition.

In partial equilibrium, the percentage of population that is strictly better off is similar across the two subsidy expansions whether we focus on the total or within groups. Note that in partial equilibrium, every working-age individual post-IVT and every retiree is indifferent because there are no altruistic gains nor general equilibrium effects.

[37] We also note that after period 50 in the deficit-financed transition, the government increases its income tax rate significantly as it begins to pay down its debt. The increase in the income tax rate after period 50 reduces the incentive for consumers to work and save, which explains the rise in the risk-free rate and the fall in the wage rates in the periods that follow immediately after 50.

In general equilibrium, the two types of transitions highlight how the magnitude of the population that is better off depends on the form of financing: in both subsidy expansions, the deficit-financed transition leads to a larger percentage of the population being better off in comparison to the balanced-budget transition. This is because the tax burden is shifted to the future, which makes more working-age individuals in the first period of the transition better off. Furthermore, in the deficit-financed transition, both subsidy expansions make roughly 75 percent of the population strictly better off, with the BA subsidy expansion estimate being slightly higher.

Table D4: Population strictly better off: total and by mutually exclusive consumer groups

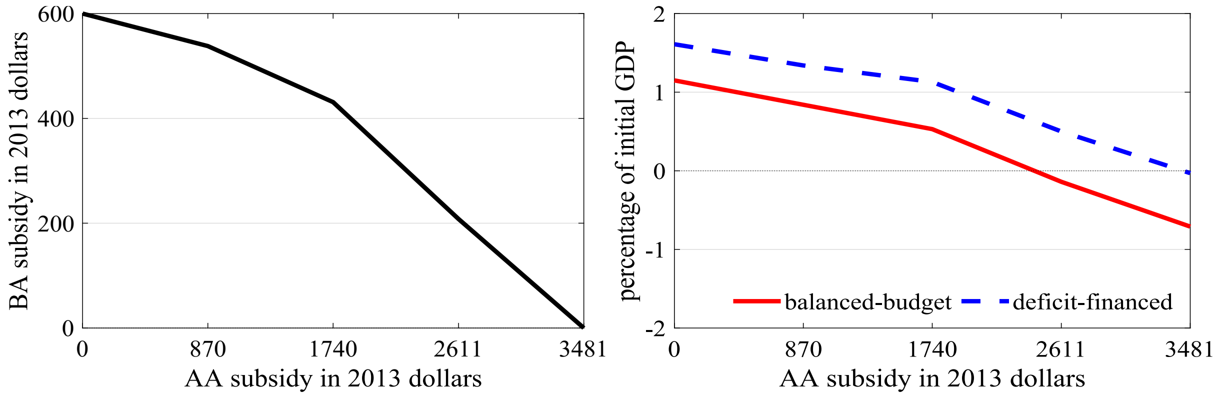
		AA subsidy expansion			BA subsidy expansion		
Consumer group	**Period**	**PE**	**Balanced-budget**	**Deficit-financed**	**PE**	**Balanced-budget**	**Deficit-financed**
All	$t \geq 1$	63	48	75	63	63	78
HS grads	$t = 1$	93	84	90	92	92	92
AA enrollees	$t = 1$	100	97	98	100	100	100
BA enrollees	$t = 1$	100	19	48	100	98	99
Pre-IVT age	$t = 1$	100	73	83	100	67	76
At IVT age	$t = 1$	100	52	80	100	95	100
Post-IVT age	$t = 1$	0	2	61	0	48	64
Retirees	$t = 1$	0	19	82	0	10	60
Future generations	$t > 1$	100	71	71	100	100	100

Notes: The table reports the percentage of population that is strictly better off from the AA and BA subsidy expansions under three cases: (1) "PE" is a partial equilibrium case in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) "Balanced-budget" is a general equilibrium case in which the income tax rate level parameter is adjusted to balance the government budget in every period; and (3) "Deficit-financed" is a general equilibrium case in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate. The statistics are reported for all consumer groups in the first row and for mutually exclusive consumer groups from the second to the last row. "IVT" refers to inter vivos transfer. "Period" refers to the transition period. Only in the partial equilibrium case, every working-age individual post IVT and every retiree is indifferent; otherwise no one is indifferent. The masses of future generations are discounted using the risk-free rate.

D.2 Optimal subsidy expansion subject to spending-equivalence

In the main text, we considered experiments in which only one of the two programs (AA or BA) is subsidized exclusively. We showed that a spending-equivalent BA subsidy is more beneficial to the economy than making the AA program tuition-free. In this section, we ask whether weakly expanding subsidies to both programs jointly subject to the spending-equivalence criterion described in Section 6.1 would alter this takeaway. We also note that the combination of tuition subsidies that maximizes total welfare is the optimal subsidy expansion policy subject to the constraint on government spending.

Figure D6 depicts, for different levels of the AA tuition subsidy, the spending-equivalent BA subsidy and the resulting total welfare changes in Subfigures D6a and D6b, respectively. Subfigure D6a indicates that, as expected, the spending-equivalent BA tuition decreases with the AA tuition subsidy. Subfigure D6b indicates that total welfare is maximized when the tuition subsidy is given exclusively to the BA program independent of whether we consider the balanced-budget or the deficit-financed transition. Therefore, subject to the spending constraint, it is optimal to exclusively expand the tuition subsidy to the BA program.



(a) Spending-equivalent BA tuition subsidy

(b) Total one-time transfer

Figure D6: AA and BA subsidy expansion: BA subsidy and total welfare change

Notes: The figure depicts for different levels of the AA tuition subsidy, the spending-equivalent BA tuition subsidy and the resulting total welfare changes in Subfigures D6a and D6b, respectively. The highest value on the x-axis is the AA tuition subsidy when the AA program is made tuition-free as described in Section 6.1.

D.3 Sensitivity analysis

In the main experiments in Section 6, approximately 70 percent of total grant expenditure was publicly funded and the remainder was privately funded (using the assets of the deceased). Our main takeaways comparing the subsidy expansions are not sensitive to whether or not grant subsidies are partially financed using the assets of the deceased (that is, private funds). To illustrate, in this section, we consider a sensitivity analysis in which the public share of grant expenditure is set to 1 and the private share of grant expenditure is set to 0 (the affected equations are (22) and (23)). This exercise illustrates the implications of subsidy expansions in a model that assumes all grant expenditure is funded by the government. Tables D5, D6, and D7 report changes in key aggregate statistics, total welfare for all consumers, and average welfare for mutually exclusive consumer groups, respectively. In each table, "Baseline" refers to the main experiments from Section 6 and is included for comparison purposes; "Grants fully public-funded" refers to the specification in which the public share of grant expenditure is 100 percent.

Panels A and B of Table D5 indicate that the AA subsidy expansion is more cost-effective than the BA subsidy expansion in subsidizing accessibility (increasing PSE enrollment) and increasing the share of graduates with a PSE degree. In contrast, Panels C and D indicate that the BA subsidy expansion is more effective in subsidizing quality in terms of increasing both labor productivity and output; in fact, the AA subsidy expansion leads to a fall in labor productivity and output. Panel E indicates that the AA subsidy expansion leads to a rise in the income tax rate, whereas the BA subsidy expansions leads to a fall in the income tax rate. Therefore, the AA subsidy expansion does not pay for itself in the long run, whereas the BA subsidy expansion does.

Table D6 shows that, in partial equilibrium, both subsidy expansions lead to total gains that amount to a one-time transfer worth more than one percent of initial GDP. Once we take general-equilibrium effects into account, the AA subsidy expansion leads to welfare losses under the balanced-budget transition and to gains that are close to zero under the deficit-financed transition; the BA subsidy expansion leads to substantial gains, above 1 percent of initial GDP, even after we take general-equilibrium effects into account. Table D7 shows that, in both subsidy expansions, the gains are dampened significantly for the many groups of consumers who are alive in the first period of the transition. The results of Tables D6 and D7 also highlight the importance of taking general-equilibrium effects and transition dynamics into account for a complete welfare analysis.

Table D5: Comparison of steady state changes after subsidy expansions in general equilibrium

Panel	Model specification	Subsidy expansion	
		AA	**BA**
A: PSE enrollment rate	Baseline	7.21	0.85
Units: percentage point change	Grants fully public-funded	7.23	0.26
B: Share of population with PSE degree	Baseline	2.16	0.83
Units: percentage point change	Grants fully public-funded	2.22	0.65
C: Average labor productivity	Baseline	−0.41	0.95
Units: percentage point change	Grants fully public-funded	−0.39	0.94
D: Output	Baseline	−0.46	0.75
Units: percentage point change	Grants fully public-funded	−0.46	0.73
E: Income tax rate at initial steady state mean income	Baseline	0.09	−0.05
Units: percentage point change	Grants fully public-funded	0.11	−0.04

Notes: The table reports general-equilibrium changes from the initial baseline equilibrium to the final steady state in the AA and BA subsidy expansions. Each panels describes the aggregate statistic. Labor productivity is computed for the working-age population as the weighted sum of individual productivities, $\epsilon(j, e, s)\,\eta$, divided by the mass of individuals.

Table D6: Comparison of total welfare changes

Model specification	AA subsidy expansion			BA subsidy expansion		
	PE	**Balanced-budget**	**Deficit-financed**	**PE**	**Balanced-budget**	**Deficit-financed**
Baseline	1.03	−0.71	−0.03	1.39	1.15	1.61
Grants fully public-funded	1.02	−0.85	0.13	1.52	1.18	1.94

Notes: The table reports total welfare changes for all consumers (total one-time transfers as a percentage of initial GDP) from the AA and BA subsidy expansions under the following cases: (1) "PE" is a partial equilibrium case in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) "Balanced-budget" is a general equilibrium case in which the income tax rate level parameter is adjusted to balance the government budget in every period; and (3) "Deficit-financed" is a general equilibrium case in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate.

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Table D7: Comparison of average welfare changes by mutually exclusive consumer groups

		AA subsidy expansion			BA subsidy expansion		
Panel	**Consumer group**	**PE**	**Balanced-budget**	**Deficit-financed**	**PE**	**Balanced-budget**	**Deficit-financed**
A: Baseline	HS grads	0.95	0.32	0.53	1.41	1.20	1.30
	AA enrollees	5.17	3.52	3.72	0.50	0.93	1.02
	BA enrollees	0.15	−0.35	−0.17	1.81	1.36	1.44
	Pre-IVT age	1.25	−0.78	0.14	1.13	1.12	1.72
	At IVT age	1.99	−0.49	0.72	1.98	1.47	2.22
	Post-IVT age	0	−1.44	−0.57	0	−0.35	0.20
	Retirees	0	−0.40	−0.07	0	−0.29	0
	Future generations	1.16	0.25	0.12	2.19	2.05	1.96
B: Grants fully public-funded	HS grads	0.91	0.26	0.53	1.52	1.23	1.39
	AA enrollees	5.17	3.43	3.71	0.56	1.00	1.16
	BA enrollees	0.15	−0.44	−0.19	2.00	1.42	1.56
	Pre-IVT age	1.26	−0.88	0.48	1.25	1.25	2.29
	At IVT age	2.02	−0.34	1.41	2.2	1.97	3.33
	Post-IVT age	0	−1.48	−0.27	0	−0.35	0.61
	Retirees	0	−0.66	−0.19	0	−0.53	−0.08
	Future generations	1.12	0.19	0.01	2.39	2.12	1.93

Notes: The table reports average welfare changes for mutually exclusive consumer groups (average one-time transfers as a percentage of initial GDP per capita 18+) from the AA and BA subsidy expansions under the following cases: (1) "PE" is a partial equilibrium case in which prices, tax rate, and transfers are held fixed at their initial steady state values; (2) "Balanced-budget" is a general equilibrium case in which the income tax rate level parameter is adjusted to balance the government budget in every period; and (3) "Deficit-financed" is a general equilibrium case in which the income tax rate is held fixed at its initial level for the first 50 years while the government borrows, and thereafter, the income tax rate level parameter is adjusted to balance the budget while the government repays its debt at a constant decay rate.

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